



Meeting of the Board
27 February - 1 March 2018
Songdo, Incheon, Republic of Korea
Provisional agenda item 15

GCF/B.19/22/Add.07

6 February 2018

Consideration of funding proposals – Addendum VII

Funding proposal package for FP065

Summary

This addendum contains the following three parts:

- a) A funding proposal summary titled “Financial Instruments for Brazil Energy Efficient Cities (FinBRAZEEC)”;
- b) No-objection letters issued by the national designated authority(ies) or focal point(s); and
- c) Environmental and social report(s) disclosure;



GREEN
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Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Programme Title: Financial Instruments for Brazil Energy Efficient Cities (FinBRAZEEC)

Country/Region: Brazil

Accredited Entity: The World Bank

Date of Submission: October 16, 2017

Contents

Section A	PROJECT / PROGRAMME SUMMARY
Section B	FINANCING / COST INFORMATION
Section C	DETAILED PROJECT / PROGRAMME DESCRIPTION
Section D	RATIONALE FOR GCF INVOLVEMENT
Section E	EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
Section F	APPRAISAL SUMMARY
Section G	RISK ASSESSMENT AND MANAGEMENT
Section H	RESULTS MONITORING AND REPORTING
Section I	ANNEXES

Note to accredited entities on the use of the funding proposal template

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal, or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

Please submit the completed form to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"[FP]-[Agency Short Name]-[Date]-[Serial Number]"

A.1. Brief Project / Programme Information		
A.1.1. Project / programme title		Financial Instruments for Brazil Energy Efficient Cities (FinBRAZEEC)
A.1.2. Project or programme		Project
A.1.3. Country (ies) / region		Brazil
A.1.4. National designated authority (ies)		Secretariat for International Affairs, Ministry of Finance
A.1.5. Accredited entity		The World Bank
A.1.5.a. Access modality		☐ Direct ☒ International
A.1.6. Executing entity / beneficiary		Executing Entity: CAIXA Econômica Federal (CEF) Beneficiary: Municipalities and private sector entities investing in public street lighting and private sector entities investing in industrial energy efficiency.
A.1.7. Project size category (Total investment, million USD)		☐ Micro (≤ 10) ☐ Medium ($50 < x \leq 250$) ☐ Small ($10 < x \leq 50$) ☒ Large (> 250)
A.1.8. Mitigation / adaptation focus		☒ Mitigation ☐ Adaptation ☐ Cross-cutting
A.1.9. Date of submission		October 16, 2017
A.1.10. Project contact details	Contact person, position	Silvia Martínez Romero, Senior Energy Specialist
	Organization	The World Bank
	Email address	smartinezromero@worldbank.org
	Telephone number	+1-202-473-0065
	Mailing address	I 5-506, 1818 H St. NW, Washington, DC 20433, USA

A.1.11. Results areas (mark all that apply)	
<u>Reduced emissions from:</u>	
☐	Energy access and power generation (E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.)
☐	Low emission transport (E.g. high-speed rail, rapid bus system, etc.)
☒	Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
☐	Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)
<u>Increased resilience of:</u>	
☐	Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)
☐	Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.)
☐	Infrastructure and built environment (E.g. sea walls, resilient road networks, etc.)
☐	Ecosystem and ecosystem services (E.g. ecosystem conservation and management, ecotourism, etc.)

A.2. Project / Programme Executive Summary (max 300 words)

Brazil's energy-related emissions have increased 130 percent from 1990-2012, against a Nationally Determined Contribution (NDC) of a 10 percent reduction by 2030. With extensive urbanization (over 85 percent of Brazil's population lives in urban areas), previous work financed by the Energy Sector Management Assistance Program (ESMAP) and others¹, led to the identification two sectors where transformative change could be made: municipal street lighting and energy efficiency (EE) in urban industrial areas. In order to overcome financing barriers and catalyze a significant shift in the trajectory of energy-related emissions, the Financial Instruments for Brazil Energy Efficient Cities (FinBRAZEEC) project, aims to unlock long term private financing for urban energy efficiency projects in Brazil by reducing the credit risk of LED street lighting (SL) and industrial efficiency projects (IEE) and enhancing their technical quality. The project will create an EE facility hosted at a local public bank, CAIXA Econômica Federal (CEF) and will support CEF in raising private financing for the efficient SL and off-balance sheet financing for IEE by incentivizing local Brazilian lenders to start lending on a "project risk" basis vs the currently dominant "corporate finance";

The project has two major components:

1. **An energy efficiency debt facility**, with a street lighting window and an industrial EE window. The facility will be initially funded by a concessional GCF loan of \$186 million, a non-reimbursable GCF grant of 5 million that will serve as the liquidity reserve, a CEF infusion of about \$180 million from its own resources and private local debt (syndicated by CEF). It is envisioned that this structure will crowd-in about \$400 million in investments from local or international private investors in debt. CEF will offer partial risk guarantees to local lenders to cover up to 50 percent of their potential losses (up to 200 million in total). Additionally, the World Bank will offer a \$200 million investment loan with deferred drawdown option (DDO²) features (heretofore referred to as "IPF contingent loan") , which will support CEF to have to make good the partial risk guarantees offered to the syndicated private lenders.³
2. **A technical assistance grant** to finance initial facility startup costs, including technical studies, operational costs, preparation of high-quality Private Public Partnership (PPP) street lighting sub-projects, and training of CEF teams in origination, assessment and monitoring of energy efficiency sub-projects in street lighting and industrial energy efficiency. The technical assistance component will be funded with a GCF Recipient executed grant of \$4 million and a World Bank \$1 million contribution (Bank executed).

FinBRAZEEC is expected to finance about 36 public street lighting and 14 portfolios of industrial EE projects, with an estimated lifetime emissions reduction of 17.4 million tons of CO₂e.⁴ See Annex 2, Annex 3.1, Annex 3.2 and Annex 14 for more details.

A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)	200 days after GCF Board Approval (estimated 15/09/2018)
Expected financial close (if applicable)	N/A

¹ Brazil Low-carbon Study although (2010) and "Inventory of Opportunities for Low-carbon Projects in Brazil" conducted with BMF-BOVESPA, the Sao Paulo Stock Exchange (2010).

² This is characterized in the financial products offered by the World Bank as an IPF contingent loan with deferred drawdown option (DDO) features. A new IPF-DDO product is being developed as a new product, to be considered for approval in the coming months.

³ Additional private equity contributions in the amount of around \$330 million are expected at the sub-project level. Sub-projects could also obtain debt from other sources, but the facility terms will likely be more competitive and offer WB supervision.

⁴ Each portfolio of projects (henceforth referred as Project-Portfolio corresponds to about 10 contracts. It is estimated that the facility will support 120 sub-projects of industrial energy efficiency. One sub-project may contemplate more than one technology intervention.

Estimated implementation start and end date	Start: <u>01/07/2018</u> End: <u>30/06/2025</u>
Project/programme lifespan	15 years 0 months



B.1. Description of Financial Elements of the Project / Programme

FinBRAZEEC offers an innovative financial structure that is designed to overcome some of the major barriers to financing of EE interventions in Brazil. After a decade of rapid growth and social progress starting in 2003, Brazil has entered a period of economic contraction and is in the midst of the deepest recession since the 1930s. Although external factors initially triggered the slowdown, political uncertainty in the wake of the *Lava Jato* corruption scandal and the impeachment of former Brazilian President Dilma Rousseff have driven investor confidence to new lows. Total private investments have fallen nearly 30 percent in the past three years, while the recession has put severe fiscal constraints on public spending at the federal, state and municipal levels. At the federal level, these constraints have sharply reduced the transfers from the government to state owned banks like CEF and the Brazilian Development Bank (BNDES), which served as major financier for municipalities and business alike.

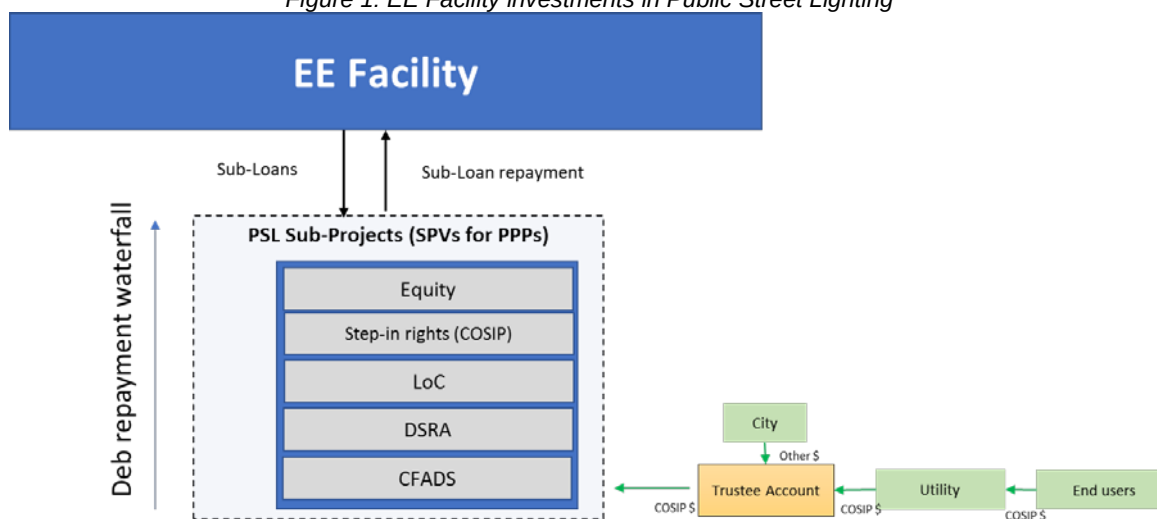
Overview of Facility Structure

FinBRAZEEC EE Facility is a debt facility that will lend to EE projects in the public street lighting and industrial sectors⁵. The objective is to develop new financial products for EE investment in Brazil that will facilitate investments at scale in these sectors, by de-risking cash flows of energy efficiency projects and making them more bankable. Once the concept is proven, the demonstration effect should reduce the need for concessional support over time and allow the concept to be rolled out to EE investments in other sectors (buildings, transport, etc.).

Market surveys conducted for the Feasibilities Studies indicate that there are very few investments taking place in these sectors. Some of the most important barriers include the lack of long-term financing in Brazil, high transaction costs associated with individual transactions, lack of off-balance sheet financing options for cities and industries, perception of risk in EE cash flows limiting the availability of financing, lack of standardization of business models, amongst others. This Facility aims to overcome these barriers by supporting project preparation to promote standardization in the market, creating new asset classes, aggregating investments into a diversified portfolio, and providing access to finance for private sector sponsors of EE projects, and implementing off-balance sheet financing approaches for cities and industries.

In the case of street lighting, the facility will primarily lend to Special Purpose Vehicles (SPVs) set up by the private sector, which will be granted concessions by the municipalities to modernize and operate the street lighting system on a Public Private Partnership (PPP) basis. See Figure 1 below.

Figure 1: EE Facility investments in Public Street Lighting



- The EE facility will make sub-loans to SPVs established by the concessionaire of EE public street lighting PPPs.

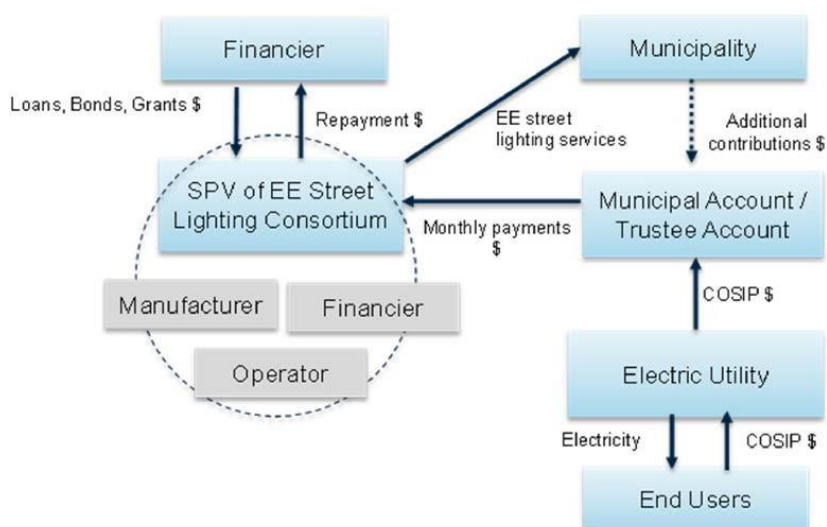
⁵ Projects are defined here in a broad sense to include SPVs, Funds, ESCOs, and other aggregators.

- Repayment of the loans will be funded by the monthly payment made by the municipality to the concessionaire, via a Trustee account, using COSIP revenues collected from end users by the utility.
- Under normal operations, the SPVs will use their Cash Flows Available for Debt Service (CFADS) to repay their debt service obligations to the EE Facility.
- If CFADS are insufficient, the SPV will use funds set aside in its Debt Service Reserve Account (DSRA), which will have a minimum required balance at all times.
- If the DSRA is depleted (or drops below its minimum balance and is not replenished within the required timeframe), the SPV will call on the Letter of Credit (LoC) that it will mandatorily purchase from a local commercial bank prior to signing the sub-loan.
- If the funds from the LoC are still insufficient to meet the SPV's debt repayment obligations, the sub-loan will go into default. At this point, the EE Facility will have remedies, including step-in rights to the Trustee Account holding any remaining COSIP revenues (in case they were not being properly transferred to the concessionaire for some reason). Lastly, the EE Facility will have limited recourse to the equity invested in the SPV.

A summary of a typical street lighting PPP model is as follows – see Figure 2 as well:

- Selection of a concessionaire through a competitive bid to whom the municipality grants a concession to deploy a wide range of responsibilities (installation, O&M) of the system over the PPP contract life;
- SPV is formed by the winning consortium (for example, an operator, lender and manufacturer), which is responsible for raising the financing for the project;
- The city remunerates the concessionaire through monthly payments using revenue raised from a dedicated levy on the electricity consumer – known as “COSIP” (or, if nonexistent or insufficient, from the municipal budget); and,
- COSIP is collected by the electricity utility directly on the electricity bills and flows to a municipal account or an escrow account. It is one of the unique aspects that make public street lighting PPP potentially bankable. COSIP will likely be an eligibility criterion for the selection of projects to be financed by CEF as part of the FinBRAZEEC facility.

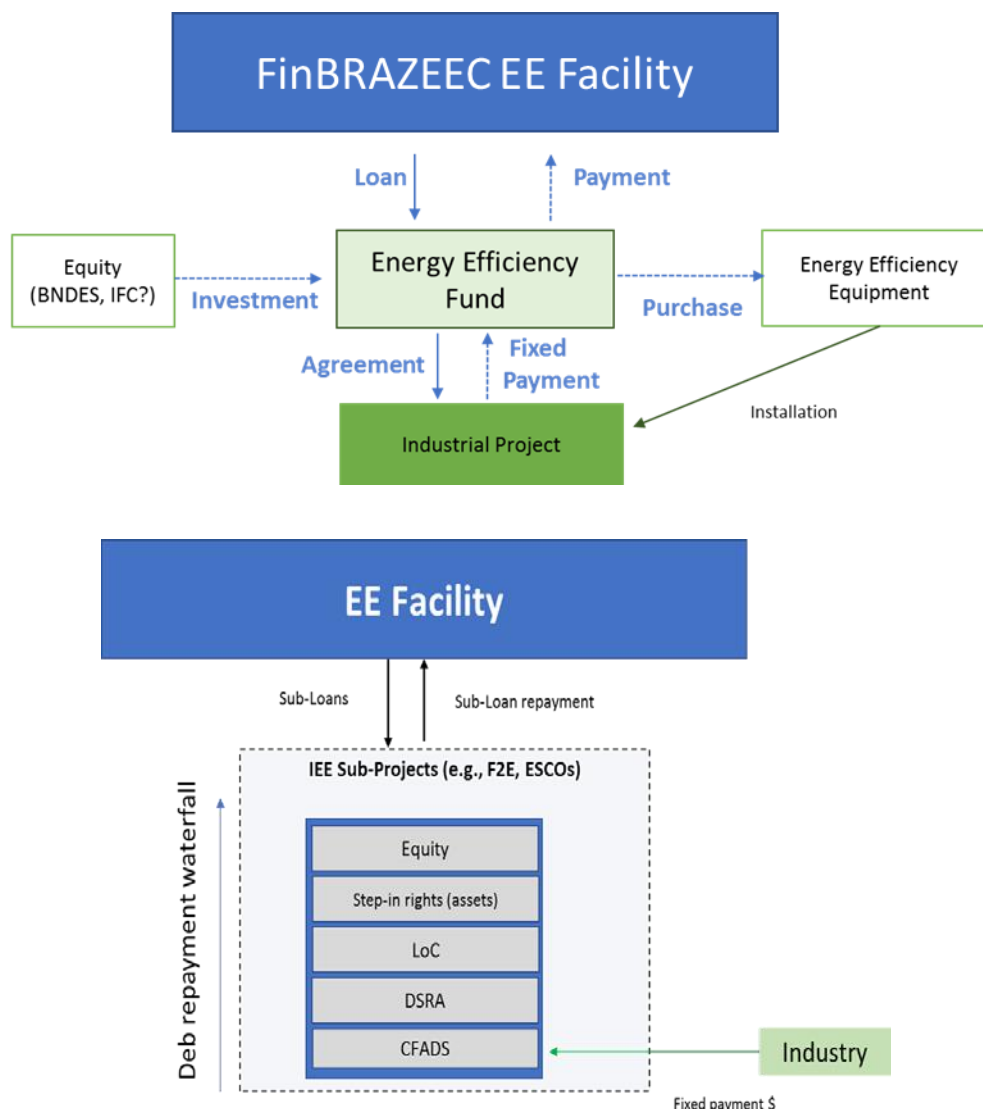
Figure 2: Example of Public Street Lighting PPP Structure



In the case of Industrial Energy Efficiency, there is not a sufficient number of institutions (ESCOs included) that offer off-balance sheet financing to industrial companies. It is still a nascent market. In the absence of this aggregation vehicle, a special account/fund needs to be created as part of the Project, which will make investments in the industrial customer's premises (host company) in exchange for a fixed payment that represents the deemed energy savings for the industrial company. See Figure 3 below.

This business model offers several advantages, the main one being the provision of an off-balance sheet financing vehicle, therefore removing one of the main barriers for industrial customers to invest in EE identified among industry companies in Brazil. Furthermore, by simplifying Measurement and Verification (M&V) and giving more certainty to cash flows, the mechanism will enable the creation of a new asset class for investments in EE, thereby attracting a wider range of potential investors, including those interested in investing in green bonds.

Figure 3: EE Facility investments in Industrial EE



The same applies to sub-loans to the IEE sector, except:

- Sub-loan borrower will be aggregators that provide off-balance sheet financing for IEE investments (F2E, ESCOs, etc.).
- The off-balance sheet aggregators will receive fixed monthly payments in exchange for installation of EE equipment. These fixed monthly payments from the industries to the off-balance sheet aggregators will be used to service their sub-loan debt to the EE Facility.
- The step-in rights in the case of IEE entail rights to recover the assets (EE equipment) installed in the Industrial companies.

Debt EE Facility

CEF is expected to raise US\$ 766 million for this Debt Facility, including US\$180 million loan from CEF, US\$ 186 million GCF loan, and US\$400 million from private sector investors through a syndication led by CEF.

CEF will be the lender of record for the syndicated loans, originating the projects. CEF and the private lenders would enter in an lender framework agreement to stipulate the maximum amount to be committed at facility level, the process to agree on the credit risk assessment for the sub-projects and the eligibility criteria of the sub-projects, among others. . CEF will blend resources from its own account, from the private lenders and the GCF, and will provide loans on a risk-adjusted basis to projects in street lighting and industrial sectors as per the framework agreement. The WB team will verify that the lending terms offered to the lenders cover all CEF's risk and that there are no hidden subsidies that could contribute to the distortion of the financial market, in agreement with WB Operational Policy OP 10.00 for Investment project financing with intermediary financing.⁶

Liquidity reserve and Credit enhancement

Given the barriers to investment for the private sector described above, the EE Facility will be designed to offer credit enhancements to attract the private sector lenders to participate in the syndication supporting the underlying projects, and enough liquidity reserves to allow the facility to be sustainable over time. Credit enhancement of the facility is offered through two means: (1) sub-projects will be asked to create debt service reserves account to cover 3-6 months of payments to the facility (2) step-in rights over COSIP (the levy on the electricity tariff collected by municipalities (3) a grant-funded "liquidity reserve" at the facility level that will absorb any first losses to the Facility, and (4) credit enhancement to cover up to 50 percent of any additional losses incurred by the private lenders in the facility offered by CEF and backstopped by the World Bank.

Liquidity Reserve at the facility level: a GCF grant (potentially complimented by a CTF grant or other climate funds) will be used to establish a liquidity reserve account within the EE Facility to improve its credit rating in order to attract private investors. The provision of liquidity at the EE Facility level addresses several risks at sub-project level, such as potential insufficiency of COSIP funds, technical performance of the equipment and systems, credit risk of industrial companies, and others that may result in delay our defaulted payments from concessionaires/industries to CEF. The Liquidity Reserve also provides flexibility for CEF to manage its risk, given that they are taking risk via their counterpart funds, their liability for the GCF loan, and the credit enhancement product offered to the private sector (described below). This is particularly important, given the fact that CEF will have to manage the risks associated with harmonizing the timing of identifying/investing in the pipeline of projects and the onboarding of capital into the fund. The Liquidity Reserve also reduces the amount of provisions that need to be put in place , thereby allowing the Facility to put a larger amount of its funding to use in investing in sub-projects.

Credit Enhancement Product: CEF will offer a credit enhancement to private sector lenders/investors that will be backstopped by the World Bank's IPF contingent loan loan. If the Facility cannot make payments in full to the syndicated lenders/investors after the other credit enhancement lines of defense have been exhausted, CEF will cover up to 50 percent of the value of the payment shortfall, drawing down on the IPF contingent loan to cover those obligations. This will cover payment shortfalls as well as losses. The credit enhancement, backstopped by the IPF contingent loan, is expected to improve the credit rating of the facility allowing it to reach a level⁷⁸ that will enable it to attract investors.

The credit enhancements are intended to improve the profile of investments in EE, enabling it to attract investors that would otherwise not be interested in investing at any price (due to the fact the Facility is investing in underlying projects and markets with no track record). The enhancements are not expected to have a substantial impact on the financing rate of these investors, but will instead allow the facility to attract a larger volume of capital.

⁶ <https://policies.worldbank.org/sites/ppf3/PPFDocuments/090224b0823064b7.pdf>

⁷ CEF needs to confirm that this will be possible for them while complying with BASEL III.

⁸ The current assumption is that the Facility could achieve a Brazilian A- rating, once the credit enhancements are considered. The facility will seek a shadow rating.

The GCF concessional loan will be critical to improving the lending terms for the facility, providing more flexibility of pricing, tenor, and grace period. All of these are critical to making the EE Facility viable. The analysis of the impact of the GCF concessional loan is provided in the subsequent section.

For Street Lighting projects, –given COSIP, its ringfencing, and the establishment of debt service reserve accounts for the sub-projects, the team estimates very low default rates, close to 1%. For industrial EE – there is no COSIP equivalent, but the underlying industrial clients are large, reputable companies and off-balance sheet financing vehicles will collateralize their payments. The team estimations with regards default rates are based on IFC experience in the IFC Industrial Energy Efficiency program in China. The first program (Chuee I) the estimated default rate is estimated about 10 percent but very low ex-post default rate. In Chuee II (therefore, first loss reduced to 5%). Therefore, assuming 1% for PSL and 5% for IEE, expected average default rate is 3%, or about US\$ 25 – 30 million.

Foreign exchange

The EE Facility will make loans denominated in BRL. The CEF and B lender loans will be denominated in BRL. However, the IPF contingent loan and the GCF loans are denominated in US\$. To limit the FX risk associated with the IPF contingent loan and GCF loans, CEF has requested the World Bank to perform a currency swap⁹ on its behalf upon disbursement of these loans. The IBRD will execute a swap transaction with a financial market counterparty. The BRL swap market is liquid at shorter and medium-term maturities (up to 10 years); beyond 10 years, a conversion can be performed on a case-by-case basis. In case a full-maturity conversion is not possible or is too expensive, a partial maturity conversion can be performed. An additional feature of the IPF contingent loan is that the repayment schedule can be adjusted at the time of disbursement, allowing CEF to better match its borrowing terms to the market conditions at that time.

As an example, as of market conditions in August 2017, the estimated cost in BRL of the IBRD IPF contingent loan and the GCF loan after the swap are as follows:

- IBRD IPF contingent loan : 15-year tenor, fixed interest rate of Libor (6m) + 0.90% in USD, would have a final costs in BRL after the swap of approximately 9.11%.
- GCF loan: 20-year tenor, fixed interest rate of 0.75% in USD, would have a final cost in BRL after the swap of approximately 5.65%.

It is important to note that the cost of the FX swap means that the impact of the benefit of the concessionality of GCF loan is reduced once it is converted into BRL. As a result, the project would benefit from higher concessional rates compared to the 0.75% interest rate assumed above.

Managing Risk

The first line of defense in mitigating risk of the facility will be to ensure a strong pipeline of high-quality sub-projects that will borrow from the facility, as well as strong capacity of CEF to conduct the due diligence of these projects.

⁹ IBRD provides, at the option of the borrower, specific instruments for managing foreign exchange risk (FX) for both the IPF contingent loan and the GCF loan. For IBRD loans, the borrower may opt for a currency conversion, which will be part of the Loan Agreement, while in the case of hedges (IBRD IPF contingent loan or GCF loan), the conditions are set out separately. The IBRD itself cannot assume the currency or interest rate risk of the local currency. Therefore, when the client requests local currency loans, IBRD must execute a swap transaction with a counterparty in the financial market. After the conversion of currencies, the principal and interest will be denominated in Reals, and payments will be made in US \$ at the current exchange rates. The cost of the hedge varies per market conditions.

The US\$ 4 million TA component of the GCF funds will be essential to supporting both of these activities. Some of the activities envisioned under the TA that will improve the quality of the sub-projects include conducting audits of industrial consumers, structuring the off-balance sheet mechanism(s) for investment in industrial EE, and identifying candidate cities and supporting the structuring of PPPs for public street lighting. In addition, the TA will support capacity building within CEF to conduct due diligence of EE projects in these sectors and implement a robust framework for risk management within the EE Facility.

In addition, the EE Facility will require that certain risk-mitigation structures be in place for the sub-projects. In the case of public street lighting, the lending agreements will include requirements for the borrowers to implement mechanisms to reduce the risk of payment defaults, for example through the implementation of Debt Service Reserve Accounts (DSRAs), step-in rights to the concession, recourse to the project equity, among others. In the case of industrial EE, loans are more exposed to credit risk from industrial companies, since there is not a cash flow based on ex-post energy savings that can be ring fenced (the role played by COSIP in public street lighting). Lending agreements between CEF and industries (directly or through an intermediary fund) will include a very thorough due diligence and some traditional credit enhancement mechanisms at sub-project level such as letters of credit and the ability to remove the equipment from the customer's premises.

In the event that the above measures were insufficient such that that EE Facility would have insufficient resources to make full payment to its lenders, qualified payment shortfalls from the Facility to its lenders would be covered by the grant-funded Liquidity Reserve (first-loss). The Liquidity Reserve would be replenished once the payment shortfall issues are resolved. Any remaining balances in the Liquidity Reserve at the end of the Project will be returned to the GCF. The terms and conditions under which the Liquidity Reserve would be drawn down and replenished will be determined during negotiations.

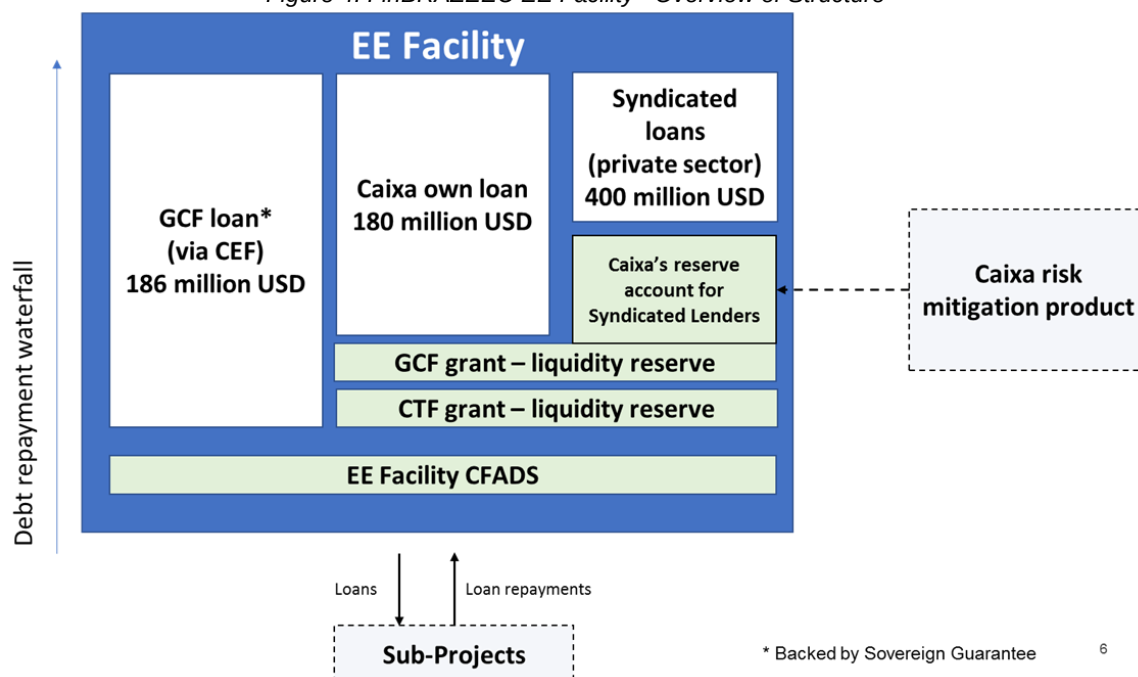
If these resources were fully depleted, any additional payment defaults would then be shared pari-passu amongst all lenders: CEF, GCF and the private sector lenders.

As in other typical syndication arrangements, CEF's syndicated loan management fees would be paid before any other repayments are made to lenders.

As part of the lending agreement with the WB, CEF will be requested to maintain a reserve account at a certain level to be determined by consultation with private lenders and CEF for payments to the B lenders CEF can replenish this facility using their own funds or requesting the disbursement from the IPF contingent loan. In case the facility runs out of funds to cover the loans repayments to B lenders, CEF will use the funds in the account and replenish the facility by requesting a disbursement from the IPF contingent loan.

Figure 4 below provides an overview of the indicative EE Facility structure:

Figure 4: FinBRAZEEC EE Facility– Overview of Structure



The breakdown of the allocation of funds to each component is shown below.

Component	Amount (for entire project)	Currency	Amount (for entire project)	Local currency	GCF funding amount	Currency of disbursement to recipient
Component 1. Financing Facility for Efficient Street Lighting and Industrial Energy Efficiency*	1,301	<u>million USD (\$)</u>	4,127	Million BRL	191	million USD (\$)
Component 2. Technical Assistance	5	<u>million USD (\$)</u>	15.8	Million BRL	4	million USD (\$)
Total project financing	1,306		4,142.8		195	

*Indicatively, street lighting is expected to absorb between 50-70 percent of the debt portfolio under the Facility. Industrial energy efficiency would absorb the remaining 30-50 percent.

Refer to Annex 2 (Feasibility study) for more information on the Street lighting and Industrial energy efficiency sub-projects.

B.2. Project Financing Information							
	Financial Instrument		Amount		Currency	Tenor	Pricing
(a) Total project financing	(a) = (b) + (c)		1,306.0		million USD (\$)		
(b) GCF financing to recipient	(i) Senior Loans		186.0		million USD (\$)	20 (15+5 grace) years () years	(0.75) % () % () % IRR
	(ii) Subordinated Loans						
	(iii) Equity				million USD (\$)	million USD (\$)	
	(iv) Guarantees						
	(v) Reimbursable grants *				million USD (\$)		
(vi) Grants *		9.0					
* Grants will be reimbursed to GCF to the extent the funds are not paid out at the end of the Facility. Economic and financial justification for the concessionality that GCF is expected to provide is provided in section F.1							
Total requested (i+ii+iii+iv+v+vi)		195.0		million USD (\$)			
(c) Co-financing to recipient	Financial Instrument	Amount	Currency	Name of Institution	Tenor	Pricing	Seniority
	Senior Loans	580.0 ¹⁰	million USD (\$)	Private Lenders, CEF	15-20 years	Est. 17.5%; 12% (CEF) (Reias)	pari passu
	Subordinated Loans	0.0	million USD (\$)				
	Equity	330.0 ¹¹	million USD (\$)	Concessionaires,	15-20 years	Est. 20-30% IRR (Reais)	senior
	Guarantees	200.0 ¹²	million USD (\$)	World Bank IPF contingent loan	IBRD terms	Libor + 0.9% (est. 9.1% in Reais)	
	Grant	1.0 ¹³	million USD (\$)	World Bank	n/a	n/a	
	Lead financing institution: World Bank, with CEF leading Syndication of private sector loans (B loans).						
* Please provide a confirmation letter or a letter of commitment in section I issued by the co-financing institution.							
(d) Financial terms between GCF and AE (if applicable)	As stated in Annex 5, the requested Accredited Entity Fee is seven percent (7%) of the GCF funding, subject to finalization of the AMA and FAA. The GCF interim policy on fees indicates the cap is seven percent (7%) of the GCF funding for large scale public sector projects/programs. It should also be noted that the Accredited Entity Fee is not included in the project budget in section (b) above.						
B.3. Financial Markets Overview (if applicable)							

¹⁰ Financing by CEF and private lenders is expected to be in local currency (BRL). The USD amounts listed here are indicative; the final USD equivalent value may vary due to exchange rate fluctuations.

¹¹ Equity contributions are expected to be in local currency (BRL). The USD amounts listed here are indicative; the final USD equivalent value may vary due to exchange rate fluctuations.

¹² This is classified as guarantee since the purpose is to provide some credit enhancement to private lenders, but in reality it is a contingent line of credit (IPF contingent loan)

¹³ This is the minimum amount that the team can obtain from different WB trust funds. At the moment, the team has managed to secure 500,000 from the Global Infrastructure facility (GIF), 250,000 from the Energy Sector Management assistance program and more than 200,000 in WB Budget support. A much larger GIF grant for PPP transaction advisory services, executed mostly by IFC is under discussion. The World Bank team could benefit partially from it to provide capacity building to CEF.

Macroeconomic setting. Brazil's economic and social progress between 2003 and 2014 lifted 29 million people out of poverty and inequality dropped significantly – the Gini coefficient fell by 6.6 percentage points in the same period, from 58.1 down to 51.5. The income level of the poorest 40 percent of the population rose, on average, 7.1 percent in real terms between 2003 and 2014, compared to a 4.4 percent income growth for the population as a whole. After this “Golden Decade” of rapid growth and social progress, Brazil's economy first stumbled and then fell into deep recession. The deterioration in both the external environment and domestic policies led to a steady decline in growth after 2010. Growth declined from an average of 4.5 percent per year in 2006-10 to 2.4 percent in 2011-14, followed by contractions of 3.8 percent and 3.6 percent in 2015 and 2016, respectively. While external factors triggered the slowdown, an expansionary policy response led to rapidly rising fiscal disequilibria and, together with rising domestic political uncertainty, to a loss of confidence and a sharp drop in investment. The *Lava Jato* investigation and the divisive impeachment process drove confidence to record low levels in early 2016. As a result, investment fell nearly 30 percent in the following 3 years.

A large structural fiscal imbalance lies at the heart of Brazil's present economic difficulties. While revenues are cyclical and have declined during the recession, spending is rigid and driven by constitutionally guaranteed social commitments, in particular on generous pension benefits. The large fiscal deficits have translated into rapidly rising public debt. General government gross debt rose from 56.3 percent of GDP at the end of 2014 to 69.5 percent by end-2016. Low foreign-currency exposure (3.9 percent of gross public debt as of March 2017), a decreasing share of fixed-rate domestic debt, and a comparatively low share of foreign-held government limit rollover risks in the near-term. The fiscal crisis extends to subnational governments, which have received a large share of lending by the World Bank and other International Financial Institutions in recent years. The problems are fundamentally the same—unsustainable welfare spending, largely mandated at the Federal level, driven by large and rapidly increasing pension obligations, in the face of declining revenues—but are made more acute because of limited own source revenues, reduced transfers from the Union, and limited access to borrowing.

The recession and tight monetary policy have brought inflation down rapidly in recent months. In 2015, the adjustment of energy, fuel, and public transport prices, combined with a food price shock and the sharp exchange rate depreciation, pushed inflation to 10.7 percent at year-end, well above the Central Bank's 4.5 ± 2 percent target. However, in 2016 rising unemployment and a rebound of the exchange rate put downward pressure on prices, with headline inflation dropping to 6.3 percent by end- 2016 and 3.6 percent by May 2017. Inflationary expectations have also fallen below the 4.5 percent target. The Central Bank tightened monetary policy sharply between 2013 and 2016 to control inflation and started easing policy only in late 2016 when there was clear evidence that inflation was falling and expectations had adjusted to the target range. The recession is also reflected in a sharp swing in the credit cycle. Real credit fell by 3.6 percent in 2015 and by 9.2 percent in 2016. The deteriorating economic situation combined with the increase in interest rates and the slowdown in credit growth has increased pressure on corporates, raising concerns about their ability to service their debt. Banking system soundness indicators remain adequate but the deteriorating economic situation and the impact of the ongoing scandals on the corporate sector may affect asset quality adversely. Credit risks have risen in the current recession with a slight deterioration of the quality of loans.

External sector adjustment has cushioned the impact of the recession. Since late 2014 the exchange rate has acted as a shock absorber and adjusted to levels in line with fundamentals. The Real fell around 30 percent relative to the US dollar in 2015 but recovered much of this loss in 2016 on an improved outlook for commodity exporting emerging markets. External balances have responded to the depreciation of the exchange rate and the sharp contraction in import demand. The current account deficit which reached 4.2 percent of GDP in 2014 (US\$104 billion) narrowed to 1.3 percent of GDP in 2016.

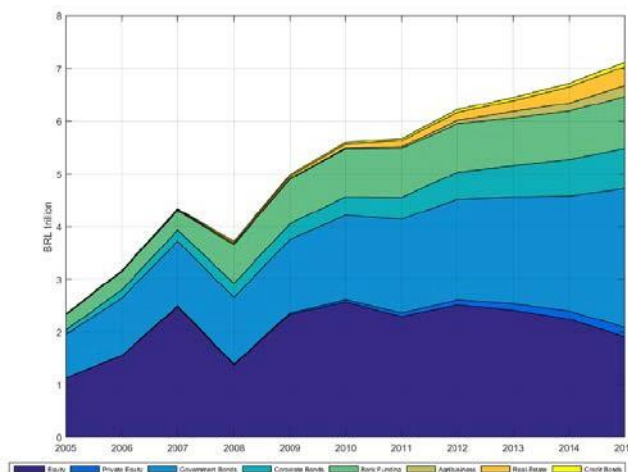
Brazil has returned to low but broad based growth in the first half of 2017. Private consumption has stabilized in recent quarters and exports have benefited from increased production of both agricultural and extractive commodities. Yet uncertainty remains high, reflected in falling investment which has further decreased to 15.5 percent of GDP over the past year (from 18.1 percent in 2015). A rebound in business confidence was capped by the political turmoil in May following new high-level corruption allegations. The unemployment rate increased to 13.7 percent in early 2017 but has receded to 12.8 percent in July, as job creation picked up in the informal sector.

Strong exports and weak domestic demand reduced the current account deficit to 0.7 percent of GDP in July from 1.3 percent at the end of 2016. Exports grew 11.3 percent over the past year, benefiting from higher commodity prices, especially for iron ore and a record soy bean harvest. Imports have begun to recover, growing 1.3 percent in the year to July. FDI inflows remained strong at 4.4 percent of GDP and reserves reached US\$381 billion (22 months of imports). Inflation has fallen rapidly, reaching 2.5 percent in August. Total credit to the economy still contracted at a real rate of 4.3 percent. Nonperforming loans remained moderate at 3.8 percent as of July. Congress approved a reform of the interest rate used for BNDES financing, linking it to long-term market rates.

A slow recovery is expected, with annual GDP growth in 2017 at 0.7 percent. Growth is expected to strengthen moderately in 2018 reaching 1.8 percent. The reduction in inflation and interest rates along with marginal improvements in the labor market should allow for a modest rebound in household consumption, while fiscal constraints will hold back government spending even in an election year. Private investment is expected to remain weak given low capacity utilization and high uncertainty in a complex election year. The recovery is expected to strengthen further in 2019, if the presidential election outcome enables reform efforts to proceed. Inflation is expected average 3.6 percent in 2017, well below the 4.5 percent target. However, the country's structural, fiscal problems remain unresolved, risking a return to economic and financial instability. On the upside, progress on structural reforms, removing protections and distortive subsidies in product and factor markets, could boost the economy's growth potential.

Brazil's capital markets. Based on the information available at the end of 2015, IMF estimated the total market capitalization of the invested global multi-asset market portfolio at BRL 7.12 trillion. Government bonds represent the largest asset class (37.04 percent), followed by equities (26.85 percent), bank funding (13.92 percent), corporate bonds (10.58 percent), and real-estate (5.10 percent). The total market capitalization of the three other asset categories is relatively small. IMF also provided historical composition of the Brazilian market portfolio going back to 2005.

Figure 5: Estimated Market values in the Brazilian Market Portfolio (2005-2015)



- Government Bonds:** With a total amount of BRL 2.6 trillion in outstanding securities, the Brazilian domestic government bond market has expanded rapidly since the mid-1990s and has become one of the most liquid and sophisticated among emerging markets. The domestic federal public debt (DFPD) is composed of a wide range of securities, including floating rate, fixed rate, inflation-linked, and dollar-indexed instruments. The Financial Treasury Bill (LFT - Letra Financeira do Tesouro), whose floating rate remuneration is based on the Selic4 rate, is the largest government security in terms of outstanding amounts. The National Treasury Bill (LTN – Letra do Tesouro Nacional), a zero-coupon fixed rate security, has expanded sharply in recent years and is now the second most important type of outstanding marketable liability. The National Treasury Notes Series-F (NTN-F - Notas do Tesouro Nacional série F), which is a standard coupon-bearing fixed rate security, has also expanded in recent years. The other NTN securities are indexed to various other indices. The NTN-B and NTN-C, inflation-indexed bonds, have increased their share of total marketable debt in recent years. Overall, the securities

discussed in this table represent 99 percent of the total domestic marketable debt of the federal government. Debt issuance by states and municipalities is modest.

- **Equity:** Brazil has one of the largest stock market capitalizations in Latin America. By the end of 2015, total equity market capitalization was BRL 1.9 trillion (US\$ 490 billion), representing 37.8 percent of GDP, with a diversified investor base including individuals, institutional investors, financial institutions, private and public companies, and foreign investors.

Brazil's equity market has grown rapidly in terms of market capitalization until 2007, when the equity market capitalization was BRL 2.48 trillion, representing 102.18 percent of the GDP. With the global financial crisis in 2008, Brazil lost equity market capitalization by about 45 percent. Its highest value over the past 10 years was BRL 2.57 trillion in 2010. Since 2012, Brazil has seen its total equity market capitalization drop by about 24 percent, from BRL 2.52 trillion to the current BRL 1.91 trillion.

Despite Brazil's large market capitalization, turnover ratio, and number of listed domestic companies, the country was considered to rank poorly in terms of transparency and corporate governance just a few years ago. In 2000, with the hope of encouraging corporations to adopt higher standards for corporate governance, transparency, and minority shareholder protection, as pre-requisites for listing, the Brazil's main stock exchange—BM&FBOVESPA, also known as São Paulo Stock Exchange.

- **Bank Funding:** The main fundraising instruments for banks in Brazil are, traditionally, demand deposits, term deposits, and savings. Over the past decade, the large and growing bank funding market has become more diversified with the introduction of the Financial Bill (LF – Letra Financeira) in 2010 and the Time Deposit with Special Guarantee (DPGE – Depósito a Prazo com Garantia Especial), in 2009. With the expansion of the menu of available financial instruments in recent years, the importance of funding through LFs has increased, although the Bank Certificates of Deposits (CDBs – Certificados de Depósito Bancário) still represent the main source of funds in this segment. The increase in the share of LFs and DPGEs partly reflects characteristics of these instruments, such as tax exemption and protection by the Credit Guarantee Fund (FGC – Fundo Garantidor de Crédito). Since 2012, state-owned banks have expanded rapidly and now make up half the banking system. They rely on subsidized financing for directed lending which accounts for over half of total credit. The authorities plan to reduce the reliance on state subsidies gradually as part of fiscal adjustment, but need to weigh the effects on corporate balance sheets and asset quality. Corporate debt constitutes almost one third of bank assets and banks' exposure to problem loans of large corporates is high. Banking system soundness indicators remain adequate but the deteriorating economic situation and the impact of the ongoing scandals on the corporate sector may affect asset quality adversely. Credit risks have risen in the current recession with a slight deterioration of the quality of loans.
- **Corporate Bonds (Debentures):** Companies do not need to issue stocks to raise financial resources. Instead, they may find it more advantageous to issue either debentures or commercial papers. These securities represent cheaper sources of financing when compared to bank credit, with more flexible maturities. Some of the most profitable bonds/debentures in 2016 include: (i) EDP @ IPCA + 8.26%, CPS @ CDI + 2.5%, Vale @ IPCA + 6.62% and sovereign bonds @ IPCA + 6% to 7.2%.
- **Credit Bonds:** Although the credit bonds market is still small in Brazil, it has grown significantly in recent years. By the end of 2015, the amount outstanding in this market was BRL 89 billion, from BRL 6 billion in 2005. The Bank Credit Note (CCB – Cédula de Crédito Bancário) has been the main instrument. The amount of CCBs increased from BRL 3 billion in 2005 to BRL 38 billion in 2005. Meanwhile, the amount of Export Credit Notes (NCEs – Notas de Crédito à Exportação) increased from BRL 1 billion in 2005 to BRL 38 billion in 2015.

Indices. In the Brazilian equity market, two benchmarks are widely used: the Bovespa Index (Ibovespa) and the Brazil 100 Index (IbrX-100). The Ibovespa is a total return index comprising the most representative companies in the market, both by market cap and traded volume. It is the benchmark index of the São Paulo Stock Exchange. For the government bonds market, the Brazilian Financial and Capital Markets Association (ANBIMA) releases several market indices

composed of fixed-income securities, which are widespread among market participants. The main indices are the IRF-M (Fixed Rate Government Bonds Index), the IMA-B (Consumer Price indexed Government Bonds Index), the IMA-C (General Price indexed Government Bonds Index), the IMA-S (Floating Interest Rate Government Bonds Index), and the IMA Global (ANBIMA Market Index).

Financial infrastructure gap. Infrastructure investments are perceived as risky, which leads to a debate within the stakeholders about who will absorb the risks: the granting authority or the concessionaire. Regarding Law no., 8.987/95 about concession contracts in Brazil, the risks related to the concession are usually private. Nevertheless, in accordance with international recommendations, the government has been making different contractual arrangements to take on some of the risks involved in infrastructure investment projects and attract more investments. The table below describes the most adequate risk allocation among stakeholders, according to Global Infrastructure Hub (GIH).

Risks	Allocation	
	GIH Developed	GIH Emerging
Land Purchase and Site Risk	Public	Public
Environmental and Social Risk	Shared	Private
Design Risk	Public	Shared
Construction Risk	Shared	Shared
Completion Risk	Shared	Private
Performance/Price Risk	Shared	Private
Resource or Input Risk	Shared	Private
Demand Risk	Shared	Private
Maintenance Risk	Shared	Private
Force Majeure Risk	Shared	Shared
Exchange and Interest Rate Risk	Shared	Shared
Insurance Risk	Shared	Shared
Political Risk	Public	Public
Regulatory/Change in law Risk	Shared	Public
Inflation Risk	Shared	Public
Strategic Risk	Private	Private
Disruptive Technology Risk	Shared	Public
Early Termination Risk	Shared	Shared

(Source: FGV)

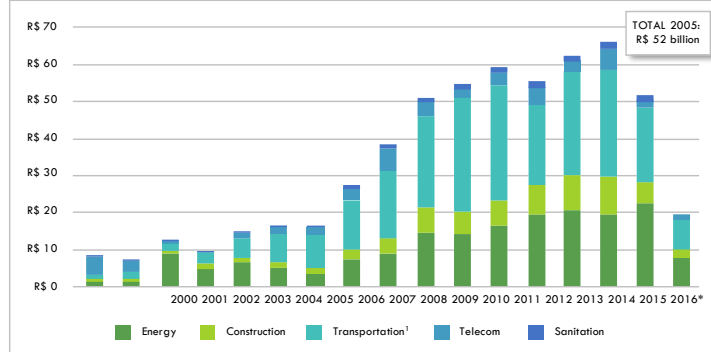
Moreover, the long maturity of infrastructure contracts involves a reasonable possibility of financial imbalances occurrences caused by Material Adverse Effect Uses in Government Action (MAGA). In such a context, a government authority that acts against the borrower or any of its subsidiaries can reasonably be expected to cause material adverse effect.

The Brazilian Development Banks, in particular BNDES and CEF have been the most important financiers of infrastructure and industrial projects in the country in recent decades. They provide funding indexed to the TJLP¹⁴, a long-term interest rate, which is far lower than regular rates from commercial banks. The BNDES and CEF together account for 62 percent of the total investments in infrastructure in Brazil in 2014.

Until 2009, the Workers' Support Fund (FAT) had been the most relevant source of funding for the BNDES. In 2009, the funding situation changed as the National Treasury assumed the role of the BNDES's main creditor. The bank's disbursements in infrastructure increased significantly after 2003, going from US\$938 million to US\$11.2 billion in 2014. However, with the macro-fiscal crisis, the National Treasury cannot anymore provide low-cost funding source to the public banks and this former financing model has come to an end and the amount of disbursement of these banks has fallen drastically as shown in Figure 6 for BNDES.

¹⁴ The TJLP depends on historical inflation, expected inflation, and country risk premium. The methodology is updated every year by the BNDES. For details, see (in Portuguese): http://www.bndes.gov.br/SiteBNDES/export/sites/default/bndes_pt/Galerias/Arquivos/produtos/download/tjlp.pdf.

Figure 6. BNDES Disbursements in Infrastructure (BRL, billions)



Source: BNDES, elaborated by FGV CERI.

Attracting commercial banks and private investors has therefore become critical to fulfill infrastructure and industry investments needs. The challenge however, is to propose investment opportunities that can compete with government bonds. Indeed, the government securities market in Brazil is widely developed and, as the basic interest rate in Brazil (SELIC) exceeds similar rates prevailing in the rest of the world, these assets are highly attractive to domestic and foreign investors. As a result, banks and pension funds invest massively in Brazilian Treasuries and have negligible participation in financing long-term assets or infrastructure. Because Brazil has not developed a secondary market for private or corporate bonds, savers have concentrated resources in government debt, which offers a profitable and much more liquid secondary market. In addition, commercial banks charge very high mark-ups, turning real interest rates of commercial loans in Brazil among the highest in the world, above 20% for corporate loans. As a consequence, there is no market reference for long-run interest rates with a duration compatible with projects-based investment. The Brazilian interest rate curve is characterized by a high level of short-term rates and upward sloping format—long-term rates do not have sufficient liquidity and short-term rates are very high.

The new government of Brazil is committed to addressing these issues and has engaged in a severe macro-fiscal adjustment, which includes reforms able to decrease and stabilize the growth of the public debt to eventually reducing short-term interest rates. Meanwhile, the share of resources that commercial banks and institutional investors have been allocating to project financing, either in loans or bonds, has remained very limited and mainly concentrated on refinancing projects that are already operational, thus with a lower risk-profile. Regarding commercial Banks, the implementation of the Basel III regulations is expected to reduce even more their ability to invest in high risk assets. Another issue is the market's ability to accept long-term bonds. Despite the need for adequate financing of investments with long maturation period, debentures issued since 2000 have maturities under six years on average. After the 2008 economic crisis, the average maturity fell even more, from six to four years.

Need to de-risk project-based investment and initially blend private resources with concessional funds

In such a scenario where the reforms initiated by the government will still require time before delivering the expected results, attracting private resources while still maintaining sustainable financial terms for investment projects requires: (a) to de-risk projects and create new asset classes, by having public development banks assuming a new role of guarantors, as proposed by the FinBRAZEEC project, and (b) blend these private resources with concessional funds. The level of concessionalism of the latter has to be sufficient to compensate for the cost of hedging the foreign currency risk. Moreover, it must be noticed that is a limited number of local Banks that have enough capital to undertake infrastructure investments (Santander, Banco do Brasil, ITAU, BRADESCO, and Votorantim).

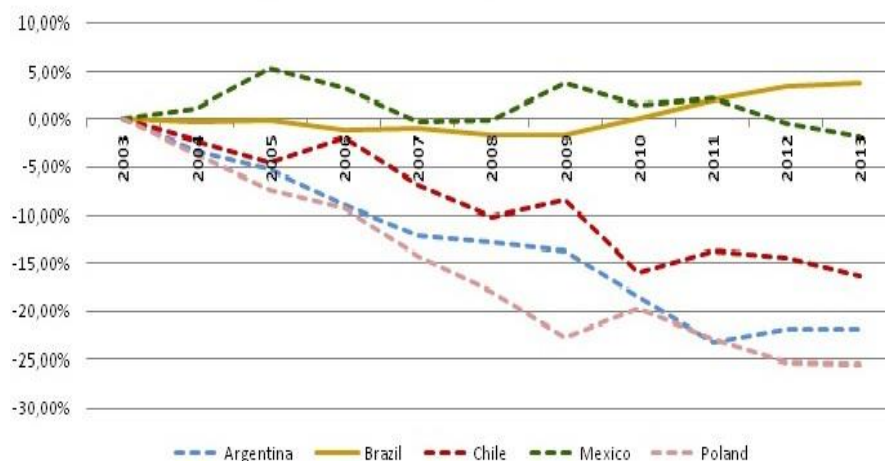
On the public street lighting side, there is both an encouraging regulatory environment that makes establishing SPVs to provide street lighting a tax-efficient and effective mechanism for providing lighting, while the limited fiscal space on municipal balance sheets makes such investments financially attractive to cities that cannot provide the large capital infusions required. Similarly, on the industrial EE front, the FinBRAZEEC facility can provide financing that would have previously been provided via a federal credit program. Notably, in partnering with a state-owned bank like CEF, the facility doesn't require new and unfamiliar modalities that may otherwise curtail lending.

C.1. Strategic Context

Brazil has a large and sophisticated energy sector. It is the 8th largest electricity generator in the world and has implemented a number of innovative energy sector policies, including successful renewable energy auctions. The country has achieved huge gains in energy access in recent decades – access rate reached 99.5 percent in 2012. Brazil has the third largest hydro installed capacity in the world, and hydroelectricity is the largest source of Brazil's electricity supply (64 percent in 2015), followed by natural gas (13 percent) and biomass (eight percent). In 2015, electricity demand was 615.9TWh, 94 percent of which was supplied by domestic production.

Although Brazil has a relatively “clean” grid due to the high penetration of hydro energy, Brazil's national energy-based emissions increased more than 130 percent for the 1990 – 2012 period, largely due to a growing dependence on thermal generation to meet peak demand. Brazil also performs relatively poorly in its measure of energy intensity (primary energy consumption / GDP). For example, when compared to LAC countries over the period 2003 to 2013, Brazil is the only country that has a positive growth rate of energy intensity (see Figure 7 below).

Figure 7: Energy Intensity
Primary Energy Consumption / GDP National Currency
(growth rate - indexed to 2003 figures)



Source: IMF and BP

Brazil has high urbanization rates (~85 percent), and thus, energy emission sources are predominately concentrated in urban areas. This is in line with global trends: cities are among the world's biggest consumers of electric energy, accountable for two-thirds of total electricity consumption and for over 70 percent of global greenhouse gases emissions.

While very clean, the Brazilian energy matrix is facing a growing challenge to meet the growth of the energy demand and comply with environmental protection requirements. Adapting hydropower to more stringent environmental rules in the Amazon region has meant forgoing multi-years reservoirs that would firm up the very variable river flows: out of 20,000MW planned to enter in operation between 2013 and 2018, only a single percent (200MW) have reservoirs; all of the rest are run-off river. Therefore, hydro storage capacity fell from 6.3 months to 4.7 months in the last 10 years and might further fall to 3.8. The reduced storage capacity has considerably increased the vulnerability of the of the hydro-dominated Brazilian power system to climate variations.

Moreover, the commitments that Brazil made at COP-21 as part of its Nationally Determined Contributions include a reduction in national GHG emissions by 37 percent below 2005 levels in 2025 (with intention to reduce its emissions to 43 percent below 2005 levels in 2030). However, the current historically low level of reservoirs that has resulted from insufficient rainfalls in recent years, demonstrates the threat that climate change represents for Brazil's energy matrix.

The few studies available on the impact of climate change on hydro-energy due to changes in rainfall patterns, point toward a significant vulnerability of the Brazilian power system. A direct consequence is the increment of the use of gas and fuel-oil-based thermal power plants. Consequently, the annual GHG emissions of the sector in a dry year are four times higher than in years with favorable hydrology.

While non-conventional renewable energy, and in particular wind energy, is finally ramping-up after lagging behind for many years compared to other countries - installed capacity increased by more than a factor of 20 between 2008 and 2014 (from 323MW to estimated 7,628MW) – wind energy represents still less than 10 percent of the installed capacity, and its intermittency and regional concentration require a complex integration planning with other sources, including thermal. In addition, the construction of transmission lines still represents a question mark to further increase the participation of wind in the energy matrix. Bioelectricity from cogeneration of heat and power (CHP) from sugarcane bagasse has also significantly increased over the last decade, from 1.1GW to 9.6 GW of installed capacity and could continue to grow if proper conditions for investing in this distributed, seasonal energy can be improved.

Against this backdrop, a quick development of energy conservation will also be needed to contain the growth emissions of the energy sector to levels consistent with the NDC, while still ensuring the supply-demand balance. Indeed, Brazil's NDC states a goal of achieving 10% efficiency gains in energy sector by 2030 (approx. 105 TWh saved by 2030). Supply-side reforms are not sufficient to reduce volatility in Brazil's complex energy matrix, nor would they be cost-effective on their own. Demand-side management interventions must play a larger role. However, initiatives in this area have been limited to date.

Brazil has two primary public financing lines for energy efficiency projects – the National Energy Conservation Program (PROCEL) and the Energy Efficiency Program (PEE). In 2011, the government implemented its first Energy Efficiency Plan, which established a specific tariff to finance utility-driven EE investments under PEE; Brazil is the only country in LAC with such a mechanism in place. These EE programs have achieved some results in a few target areas, but the overall cost-effectiveness is still a point of concern. For example, resources have been prioritized for lower income groups at the exclusion of other large and low-hanging fruits (e.g., industrial EE). The electricity sector regulator, ANEEL, has been working to make the efficient use of PEE resources by introducing a competitive process for EE projects. Results are still being assessed. Finally, the use of EE has not been considered as an adaptation option in the recent years. Despite the successful previous experience with EE measures to respond to the 2001 energy crisis, no demand-side actions were implemented during the more recent 2013-2014 power crisis, resulting in the continuous dispatch of thermal generation.

Regarding the demand of fuels in the industry, the only program implemented so far is the limited National Program for the Rationalization of the Use of Petroleum Products and Gas (CONPET), created on July 1991 by a presidential decree. CONPET depends on resources allocated and mainly executed by Petrobras, amounting only around \$3 million per year on average and focusing mostly on soft measures, like labeling and educational programs in schools.

Therefore, Brazil has significant opportunities to reap economic efficiencies through EE, particularly given its high energy intensity and energy supply/price volatility due to heavy reliance on hydrology. With the high rates of urbanization, improving urban energy efficiency will be particularly important. However, the barriers to invest in urban EE are often even higher than typical EE projects, such as higher transaction costs or municipal credit risk, among others.

In a first attempt to identify concrete opportunities to reduce energy consumption and GHG emissions in the urban context in Brazil, the World Bank team embarked on a strong engagement with Brazilian cities under the cross-sectoral US\$1.3 million Brazil Energy Efficient Cities (Brazeec), a non-lending technical support program funded by the ESMAP. Brazeec has examined several EE and GHG reduction opportunities, including street lighting, industrial EE, public buildings, and transport. Two sectors were identified for their potential impact and scalability of investments in urban EE in the near-term: Public Street Lighting and urban industry.

Public Street Lighting

The World Bank estimates that Brazil has more than 18.5 million points of light, with a service penetration of approximately 95.5 percent of residential areas. Public street lighting represents about 3-4 percent of total electricity consumption in Brazil and close to 5 percent of peak consumption. The installed public lighting footprint is predominantly comprised of high-pressure sodium mercury vapor lamps. LEDs' technology penetration is quite modest, although many cities have pilot projects in the pipeline to implement this technology. By replacing mercury and sodium lamps with LEDs, cities can achieve electricity savings ranging from 50 to 80 percent and Operation and Maintenance (O&M) savings of about 40 percent.

A number of factors have come together to make EE investments economically viable in Brazil: (i) the quality and costs of LEDs have reached a threshold where the economics of projects in Brazil are viable; (ii) new regulatory changes require all electricity distribution companies to transfer public street lighting assets to municipalities, creating opportunity for investments in EE through Public Private Partnerships (PPPs); (iii) the tariff increases in 2015; and (iv) existing legal framework for creating tax-efficient, special purpose entities by cities to invest and recover street lighting costs.

The World Bank's "Lighting Brazilian Cities" report (Annex 12) provides an in-depth analysis of this sector.

Despite the favorable conditions described above, investments in EE public street lighting have been slow to take off. Only a few of PPPs have been signed or are at advanced stages for having concessions granted (from which, only one is a large city, Belo Horizonte).

Table 1: PPP streetlighting projects in planning or implementation

Date	Name	Status	Amount
9-Oct-17	Salvador street lighting PPP project	Project in tender	635
19-Jul-17	Açailândia street lighting PPP project	Project signed	303
5-Jul-17	Cuiabá street lighting PPP project	Project canceled	
14-Mar-17	Porto Alegre street lighting PPP project	Project in planning	
15-Dec-16	Mauá street lighting PPP project	Project signed	60
2-Dec-16	São José de Ribamar street lighting PPP project	Project signed	93.78
17-Oct-16	Guanambi street lighting PPP	Project in planning	38.6
20-Sep-16	Uberlândia street lighting PPP project	Project in planning	
1-Aug-16	Guaratuba street lighting PPP project	Project signed	60
22-Jul-16	Belo Horizonte street lighting PPP project	Project signed	tbc
20-Jun-16	Barbacena street lighting PPP project	Project in planning	
8-Jun-16	Aparecida street lighting PPP project	Project in planning	
13-May-16	Street lighting PPP in Sao Paulo	Project in tender	
3-May-16	Uberaba street lighting PPP project	Project in tender	32
31-Aug-15	São Bernardo do Campo street lighting PPP	Project in planning	
29-Jul-15	Caraguatatuba street lighting PPP project	Project signed	20
16-Jul-15	Vitória city street lighting PPP project	Project in planning	

(Source: Infra PPP market Intelligent), 2017

Market research indicates the following barriers:

1. Constrained municipal balance sheets, including the Fiscal Responsibility Law on municipal indebtedness, which imposes strict borrowing limits on cities;
2. Municipal credit risk, as demonstrated by the recent case of the Sao Paulo PPP for EE street lighting, in which all major international firms pulled out from the tender process due to lack of acceptable guarantee mechanisms for repayment by the municipality;
3. The challenging macroeconomic situation in Brazil leading to high interest rates and high hedging costs due to exchange rate volatility; and,
4. High project preparation/transaction costs involving a new technology and low municipal capacity to structure PPPs.

Industrial Energy Efficiency

The potential for efficiency in the industrial sector is enormous. A study from the American Council for an Energy-Efficient Economy (ACEEE) analyzed the performance of the 16 largest economies. Brazil was ranked #16 (last) on industrial energy efficiency. Brazilian industrial associations (National Confederation of Industry Brazil- CNI and Associação Brasileira de Grandes Consumidores Industriais de Energia e de Consumidores Livres-ABRACE) have carried out several studies demonstrating the high potential for industrial EE and GHG emissions reduction, in fossil fuel-based thermal processes (potential savings range from 8 to 40 percent of current consumption depending on the sector). The World Bank also published two reports, which detail the potential for mitigation options related to energy generation and use, with specific results on the potential related to energy efficiency in the industry¹⁵.

While a large share of Brazilian industrial companies is in urban industrial districts, cities often leave industrial districts out of their citywide plans for emission reductions, largely because they are outside of their direct purview, in contrast to areas like public buildings, waste, street lighting, etc. As example is the Campo Grande industrial district in the metropolitan area of Rio de Janeiro, the 4th largest in the country and accounting for 33 percent of Rio's urban emissions.

More than 150 projects with high-return and short payback periods have already been identified among large industrial energy users. Projects encompass a variety of sizes and technologies, such as automation in furnaces, regenerative burners, heat recovery, efficient motors, frequency controls in pumping, etc. Some investments in EE projects present very low paybacks of 1-3 years, and internal rates of return ranging from 30 to 90 percent. More capital intensive projects, such as co-generation, may present larger paybacks of 7 years and IRRs of about 20 percent. Potential for EE should also be significant in medium and small industries and commercial establishments.

An assessment conducted by CNI and ABRACE members, revealed that the current high level of indebtedness and lack of room in balance sheets prevents industrial companies from investing in anything outside of their "core" business (such as EE), despite many of these companies having sufficient net operating incomes that could be used to pay for efficient equipment and EE upgrades over time.

Investments are also constrained by the lack of specific government programs targeting industrial energy efficiency. PROCEL (Eletrobras), and PEE (ANEEL) - the largest government-led EE programs in Brazil, have primarily targeted EE investments in low-income residential, commercial, street lighting, and small industries. Moreover, private sector financiers in Brazil (direct financing or via the ESCO market) typically shy away from providing upfront financing for large industrial projects because they are very capital intensive and viewed as risky, since banks are not sufficiently familiar with the technologies or EE cash flow streams.

¹⁵ "Brazil Low-carbon Scenario" – Energy technical report, de Gouvello et alii, The World Bank, 2011, 199 p.
"Levantamento de oportunidades concretas de projetos de baixo carbono no Brasil", The World Bank and BMF-BOVESPA, 2011, 242 p.

To overcome these barriers, off-balance sheet solutions leveraging private sector capital are required, FinBRAZEEC seeks to catalyze investments providing a ready source of funds to projects.

FinBRAZEEC will be implemented in parallel and in close coordination with the broader, ongoing dialogue between the World Bank and the Government of Brazil on energy sector reform, including on measures to promote energy efficiency. The World Bank currently has a number of technical assistance activities underway in this regard, including:

- a. The WB is currently working with the government of Brazil to support the reform agenda both in the gas sector and the power sector, including distributed resources like demand response. This support is provided under the ongoing Advisory Services and Analytics (ASA) project “Revisiting the Power and Gas Sector Reforms in Brazil”.
- b. This ASA has also informed a restructuring of the large \$50M lending Technical Assistance program (“META” project), to implement the energy reform agenda, including activities related to planning and regulatory framework for EE.
- c. Under the \$3 million Technical Assistance activity under the World Bank’s Partnership for Market Readiness (PMR), the World Bank is supporting the government in designing new policy instruments that combine existing policy instruments in the energy sector with carbon pricing instruments, with the objective of creating a new incentive framework for energy efficiency to help achieve the Brazilian NDC target.

Therefore, FinBRAZEEC will serve as a financial instrument to finance the investments required to respond to the new incentive framework being implemented by the government with the World Bank’s support.

C.2. Project Objective against Baseline

Public Street Lighting

Identification of the Baseline Scenario

The analysis assumes that, without this project, municipalities would slowly replace burned out lamps in the existing inventory with LEDs. However, given the municipal constraints on financing (under the Fiscal Responsibility Law, described in detail in the Lighting Brazilian Cities report), without the financing of these projects through PPPs by the EE Facility, cities would be limited to using only surplus revenues available from their street lighting budget, after paying for all electricity and O&M costs. For many cities, the source of revenue to fund public street lighting is known as COSIP, a tax levied on end-users via their electricity bill.

Under the Baseline Scenario, the Feasibility Study analysis estimates that an “average” city would eventually fully convert its inventory to LEDs; however, due to its financing and budget constraints, the process will take place over a long period of time. By the end of 13 years, the Baseline Scenario assumes that 30 percent of the existing inventory has been replaced by LEDs.

Project Scenario

The Project Scenario, as described above, involves the establishment of a PPP that will finance the upfront cost of modernization of the entire inventory of public street lights with LEDs. By bringing in private sector financing through the FinBRAZEEC facilities, the project is able to overcome the one of the key barriers for implementing a LED modernization project: upfront financing costs under constrained budget and balance sheet conditions of municipalities. The project also overcomes other barriers, including improving municipal capacity and reducing the transaction costs involved in the preparation of projects through standardized approaches, and identifying investors willing to invest in these projects through a de-risked facility.

Industrial Energy Efficiency

Identification of the Baseline Scenario

The baseline scenario assumes that energy efficiency projects in the industrial sector will end up being developed, but it will take significant time for the implementation of proper off-balance financing mechanisms accessible to industrial companies. Due to the barriers currently faced to finance industrial EE projects, it is assumed that new efficient technologies would only be implemented in eight years.

Project Scenario

FinBRAZEEC's EE Facility will facilitate off-balance sheet financing to industries – via an EE fund (F2E) or other off-balance sheet solutions - thereby significantly accelerating the pace of investments in this sector. F2E involves a wide variety of projects and technologies. For simplification, it was assumed that the implementation of the IEE program will involve the scaling up of a series of notional “Portfolio-Projects.” Each one of them encompasses about 10 different technology/process interventions (e.g. HVAC, energy efficient motors, heat recovery, variable speed pumps, etc.) developed for a large number of clients and facilities. It is estimated that every Portfolio Project entails investments of US\$ 39 million. It is also estimated that the entire IEE program will contemplate about 14 Projects-Portfolio. Under the project scenario, these investments will take place over a period of three to five years.

C.3. Project / Programme Description

In order to overcome the challenges to investment in EE in the urban industrial and street lighting sectors, FinBRAZEEC proposes the creation of a EE Facility that will offer financing (concessional loans and potentially equity investments) to EE projects in these sectors. The objective of this facility and underlying financial arrangements is to create a new asset class for energy efficiency that can attract the interest of a wide range of investors and bondholders. The EE facility will be managed by a local public bank. The World Bank team has an in-depth engagement with CEF, which is expected to play this role.

The concept of new asset classes for EE (industrial and PSL) will be the single largest innovation, enabling a significant scaling up of investments in this area, by attracting new groups of financiers and equity providers

Macro-economic crisis. The project will also move from pilots to achieve scale. The sub-programs (i.e. PSL and industrial energy efficiency) have already sufficient scale in absolute terms to prove their feasibility beyond the proof of concept. In relative terms, they are still small compared to the size of the market. However, all the basic ingredients for scaling up will be already developed and tested in this first phase, enabling a smooth scaling up

The project will use a variety of instruments to capitalize the EE facility and increase the quality of the EE projects, including:

- \$186 million GCF concessional loan to the EE facility;
- \$5 million GCF grant for liquidity facility for the EE facility;
- \$180 million in counterpart funding from CEF's own funds;
- \$400 million in debt investments from private financiers;
- \$330 million in equity at the sub-project level;
- \$200 million loan offered under the IPF contingent loan from the World Bank;
- \$4 million technical assistance grant from the GCF; and,
- \$1 million for technical assistance in a combination of grants and loans from the World Bank and other trust funds housed at the World Bank, including ESMAP and the Global Infrastructure Facility (GIF).

The total estimated financing for the FinBRAZEEC project is \$1,306 million, with about \$776 million directly funding the Debt EE facility, \$200 million backstopping the facility to its creditors, and \$5 million in technical assistance. There is a good justification for the combination of various instruments in this project:

- Concessional lending to private partners is required in the current context of high local interest rates in Brazil, the high volatility of the exchange rate which triggers high hedging costs, as well as municipal credit risk in the case of public street lighting. Working with GCF also attracts the interest of local financial institutions and provide a “seal of approval” on the quality and perception of climate-friendliness of investments, attracting interest from other financiers.
- The World Bank IPF contingent loan is needed to improve the credit rating of the facility to attract a wider range of investors and to reduce the facility’s cost of funding (including loans and potentially green bonds). It will provide liquidity to the facility and help with the crowding-in of investors when CEF seeks other financiers to participate in the facility. In this facility, the IPF contingent loan plays the role of a guarantee, but with additional flexibility – a more agile process within the World Bank, faster disbursement in case of a lack of liquidity at the facility, and more flexible financial terms to better suit the specific needs of the borrower at the time of disbursement.
- Grant funding of \$5 million is needed to support essential technical assistance (including training for CEF personnel) and early operational costs, for which no bank or private investor is willing to invest. In the case of Industrial EE, these funds will be used to support technical studies and operational costs during the initial labor- and investment- intensive months of operation, before revenues are sufficient to support these functions. In the case of street lighting, grant funding will be used to support cities’ preparation of high-quality sub-projects. The World Bank will contribute \$1 million to support these technical assistance activities, with the remainder being provided via a GCF grant.
- The participation of private financiers is an important outcome of the FinBRAZEEC project, in order to demonstrate the viability of private sector investment in urban EE sectors in Brazil at risk-adjusted returns and create a new asset class for energy efficiency.

Component 1: Financing Facility for Efficient Street Lighting and Industrial Energy Efficiency, housed at CEF.

The goal of the EE facility is to directly provide financing to public street lighting and industrial energy efficiency sub-projects. The facility will be housed at CEF, a state-owned bank with extensive relationships with municipalities and businesses. CEF will be responsible for identifying, appraising, analyzing credit risks, approving, and investing (or providing credit enhancements) to a pipeline of EE sub-projects in the industrial and public street lighting sectors. CEF may delegate part of those roles to a third-party selected competitively (i.e. fund management, pipeline identification) where the third-party has a competitive advantage. CEF is well positioned to manage the EE facility given its knowledge of the World Bank lending process, its extensive reach (which allows it work across the country with municipal governments and small industries), its familiarity with many of the facility’s potential clients, and the fact that CEF has decentralized technical skills to identify and appraise projects.

The GCF concessional loan will flow through the World Bank to the EE facility housed at CEF. The GCF concessional loan will allow the EE facility to reduce its financing costs, thereby reducing its lending rates to the final beneficiaries. The final beneficiaries of the FinBRAZEEC project will be cities and private sector entities that borrow from the facility to invest in public street lighting and industrial EE projects.

The GCF grant (potentially complemented by a CTF grant or other climate funds) will be used to establish a liquidity mechanism within the EE Facility to improve the credit rating of the facility in order to attract private investors. It also provides flexibility for CEF to manage its risk, given that they are taking risk on their own loan, the GCF loan (since they are the borrower). The liquidity facility also reduces the amount of provisions the EE Facility will need to make, thereby allowing the Facility to put a larger amount of its funding to use in investing in projects. Moreover, it reduces the risk of delays in the Facility’s investment in projects. The facility will cover payment shortfalls as well as losses.

The World Bank IPF contingent loan will be used to backstop a credit enhancement product¹⁶ that CEF will offer to private sector investors. If the Facility cannot make payments in full to the syndicated lenders, CEF will cover 50 percent of the

¹⁶ Credit enhancement may be offered using an existing guarantee fund currently managed by CEF.

value of the payment shortfall, drawing down on the IPF contingent loan to cover those obligations. This will cover payment shortfalls as well as losses. The credit enhancement, backstopped by the IPF contingent loan, is expected to improve the credit rating of the facility allowing it to attract a new class of investors. It is important to note that the World Bank IPF contingent loan is designed to ensure the long-term viability of the facility and not of any individual sub-project; losses at the sub-project level will not be covered directly, but only if and when the facility as a whole is unable to pay a creditor due to the losses at the sub-project level.

CEF will leverage the concessional funds and the guarantee backstopped by the World Bank IPF contingent loan to raise additional debt and equity from the public and private sectors, as well as provide about \$180 million in counterpart financing from its own account. These funds will be managed as a single facility, and be used to finance EE sub-projects in the industrial and street lighting sectors. It is envisioned that CEF will lead a syndication arrangement with multiple lenders.

Ultimately, street lighting EE investments will benefit the population in general, which will receive a better quality of lighting services which is expected to result in enhanced perception of safety, improved mobility and increased local economic activity and wellbeing of the population at large. Thanks to the industrial EE, the urban industries will become more competitive and improve their ability to preserve existing jobs. Both components will have significant environmental co-benefits, such as cleaner air and water resulting from less industrial pollution.

In the case of street lighting, the facility will primarily lend to Special Purpose Vehicles (SPVs) established by the private sector, which will be granted concessions by the municipalities to modernize and operate the street lighting system on a PPP basis. Below is a summary of a typical street lighting PPP model – see figure 2 above as a reference.

- Selection of a concessionaire through a competitive bid to whom the municipality grants a concession to deploy a wide range of responsibilities (installation, operation & maintenance) of the system over the PPP contract life;
- SPV is formed by the winning consortium (for example, an operator, lender and manufacturer), which is responsible for raising the financing for the project;
- The city remunerates the concessionaire through monthly payments using revenue raised from a dedicated levy on the electricity consumer – known as “COSIP” (or, if nonexistent or insufficient, municipal budget); and,
- COSIP is collected by the electricity utility directly on the electricity bills and flows to a municipal account or an escrow account.

In the case of industrial EE, a special account/fund will be created within the EE facility, which will make investments in the industrial customer's premises in exchange for a fixed payment that represents the deemed energy savings for the company. This business model offers several advantages, the main one being the provision of an off-balance sheet financing vehicle, therefore removing one of the main barriers for industrial customers to invest in EE: the lack of certainty in the stream of payments deriving from EE gains. Furthermore, by simplifying measurement and verification (M&V) and giving more certainty to cash flows, the mechanism will enable the creation of a new asset class for investments in EE, thereby attracting a wider range of potential investors, including those interested in investing in green bonds.

Component 2: Technical Assistance

The technical assistance component of this project will support essential technical assistance and early operational costs, for which no bank or private investor is willing to invest. In the case of industrial EE, these funds will be used to support technical studies and operational costs during the initial labor- and investment- intensive months of operation, before revenues are sufficient to support these functions. In the case of street lighting, grant funding will be used to support cities' preparation of high-quality PPP sub-projects.

To date, the team has already secured US\$0.2 million from ESMAP and US\$0.5 million from the Global Infrastructure Facility (GIF). ESMAP funds are already being used to support technical work that is informing the design/structure of the

facility. The GIF resources already secured are in the form of a PDA (Project Definition grant), with the objective of support the preparation of a strong pipeline of public street lighting PPP projects that could ultimately request financing from the FinBRAZEEC EE facility. More specifically, GIF resources are being used to support work co-led by the World Bank and CEF to (i) conduct pre-feasibility studies on a short-list of cities interested in pursuing a PPP to modernize their public street lighting systems, and (ii) complete the first stage of preparing a toolkit for PPP street lighting in Brazil. Ultimately, this work will identify approximately 10 cities from the current long-list (see Annex 5) that are strong candidates to move to the full feasibility / PPP structuring phase.

The team has already initiated discussions with the GIF to apply for the second phase of PPP support via their PPSA (Project Preparation & Structuring) resources – which are reimbursable upon successful PPP signature - for the full feasibility study, transaction advisory services and procurement. The GIF PPSA will be implemented in coordination with CEF and IFC Advisory Services.

The GCF technical assistance resources will be used to support the remaining work required to successfully implement the facility and ensure a strong pipeline of projects from both sectors. This will include the costs associated with structuring and operationalizing the facility, confirming the portfolio of projects for the industrial EE window, structuring the off-balance sheet vehicle for industrial EE investments, finalizing the toolkit for public street lighting PPPs and additional market studies. Please refer the attached procurement plan for more details on the scope.

C.4. Background Information on Project / Programme Sponsor (Executing Entity)

The project's overall financing structure involves a FIL to CEF as a financial intermediary which will on-lend to SPVs operating in PPP agreements with municipalities or urban energy efficiency funds and service providers to implement urban street lighting and industrial energy efficiency projects.

Other Brazilian financial institutions were considered as alternatives to CEF. However, it was determined that CEF had comparative advantages as it is the country's leading financier for public infrastructure such as water and sanitation projects, with extensive experience with private sector financing of municipal projects. It also has the largest market penetration in municipalities of all Brazilian institutions. CEF is very well positioned to provide intermediary financing and debt syndication services as part of its regular financing package to municipalities and interested private companies.

CEF is a financial institution constituted by Decree-Law 759/1969, in the form of a public company and linked to the Ministry of Finance, with headquarters in Brasilia. Today CEF is the third largest bank in Brazil in assets and one of the five largest in Latin America. As part of the national financial system, it assists in the execution of the federal government's credit policy, and is subject to rules and decisions of the federal government competent bodies. CEF implements its main banking activities through the collection and application of resources in commercial, foreign exchange, real estate and rural operations, in the provision of banking services, including the administration of investment funds and portfolios, as well as to implement social public programs. In addition to the intermediation of securities, debit and credit cards, insurance, private pension, capitalization and administration of consortia through equity interests in CEF Seguridade and CEF Participações S/A (CEFPAR).

The Brazilian legal framework for public financial management is well-developed at all levels of government, which is underpinned at the Federal Constitution, and two main legal frameworks: The Public Finance Law of 1964, No.4,320 that regulates financial controls, budgeting and reporting and the Fiscal Responsibility Law of 2000, Supplementary Law No. 101 that dictates macroeconomic and fiscal discipline. Other reforms are being introduced (including a proposed revision to the Public Finance Law) to help strengthen accountability and transparency in the management of public finances in Brazil, including: the establishment of standards for the selection and evaluation of investment projects; the introduction of a medium-term fiscal framework; measures to minimize the impact of expenditure commitments (Restos a Pagar); the adoption of the International Public Sector Accounting Standards, IPSAS; and strengthening rules and regulations of internal and external oversight control institutions.

The existing Public Financial Management System has satisfactory internal rules and controls, with a clear definition of responsibilities and institutional arrangements. The Ministry of Planning, Development and Management is responsible for budget preparation; the Secretariat of the National Treasury (STN) takes responsibility for treasury management and accounting functions, the Office of the Comptroller General of the Union (CGU) carries out internal audits overseeing the budget, and is responsible for fighting acts of corruption and enhancing transparency within the federal public administration. The Federal Court of Accounts (TCU) (i.e. the supreme audit institution) has the mandate to audit all public expenditures. Both the TCU and CGU have sufficient autonomy and their recommendations are generally implemented.

CEF is subject to the Federal Government internal and external control oversight bodies, CGU – General Controller of the Union) and TCU (Supreme Audit Institution) policies and procedures. Its internal audit department AUDIT follows international standard practices issued by The Institute of Internal Auditors (IIA). CEF adopts the Brazilian Corporation Law no. 6.404 / 76, modified by Law no. 11.638/07 and the guidelines of the Brazilian Securities Commission (CVM/SNC/SEP), the Central Bank of Brazil and specific regulations applicable to the finance institutions, under the National Monetary Council. The consolidated financial statements are under the responsibility of its Board of Directors and have been prepared in accordance with International Financial Reporting Standards (IFRS), issued by the International Accounting Standards Board (IASB). Ernst & Young Auditores Independentes issued an unmodified opinion over the latest available (FY2015) consolidated Financial Statements, however, due to ongoing investigation by the federal public authorities of possible non-compliance with laws and regulations by CEF employees and administrators, it is not possible to predict their effects on the consolidated financial statements.

CEF has sufficient capacity to fulfill its financial management responsibilities; is adequately exposed to the Bank's financial management procedures having adequately implemented Bank funded operations (P106702) Integrated Solid Waste Management and Carbon Finance Project). However, based on previous project lessons learned, to mitigate implementation delays, at least two dedicated qualified financial management staff should be appointed no later than the project signing to support the multiple agencies and entities involved throughout project implementation. In addition, CEF's corporate system should be adjusted to properly reflect, account project transactions (loans and guarantee operations) and generate reports.

CEF's National Office of Financial Products for Governments and Infrastructure (GESAN) will be responsible for project implementation and will coordinate this project and interface with the World Bank's project team. In particular, is creating a new fund development/fundraising department under GESAN that will be responsible for the Signing of agreements and agreements for the collection of resources; Coordinate the process of technical analysis related to abstraction carried out by other units of CEF (ENGINEERING, LEGAL, SOCIO-ENVIRONMENTAL, FINANCIAL, INTERNATIONAL AREA); Analyze and prepare, together with the IBRD and the Legal Area, contractual drafts; Relationship with federal agencies for the internalization of international resources.

Besides providing financing for the implementation of projects by the public or private sector, CEF will manage a technical assistance fund to provide support to municipalities in project preparation. Based on a Memorandum to be signed between CEF and a public beneficiary, consultants will be contracted by CEF to help cities and states in preparing street lighting PPPs, especially regarding socio-environmental, financial, institutional and bidding aspects of the sub-projects. Specific criteria have been prepared for sub-project selection.

C.5. Market Overview (if applicable)

There is essentially no market for investment in EE projects at scale in Brazil. Furthermore, there is very limited off-balance sheet financing vehicles to support industrial energy efficiency projects.

The Brazilian power sector has undergone a series of major reforms in the last two decades. The first occurred in 1995, which resulted in unbundling and privatization of a large portion of the power sector. The second took place in 2003, with strengthening of energy planning and implementation of energy auctions. The last one happened in late 2012. This legislation authorized the Executive Power to renew the terms and conditions of generation, transmission and distribution concessions that would expire between 2015 and 2017, with the objective of reducing tariffs to end users. This last reform had unintended consequences, one of them being a price reduction in times of hydro scarcity, leading to a considerable increase of the use of thermal plants. As a result, operational costs have skyrocketed while tariffs were initially maintained, creating a huge cash shortfall and considerable financial stress on the sector. Energy efficiency has never been in the “drivers’ seat” of the reforms.

Electricity tariffs have since been adjusted and the current tariff structure is largely cost reflective, creating the necessary price incentives to invest in energy efficiency. Regarding fossil fuels used in industry, prices are also cost effective, although the gas market is not yet sufficiently competitive.

The analysis conducted under the Feasibility Studies (Annex 2) show that EE projects are viable at current electricity prices for the street lighting and industrial EE sectors. Despite the fact that price signals create incentives to invest in EE, investment in demand side EE has been limited to date. As described in Section C.1 above – the main barrier to financing EE projects is the lack of adequate business models and access to financing.

Brazil has consolidated a few energy efficiency funding mechanisms, such as PROCEL (National Electricity Conservation Program), and a regulatory obligation imposed on utilities to invest 0.5% of their revenues in energy efficiency projects (the so-called PEE - Programa de Eficiência Energética das Concessionárias de Distribuição de Energia Elétrica, regulated by ANEEL).

PROCEL, which was instituted on 30 December 1985, is a Brazilian government plan coordinated by the Ministry of Mines and Energy (MME) and executed by the Eletrobrás. It aims to promote the efficient use of electric power and reduce waste. The results created by the program contribute to the efficiency of goods and services and enable the postponement of investments in the electricity sector, thereby reducing environmental impacts. Specifically, the PROCEL conducts the following activities: identifying and assigning an energy efficiency label for equipment and buildings; disseminating information on energy efficiency through educational action in the formal education system or publication of books, software, and technical manuals; provisioning support to municipalities for planning and implementing equipment substitution projects and improving public lighting and traffic lights; and providing training and assistance for the planning and implementation of projects that reduce energy consumption in municipalities and promote the efficient use of electricity and water in sanitation. Between 1986 and 2015, the PROCEL invested USD \$808 million, which led to electricity savings of 92.2 TWh.

PEE has been investing resources heavily among low-income customers while PROCEL has prioritized modernization of public street lighting (replacement of mercury lamps by sodium ones) using funds from RGR (Reserva Global de Reversão). With the 2012 reform, RGR was terminated and so the funds to modernize street lighting, exactly at a juncture when LED technologies became available. Municipalities alone will have very limited financial space to modernize their street lighting systems (they continue to be limited by Fiscal Responsibility Law and covenants resulting from debt negotiation with Federal Government. Since the Modernization of public street lighting is a very capital-intensive activity, It is necessary to create conditions to attract the private sector via PPP (administrative concessions).

In light of the above, developing Public-Private Partnerships (PPPs) and innovative financing and business modalities to attract private investment have become a priority of the Federal Government. This led to the creation of a new secretary,

directly reporting to the President of the Republic, called the Investment Partnership Program (Programa de Parceria de Investimento – PPI), which is one of the main governmental counterparts of the FinBRAZEEC project. The PPI is acting as catalyst of policy makers, regulators, governments’ decision makers, financial institutions and other stakeholders to attract private investment.

Street lightening projects in Table 1 (strategic context) are examples of comparable transactions to those proposed by FinBRAZEEC. However, while there a number of PPP projects under preparation, very few have reached financial closure.

Regarding industrial energy efficiency, it has never been a priority among government EE programs in Brazil. Due to the fact that industrial EE represents the “biggest bang for the buck” in terms of GHG abatement, additional work to overcome barriers to investment in this sector are urgently needed.

Many of the key barriers for investment in EE in public street lighting and industry will be addressed through the EE Facility; however, other policy and regulatory reforms could help support the acceleration of investments. Information regarding the World Bank’s engagement with the Government of Brazil in this area is provided in Section C.1.

C.6. Regulation, Taxation and Insurance (if applicable)

Regulations

A PPP on street lighting is a form of administrative concession. It requires the granting authority (i.e. the municipality) to carry on a competitive tender to select the concessionaire which will perform the modernization of street lighting services. In principle, each individual municipality should grant concessions for street lighting PPPs. Consortiums of municipalities is also possible, and have been implemented in waste management in Brazil. In the case of industrial energy efficiency, there are no obligation imposed on industries to achieve efficiency levels (like the PAT program in India), GHG reductions targets (like in China) on EEO (Energy Efficiency Obligations, like in Europe). Industrial energy companies in Brazil has almost no regulatory incentives to make investments to make their processes more efficiency, and decisions are purely based on economics, and have to overcome a series of well-known barriers in the EE space.

Taxation

In the case of street lighting, about 70 percent of the equivalent value of the luminaire is imported. Import taxes are about 70 percent. With the increasing use of LEDs, local manufacturing facilities should be established, reducing the imported content and making the cost of the equipment more competitive. In the case of industrial energy efficiency, this percentage will vary widely depending on the nature of the project and adopted technology.

In the case of public street lighting, the manufacturer should provide up to a 10-year warranty for the quality of the product. Typically, this guarantee does not include the labor cost to replace defective units, which must be borne by the concessionaire. To reduce its risk, the concessionaire may decide to buy insurance to backstop the technical quality of the equipment. Some insurance companies are entering this space.

In the case of industrial energy efficiency, the EE facility will have the ownership of the assets installed in the industrial company’s premises. Therefore, the facility should require the host industrial company to buy insurance against malfunctioning and poor operation. Alternatively, the EE facility may buy insurance and transfer this cost to the host company, if this is mutually agreed in the service contract.

Accounting Rules.

Currently, off-balance sheet financing in the energy efficiency space is considered tantamount to an operational leasing, and as such is subject to FASB rules and constraints, at least for companies which adopt this accounting standard. It is expected that GAP and FASB rules will converge over time. It is also expected that large industrial companies (the initial target market for the IEE fund) will adopt FASB accounting principles. It is envisioned that IFRS 16 (which deals with Leasing in particular) will undergo significant changes as of 2019, reducing the scope of off-balance sheet financing. The

impact on the projects contemplated by the IEE fund has yet to be determined, and will depend on how EE savings will be measured (e.g. deemed savings, ex-post monitoring, etc.), and to what extent payments for EE services will be adjusted to the level of those savings. It is possible that some projects will no longer qualify for off-balance financing if contracts are structured as pure fixed payment, irrespective of the savings. The exact impact of IFRS 16 on IEE portfolio and nature of contracts will be examined as part of the technical assistance provided by FinBRAZEEC.

Arbitration

Arbitration is a key issue in the context of attracting investments to Brazil. It is regarded as a means of building investors' confidence, by providing an agile, impartial, confidential, and highly specialized mechanism for dispute resolution. In this sense, the Brazilian government is attempting to increase the role of arbitration in the country as part of a strategy to attract investments to the infrastructure sectors. However, from the investors' perspective, there are relevant risks that need to be addressed. Such risks can be divided into two main issues: (a) uncertainty about what matters may be settled by arbitration; and (b) uncertainty about whether investors will ultimately benefit from an agile and highly specialized forum, often threatened by judicialization. Especially in heavily regulated sectors, such as infrastructure, the perception of such risks is enhanced. This is mainly because some disputes will involve the participation of a public (or quasi-public) entity as a party. This is the case of public street lighting.

Integrity

The insertion of clauses intended to prevent contractors from committing crimes against the public administration (active corruption, influence peddling, and so on) is becoming more frequent in contracts between private individuals who have some kind of possible interaction with the public power. In general, such clauses provide for reciprocal obligations of (a) abstention from acts that may constitute such offenses (payments, gift offers, promises of advantages, rewards, and so on); (b) constitution and execution of compliance programs; and (c) submission to different types of audits, from part to part, to verify compliance with previous obligations ('interference clauses').

Currency risk

Brazil has made significant effort to develop and design mechanisms to attract foreign investors to the infrastructure sector. Currency risk mitigation mechanisms have been piloted in the transport sector auctions, with optional adhesion. These mechanisms enable that changes in the exchange rate in a period of one year that exceed a reference component (variation in the inflation rate and country risk component) be offset by a reduction or increase in the amount to be collected as a grant. The mechanism establishes two ways to vary the grant to compensate for currency rates changes. The use of solutions already available in the financial markets could lead to better results. This project will make use of such solutions to mitigate currency risk.

C.7. Institutional / Implementation Arrangements

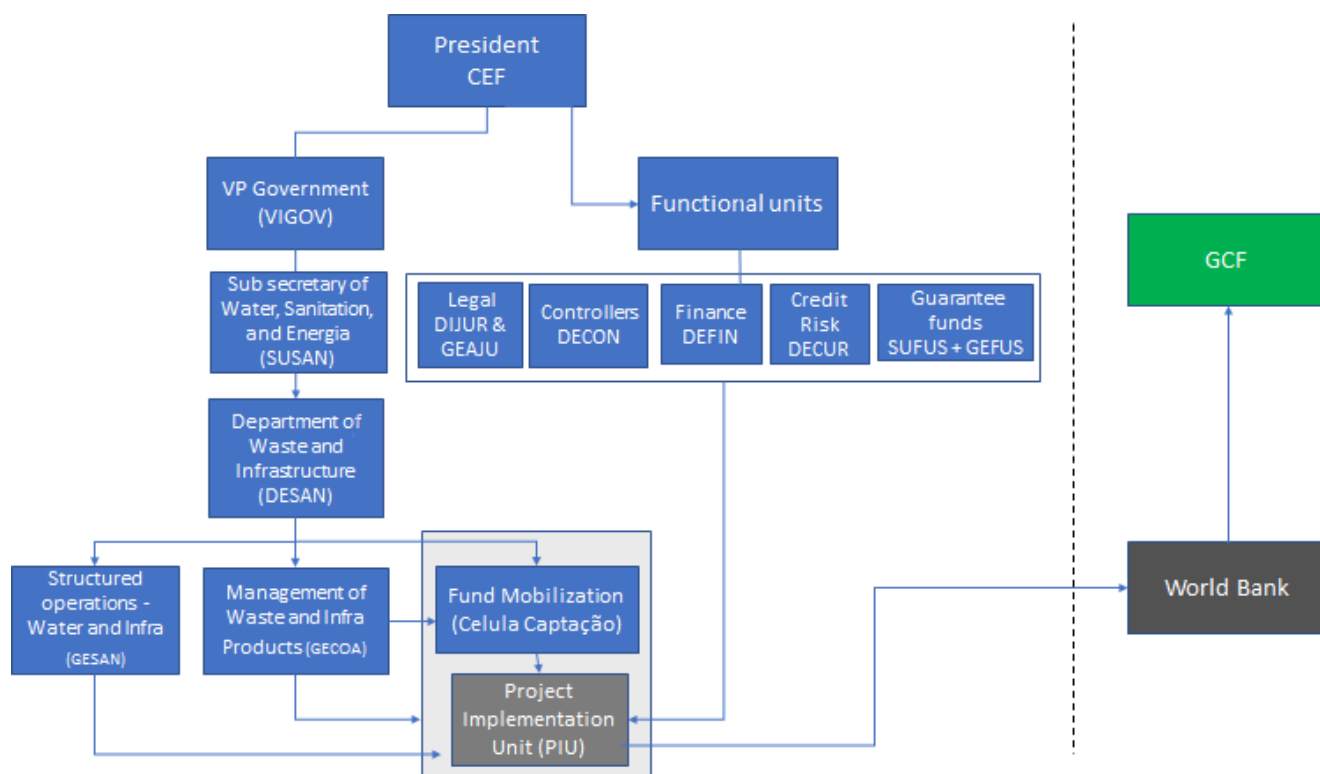
Governance

The project will be implemented over seven years, with an option to be extended if demand is higher. Overall responsibility for the facility lies with CEF, as the administrator of the EE facility. CEF will be responsible for identifying, appraising, analyzing credit risks, approving, and investing (or providing credit enhancements) to a pipeline of EE sub-projects in the industrial and public street lighting sectors. CEF may delegate part of those roles to a third-party selected competitively (i.e. Fund management, pipeline identification) where the third-party has a competitive advantage. CEF will select an independent external auditor to conduct the annual project audit; this will be financed from the ta component. The CEF has the main responsibility for signing the contract and coordinating the auditor's work.

CEF will have full responsibility for the EE lending process and approvals, following the agreed om, and will bear the associated credit risks. CEF will form a project implementation unit (PIU) with dedicated teams supported by technical, environmental and social and procurement experts. The project implementation unit will implement the lending and IPF contingent loan disbursement activities and act as CEF focal point to interact with the world bank and other stakeholders as needed.

CEF will supervise/monitor all loans to ensure they are implemented according to Brazilian and bank requirements and provide periodic reports, including fiduciary and safeguards reports to the ministry of finance and the bank. Independent auditors will be selected to conduct annual project audit on the performance of CEF.

CEF will establish a PIU that will be responsible for coordinating all of the project preparation, implementation and supervision. The PIU will be housed within the “Captação” (Fund Mobilization) team, which sits within the Government Vice Presidency (VIGOV) within CEF. The PIU will be comprised of member(s) of all the key units within CEF to prepare, implement and supervise the project (including, but not limited to, Legal, Controllers, Finance, Credit Risk, Guarantee Funds). The diagram below provides an overview of the governance structure within CEF.



A detailed project operation manual (OM) covering technical, fiduciary, safeguards, and management requirements and procedures will be prepared for the IBRD loan. The OM will be incorporated into the project agreements between the World Bank and CEF. The OM will define the detailed eligibility criteria for subprojects, for example, the minimum energy savings to be achieved, and exclusions linked to world bank safeguards, such as exclusion of World Bank Category A sub-projects or exclusion of projects with potential impact in international waterways.

Key arrangements at sub-project level

In the case of street lighting, the concessionaire will be fully responsible for the design, procurement, installation, and maintenance of the system. The municipality will monitor quality and make a monthly payment based on deemed savings and adjusted for the quality of service rendered. A key tripartite contractual agreement involved the utility, the municipality and the concessionaire should be signed. This operational agreement should specify the conditions for the delivery of electricity, including some basic technical specifications that may have an impact on the physical integrity of the equipment and the quality of lighting services provided by the concessionaire. The granting authority will monitor

1. Funded Activity Agreement between GCF and IBRD: Providing for the \$186 million and the \$4 million grant for the start-up costs for the EE Facility and TA activities; and the \$5 million reimbursable grant for the liquidity fund as part of the EE Facility, and all in accordance with the Accreditation Master Agreement (AMA). The GCF loan will be in the form of a trust agreement, for the World Bank to provide the GCF Proceeds in the form of loan and grant to the Executing Entity (CEF).
2. Subsidiary Agreement between IBRD and the Executing Entity (CEF) for the GCF-funded loan (\$186M): defines terms under which WB will provide the GCF Proceeds that will finance the Facility's sub-loans.
3. GCF Grant Agreement between IBRD and Brazil: Providing for the downstream transfer of the funds to CEF.
4. IBRD IPF contingent loan agreement between IBRD and the CEF as IPF contingent loan beneficiary: defines the terms and conditions of the IPF contingent loan offered, including covered risks, IPF contingent loan amount and terms, payout procedures etc.
5. Sub-loan agreements between CEF and street lighting SPVs, and between CEF and F2E (Fund for Energy Efficiency): defines the terms and conditions of the sub-loans, and the reporting conditions by the borrowers with respect to CEF (an acceptance of conditions agreement will be defined in order to ensure adequate reporting to CEF that ultimately will allow the World Bank to obtain information to monitor the project indicators).

D.1. Value Added for GCF Involvement

A financial analysis of the Facility was conducted to determine the benefits that the GCF concessional loan and grant offer to the project.

Key assumptions

The key assumptions regarding the design of the EE facility are as follows:

- This analysis uses the same assumptions of an “average” street lighting and industrial projects, described in the Feasibility Study (Annex 2). This is a notional concept adopted in the early stages of design of facilities of this nature when sub-projects have not been fully identified.
- The share of capital invested in public street lighting and industrial EE projects (“sub-projects”) will be approximately equal (fifty percent each).
- As described above, the facility is expected to be funded with loans from CEF (approx. 24 percent), GCF (approx. 24 percent), and private sector lenders (52 percent). See Figure 8 below.

Figure 8: Share of sub-project sector investments

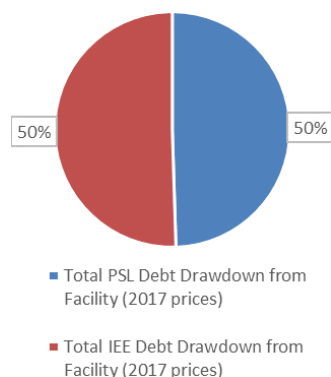
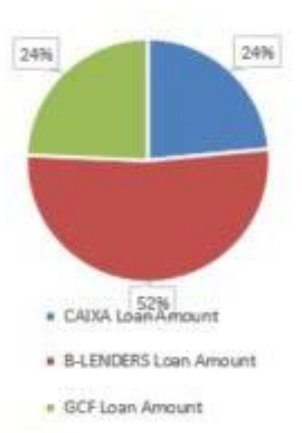
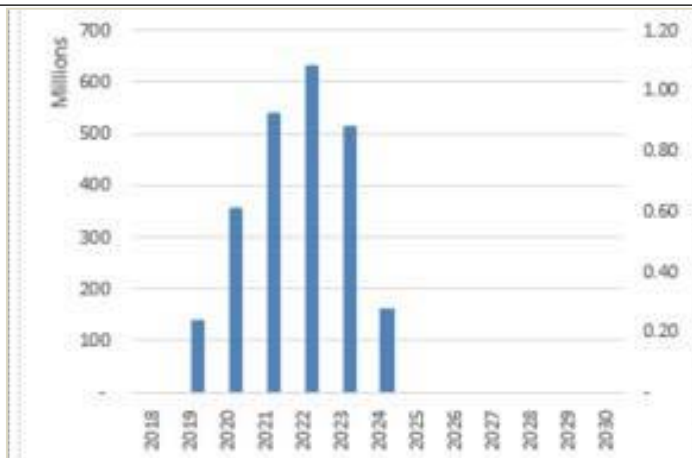


Figure 9: Funding Sources



- The sub-projects are assumed to have debt-to-equity ratios of 70:30. The facility is presumed to be the sole lender for eligible projects¹⁷.
- The Facility’s disbursement schedule was designed based on the financial models developed for the Feasibility Studies conducted for the PSL and Industrial Sectors (See Annex 2). Combining the investment profile of the projects and the assumed timeframe to prepare projects, the Facility is expected to invest over a period of 6 years. See Figure 10 below.

Figure 10: Facility Disbursement Schedule



- The facility will try to match, as much as possible, the funding sources with disbursements to sub-projects. Cash surpluses, if any, will be reinvested in one year Government Bonds (or equivalent instruments, if they have sufficient liquidity). For the purpose of this analysis, the presumed interest rate on these investments are 7.5% p.a..
- Since this interest rate is lower than the blended cost of funding for the Facility (more so for the cost of B-lenders), some financial losses are modeled for cash balances held in BRL. In practice, the Facility will optimize cash management to minimize those losses as much as possible (for example, by keeping disbursements from GCF in US\$ (at the Central Bank) and converting to BRL only when necessary).
- The cost of funding of the Facility is calculated based on the blended cost of lending of the Facility's sources of funds according to their individual costs share in the Facility's capitalization.
- This analysis assumes the interest rate of the GCF loan will be 0.75 percent in USD. However, as shows in the Feasibility Study (Annex 2), further concessionality could provide critical flexibility to allow the EE Facility to lend at lower rates to sub-projects, improving their viability and expanding the types of projects to which the EE Facility could lend (e.g., projects associated with smaller cities, which will face higher equipment costs due to smaller scale).
- After taking into account the estimated cost of the swap (which will vary according to market conditions and the weighted average term of the loan), the GCF loan interest rate is estimated to be 5.65% in BRL¹⁸.
- The private sector cost of funding is benchmarked to the price of a low- to mid-grade Brazilian infrastructure bond (A-). The reference is the 10-year debenture of a Brazilian company CCR company debenture, which is trading at 17.5%. This cost is comparable to the remuneration of debentures issued by large companies in Brazil (such as EDP, Vale).
- CEF's cost of funding (loans from its own-account) is estimated based on the mix of resources available for those kinds of infrastructure projects, which may include, inter alia, access to funds from FI-FGTS (exact figures to be discussed with CEF).
- A summary of the terms of the three different lenders are shown below:

Table 1: Cost of funding

¹⁷ For example, eligible projects will be economically feasible, meet credit requirements, have a GHG impact commensurate to similar projects. Project eligibility will be defined in collaboration with CEF during negotiations.

¹⁸ Based on information available in the GCF – Terms and conditions as of Feb 17 2015, and based on market conditions as of August 2017. The actual cost of the swap will be determined based on market conditions at the time of conversion.

Funding source	Loan amt (USD)	Cost (BRL)	Grace Period	Repayment Period
Caixa	180,000,000	12.00%	5	15
Syndicated Loans (B loans)	400,000,000	17.50%	2	6
GCF	186,000,000	5.65%	5	15
TOTAL	766,000,000			
Weighted average funding cost		13.33%		

- CEF is estimated to a disbursement fee (2.5% of disbursed amounts) and a syndication fee (0.8% on the CEF and Syndicated Loan drawdowns).
- To match the estimated investment needs of the envisioned portfolio of investments shown in Figure 6, this analysis assumes a capitalization schedule of the Facility over a six-year period from 2019 – 2024. Table 2 shows the capitalization schedule by lender, and Figure 12 shows the profile of the Facility's annual capitalization alongside disbursements.

Table 2: Capitalization Schedule by Lender

LENDER	TOTAL	2018	2019	2020	2021	2022	2023	2024
CAIXA Capitalization Schedule	100.0%			12.5%	20.0%	30.0%	27.5%	10.0%
Private Sector (B-LENDERS) Capitalization Schedule	100.0%			12.5%	20.0%	30.0%	27.5%	10.0%
GCF Capitalization Schedule	100.0%		25.0%	25.0%	30.0%	20.0%		

Figure 12: Total Capitalization and Disbursement Schedule



- The average financing price that the Facility offers to sub-projects is estimated to be 14.6%, or SELIC (the Brazilian interbank rate) + 710 bps. This price was estimated based on an attempt to provide the minimum level of concessionally compared to commercial financing rates (equivalent to B-lender rates) required to make the

sub-projects viable. At pricing above these rates, the projects in the public street lighting sector are expected to encounter cash flow problems (low DSCRs) and the returns to equity may be insufficient. As described above, lower rates could allow the EE Facility to invest in a larger range of sub-projects, including smaller cities with lower economies of scale (and where the development impact could be large).

- The average terms for public street lighting projects are assumed to be 1 year grace period with 7 year-repayment. For industrial energy efficiency, the assumption is a 1 year grace period with a 6-year repayment (on the basis that industrial projects will have a slightly faster payback period).
- Actual lending terms from the facility to sub-projects will be done on at project-by-project basis at risk-adjusted rates.

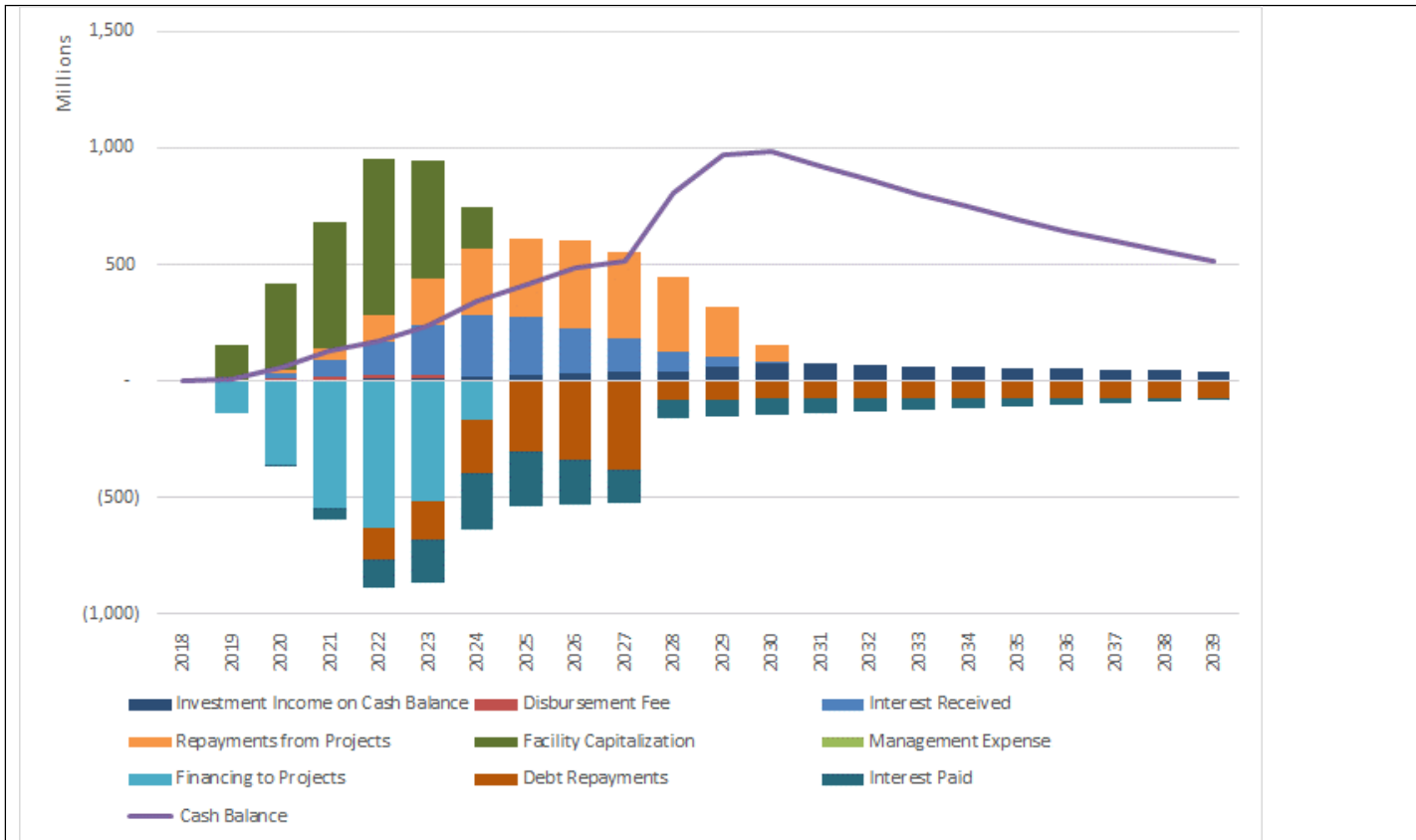
Results

The findings from the financial analysis of the EE Facility show that the GCF concessional loan and grant are critical for the initial phase of the Facility's operation. Once the concept of the Facility is proven over time, the it can become a sustainable debt facility, with a proven model of de-risked EE projects, without the use of credit enhancement mechanisms.

As shown in Figure 12 below, the Facility is able to maintain a positive cash balance over its life. During the period from 2019 – 2030, the Facility is actively investing in new projects and at the same time receiving debt repayments. After 2030, the Facility has received all debt repayments and reaches a maximum cash balance.

Although it is not shown here, the Facility would be able to use its cash balance to invest in a second round of loans to sub-projects, thereby recycling the Facility's capital. A portion of accrued surplus capital could also be used to replace the credit enhancement mechanisms currently supported with GCF grant and the World Bank IPF contingent loan.

Figure 12: Facility Cash Balance (BRL) – base case



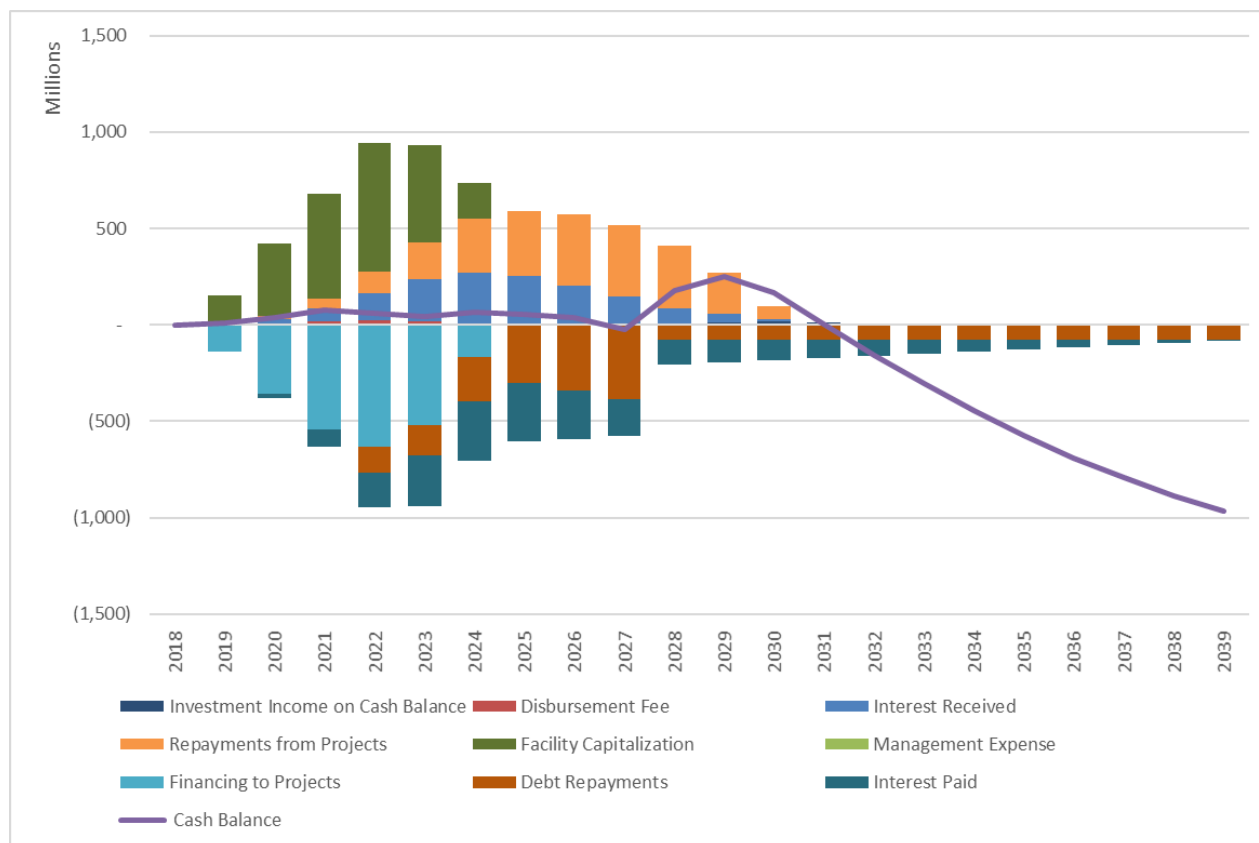
As currently structured, the Facility can manage a default on 5-10% of its payments and still meet its debt obligations. Payment shortfalls incurred beyond this level would be covered initially by the Liquidity Reserve. The analysis shows that this will offer benefits to the private lenders/investors by providing coverage for 3-9 months of payments of the fund's annual payments. In the event the sub-project-level protections and the Liquidity Reserve were fully depleted, CEF then could trigger a drawdown of the IPF contingent loan to cover another US\$ 200 million in losses to the private lenders/investors.

An analysis was undertaken to show the impact that GCF concessional funding has on the viability of the Facility, as described below.

Without the GCF concessional loan, the cost of funding would be significantly higher. If the US\$195 million was instead provided by the private sector, the cost of funding of the facility is estimated to increase from 13.3 percent to 16.2 percent, equivalent to a 22 percent increase in the funding costs. Analysis at the sub-project level shows that investments in these projects would not be viable at this higher rate. This is particularly the case for medium-sized projects in PSL (under 100,000 lamps), which exhibit a higher unit cost per luminaire replaced due to diseconomies of scale.

As shown in Figure 13 below, without the GCF loan, the financial analysis shows the Facility would have a marginal to negative cash balance in most years of its operation and would not be able to meet its debt service obligations at many points throughout its operation. In light of this, without the GCF funding, the Facility would have to be redesigned at a much smaller scale, with a much slower rate of investment in projects, and possibly exclude street lighting projects in small and medium-sized cities.

Figure 13: Facility Cash Balance (BRL) – without GCF loan



The GCF loan also provides essential benefits in terms of grace period and tenors, allowing the Facility to similarly offer grace periods and longer tenors to the sub-projects, reducing their financing risk and improving their overall bankability. As mentioned above, these features also support CEF in mitigating the cash management risks of the facility in terms of the timing of capitalizing the Facility vis-à-vis the disbursement of loans to sub-projects.

Without the GCF grant for the first-loss Liquidity Reserve, CEF may perceive their risk exposure to be too high, given that this will be a new business line and will entail a new approach to financing by leveraging their resources to bring in the private sector. Without this feature, CEF and other private sector lenders may not be interested in investing, or their amounts of investment could be much smaller. Moreover, the private sector investors are expected to highly value the provision of the Liquidity Reserve from the GCF implemented via the World Bank, an AAA credit rated institution, thereby increasing the number of private sector investors willing to lend to the Facility.

On this basis, the analysis concludes that GCF loan and grant are essential to enable the EE Facility to achieve its goals of attracting private sector capital to create a new and sustainable financial product for EE investments in Brazil.

D.2. Exit Strategy

Based on the Financial Analysis described in Section D.1 above, the Facility is envisioned to be structured so that it can build up a positive cash balance over its life. The Facility would be able to use its cash balance to invest in a second round of loans to sub-projects, thereby recycling the Facility's capital and creating a sustainable business model. A portion of accrued surplus capital could also be used to replace the credit enhancement mechanisms currently supported with GCF grant and the World Bank IPF contingent loan.

On the project side, as the program reaches scale and confidence among financiers, operators and manufacturers, one would expect the installation of manufacturing plants for luminaires, LEDs, lenses, and electronic parts and systems in Brazil, reducing the imported content and the overall price of the system. This will improve the risk perception and CAPEX of the EE investments in public street lighting and industrial EE, making investments in these sectors more attractive over time and allowing the mechanism financially self-sustaining eventually without concessional financing.

The project will help create new asset classes for street lighting and industrial energy efficiency, putting the right mechanisms in place and building confidence among investors. The idea is that the project, as originally designed will tap into a small fraction of the market, but has the potential to be scaled up involving a larger number of municipalities and small and medium-sized industrial companies.

In addition to the benefits from Component 1, Component 2 will be essential to ensuring the sustainability of the business model. By providing capacity building, technical support, and standardized models, the program will create the conditions for the continued preparation of high-quality projects that can be financed by a second round of the FinBRAZEEC facility and other market players. These benefits will long outlast the program itself.

E.1. Impact Potential

Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas

E.1.1. Mitigation / adaptation impact potential

Combining the investments of the Facility in the two sectors (sub-programmes), the project is estimated to reduce 17.4 MtCO₂e. This considers 8 years of emission reductions for IEE projects and 13 years of emission reductions for PSL projects. Description of the baseline methodologies is provided in Section E.6.6.

The cost per tCO₂e of the total investment (US \$1.1 billion) is \$ 63.

The GCF share of the investment cost will be US\$195 million. Given that GCF funds are essential to make the project viable, the real impact of the GCF loan can consider the GCF investment per total tCO₂e reduced by the project, which is equal to US\$ 10.7/tCO₂e.

It is important to note that given the nature of the Project to support a number of SL and EE sub-projects through, the ex-ante estimates of energy saving and GHG emissions avoided would not be accurate and could change upon the composition of EE lending portfolio supported throughout the implementation. The calculation methodology takes into account this uncertainty and deploys somewhat conservative approach. Details are in section E.1.2.

E.1.2. Key impact potential indicator

Provide specific numerical values for the indicators below.

GCF core indicators	Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (Mitigation only)	Annual	2.1 MtCO ₂ e (assume average over 13-year period for PSL and 8 years for IEE)
		Lifetime	17.4MtCO ₂ e (assumed 13 years for PSL and 8 years for IEE)
	- Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience); - Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)	Total	N/A
		Percentage (%)	N/A

Other
relevant
indicators

Annex 2 provides more details on methodologies and assumptions used in Street lighting and Energy efficiency in the Industry.

E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyze impact beyond a one-off project/programme investment

E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)

In both public sector lighting and industrial EE, the project is proposing to develop new asset classes. Once those new asset classes are accepted in the market, the potential for replication is huge. In the case of street lighting, other municipalities can develop their purchase agreements and the concessionaires will be creating their own asset classes which can serve as collateral to raise funds in the market and therefore expand the business into new municipalities, since the regulatory framework and the business models may be very similar across cities.

In the case of industrial energy efficiency, the opportunities for replication derive from the significant market potential of EE. Materializing this potential depends on the existence of sustainable off-balance financing mechanisms. Projects will not be as much standardized as in the case of street lighting, but if a good asset class is created (de-risked with cash flows and risks understood by financiers), the potential for scaling up is enormous. Initially, the facility will focus on large industrial customers, but it can expand in the future to contemplate medium-sized industrial and commercial facilities.

E.2.2. Potential for knowledge and learning

For street lighting, giving its relative standardization of business and financial models (therefore replication), the opportunities for municipalities in general (and to potential operators and financiers) to learn from success cases is enormous.

In the case of industrial energy efficiency, the EE facility will enhance its technical and marketing skills over time on how to identify, screen and assess projects, and refine its procurement and construction skills, opening the off-balance sheet financing mechanisms and the technical and financial knowledge to other industries. Given the innovative aspect of the facility, the idea of creating EE asset classes can be tested in other countries in the region.

E.2.3. Contribution to the creation of an enabling environment

Immediately, the design and funding structure of the facility are designed to crowd-in green and climate-conscious investors by offering a relatively de-risked, highly scalable investment opportunity. Municipalities and industries that have limited working capital, but can benefit from making efficient investments have a new funding source that is reasonably independent of the broader macroeconomic trends. Longer term, the design and implementation of new asset classes for public street lighting will create an enabling environment for viable but still untapped EE opportunities, which can provide stable investments domestically or regionally if replicated.

E.2.4. Contribution to regulatory framework and policies

In the case of street lighting, the structuring of the first PPPs will help identify the key legal and regulatory aspects that need to be put in place. Given the nature of the business and its degree of standardization, this cross-fertilization will help any other city interested in developing PPPs in preparing the regulatory framework and policies for a sustainable modernization and expansion of the street lighting business.

For industrial EE, the project opens up the possibility to expand emission control targets to urban industrial districts that currently are outside of the purview of municipal governments, by providing a funding mechanism for efficiency improvements. Given the extent to which such urban industrial areas contribute to the total urban emissions, it is possible to significantly alter the emissions trajectory in perpetuity.

E.3. Sustainable Development Potential

Wider benefits and priorities

E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

Economic co-benefits

- Total number of jobs created
- Amount of foreign currency savings
- Amount of government's budget deficits reduced

Reduction in municipalities' fiscal deficit. Energy is the second largest item in a municipality budget. New LED will enable between 50-80% of energy savings (depending on the baseline) and 40% in O&M costs. Using PPPs, city can benefit from better services without incurring any upfront investments, therefore reducing pressure on its fiscal budget.

Social co-benefits

- Improved access to education
- Improved regulation or cultural preservation
- Improved health and safety

Public street lighting leads to improved safety and perception thereof, resulting from a better luminosity and color rendition, which is a deterrent to robbery and crimes to individuals and businesses.

Replacement of existing high pressure sodium, mercury vapor, and metal halide luminaires by 1.44 million LEDs eliminates mercury-laden lamps in the system, resulting in a reduction of about 146 kg of mercury equivalent. This metal will no longer be improperly disposed in landfills, thus avoiding its transformation in toxic substances and resulting contamination of air and water beds. Elimination of mercury also creates better health conditions for landfill workers (formal or scavengers) who get heavily exposed to the toxic fumes.

Industrial EE improvements go in tandem with improvements in process efficiency, resulting in less waste generation, particulates and gas emissions such as NOx and SOx.

Environmental co-benefits

- Improved air quality - less airborne mercury from lamps, less particulates and gases, due to improved industrial energy efficiency
- Improved soil quality - less mercury seeping into the soil and water beds (leachate), as well as less waste generated from industrial processes which may contaminate soil and water
- Improved biodiversity – mercury has a cumulative impact on the food chain. Qualitatively speaking, less mercury in the air and water will help maintain a more balanced biodiversity.

Gender-sensitive development impact

- Proportion of men and women in jobs created

Women and children are the primary beneficiaries of improved safety (or perception thereof) resulting from better public lighting. The fear of being robbed or raped in dark streets and alleys prevents women from carrying on economic activities, or from children/adolescents (in particular girls) from attending school at night, limiting their possibilities to get secondary or technical education

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

Given the nature of the project, and since the municipalities that are going to be supported by the TA component of the project haven't been selected yet, the team doesn't have sufficient information to evaluate this risk at this stage and will need to gather more information to evaluate the climate and geophysical hazards and their relationship to the project.

Large regions of Brazil are subject to different climate change related risks, such as extreme temperatures and extreme precipitations and droughts that could have an impact in the street lighting and energy efficiency projects supported by this project. Over the past two decades, Brazil has experienced a significant increase in flooding events from 20 in the 1990s to 34 in the 2000s. River and flash floods combined with landslides are the most frequent and costly natural disasters. Around 19.3 million people, 9.6% of the total population, live in areas prone to river flooding. The southeastern, southern and northeastern states of Brazil are most likely to be affected by flooding.

The semi-arid and dry sub-humid regions, mainly located in the Northeast region and in the north of the states of Minas Gerais and Espírito Santo are highly prone to desertification. Areas prone to desertification cover between 940 thousand and 1.3 million km² in total, concentrating 1,482 municipalities and nearly 32 million inhabitants.

Droughts have caused significant rainfall anomalies and hydrological stress. The affected region in 2010 was approximate 3,000,000 km² of rainforest, compared to 1,900,000 km² in 2005. The Brazilian Northeast region has a high potential for water evaporation due to high temperatures and solar strength. Strong drought and dry cycles tend to afflict the region in intervals ranging from a few years up to decades. Temperature increases projected could cause major evaporation of lakes, ponds and reservoirs. As previously mentioned, droughts have also generated power sector crisis due to the hydro power dependence of the power sector leading to rationing in the past. While power rationing could also affect the projects financed by the project, the use of energy efficiency and demand response could actually help to mitigate the impact of future droughts without adding new fossil fuel based thermal power capacity to the system.

Contribution to of the project to Climate change adaptation. The INDC pledges to unconditionally reduce greenhouse gas (GHG) emissions by 37% below 2005 levels by 2025; and includes a subsequent indicative unconditional contribution of 43% below 2005 levels by 2030. The INDC has committed to an absolute reduction of emissions from a base year, which is a more progressive and transparent target type than reductions based on projected emissions. The key sectors targeted to reduce emissions are land use change and forestry, energy, industry and transportation. Energy efficiency measures in electricity system and demand side management have been identified as key areas of action in the energy sector.

E.4.2. Financial, economic, social and institutional needs

Municipalities in Brazil are financially constrained and have very strict limits on borrowing, in particular now when the country is imposing an austerity program to reduce fiscal deficits at federal, state and municipal levels.

Industrial companies in Brazil, like in other parts of the world, are reluctant to borrow or use equity to invest in non-core business activities, like energy efficiency. Public funds from BNDES and other development banks in the country are also constrained as part of the ongoing fiscal austerity program. There is a sound capital market in Brazil, but there is very little interest in investing in energy efficiency. Despite their proven financial feasibility, investments in EE are diffuse, diverse, and have not been properly de-risked. Capital markets discount EE benefits heavily and require large collateralization levels due to the uncertainties (in part attributable to lack of knowledge of EE cash flows). Therefore, it is vital to create new asset classes that the financial markets can understand and invest in confidently. Developed countries have already consolidated financial structures via guaranteed savings, and EE contracts have been collateralized and even traded in the secondary market. However, nothing of this sort has happened in developing countries, and in Brazil in particular. The technical assistance grant will also strengthen the pipeline of potential investment targets for the facility by ensuring a strong set of sub-projects are ready when the facility is fully capitalized and ready to make investments.

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

Both projects have a significant impact on energy efficiency, which is one of the pillars of the Brazil NDC. Furthermore, projects on industrial energy efficiency have an impact on the use of fuels, resulting in significant emissions abated for each barrel of oil equivalent saved.

E.5.2. Capacity of accredited entities and executing entities to deliver

CEF will be the primary executing entity and is a very solid government owned financial institution in Brazil. Together with BNDES, CEF is one of the largest development and infrastructure financiers in Brazil. BNDES (which may also be one of the possible financiers for the facility), generally focuses on large scale infrastructure projects, while CEF's interventions focus on the needs of municipalities –including housing, waste management, water and sanitation, and more recently public street lighting. CEF has decentralized technical teams able to assess projects. CEF also has a strong reputation as a fund administrator, and is responsible for the management of FGTS, the national mandatory pension fund for all workers in Brazil and one of the largest pension funds in Brazil. CEF has successfully worked with the World Bank in the past, and has expressed interest in promoting the development of new financial products in the municipal and EE spaces.

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

The development of both public street lighting and industrial EE programs have been a multi-stakeholder engagement from its very beginning.

For street lighting, the initial pre-feasibility assessments were carried out for the municipalities of Rio de Janeiro and Belo Horizonte. The WB provided technical assistance working hand in hand with the municipal secretariats involved in the provision of street lighting, agreeing on the assumption, discussing the findings and recommendations. The work has subsequently expanded to other cities. In June 2016, the Bank organized a two-day workshop in Sao Paulo when more than 200 stakeholders in the public street lighting sector were invited to discuss the issues, options and challenges to modernize the PSL systems in Brazil. In this occasion, the Bank launched a report entitled Modernizing Street Lighting Services for Brazilian Cities and the CityLED pre-feasibility tool, which generated a lot of interest from multiple municipalities. The WB and IFC have been invited to speak in several fora to disseminate the concepts. Both institutions have been approached by several municipalities interested in learning more, and to discuss WBG contribution. IFC has just been retained by the Mayor of Rio de Janeiro to structure a PPP for the modernization of the entire PSL in the city (about 430,000 lighting points). The WB has been discussing with financial institutions in Brazil and probing their interest in playing a more active role as a financier and fund manager for the FinBRAZEEC. Current discussions with CEF are progressing well.

In the case of industrial energy efficiency, the WB was approached by the Association of Large Energy Consumers (ABRACE) and National Confederation of Industries (CNI) about 18 months ago to discuss alternatives for improving efficiency among large industrial users and share WB international experience. The concept proposed by CNI and ABRACE was very innovative, but has required some fine tuning. Significant work has been carried out together with CNI and ABRACE in the design of the industrial EE window organization, processes, systems, governance, pricing criteria, and a detailed technical, economic and financial feasibility analysis has been jointly carried out, taking real case energy efficiency opportunities among CNI and ABRACE industry members. The interest of being an agent and financier for industrial EE has also been tested with local financial institutions such as Banco do Brasil, CEF, and BNDES. Once the business models and role of local financial institutions are agreed upon, the idea is to reach out a wider range of investors (banks, funds, MFIs) to probe their interest in financing the expansion of industrial EE measures.

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

E.6.1. Cost-effectiveness and efficiency

The EE facility offers strong economic benefits at the sub-project level, with high economic rates of return. The facility provides additional benefits of attracting new, private sector investment, in order to maximize the benefit of the public sources of funding. The EE facility can eventually become sustainable without the use of concessional terms and with reduced credit enhancements, thereby creating a new and sustainable asset class for EE investments in Brazil.

For more details, see the economic and financial analysis provided in Section D.1 and the Feasibility Study in Annex 2.

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

The total estimated leverage of the GCF loan is about than 1:5 (i.e., every dollar of GCF concessional funding will secure \$5.00 of additional funding for the project from both public (35%) and private (65%) sectors. Private sector is fundamental in order bring scale to investments in EE in these sectors in Brazil.

E.6.3. Financial viability

See the financial analysis provided in Section D.1.

E.6.4. Application of best practices

Experience in providing resources to energy efficiency projects by the private sector via financial intermediaries illustrates the importance of a well-designed and targeted program, with significant technical support provided to several areas of the partner banks.

EE Projects in IFC are a case in point. One of the most successful EE programs in IFC is the so called CHUEE project, which started in China in 2006 and is now going through its fourth round. The first two rounds provided sophisticated credit enhancement mechanisms (via an IFC Risk Sharing Facility), (together with a massive technical assistance program) to de-risk investments to be made by partner banks to industrial companies. Project eligibility was clearly defined, and was targeted at large industries, typically the most polluting ones and with the most cost-effective programs in terms of GHG abatement. CHUEE (rounds 1 and 2) were a great success. The Risk Sharing Facility size was larger than US\$ 525 million, having GEF contributing to the first loss and advisory budget. As a result, 178 loans were disbursed, with a total loan value of >US\$783 million. More than US\$1.7 billion of private sector investment was catalyzed. GHG emission reduction achieved 19 million tons CO₂e/year, with annual energy savings of 44.2 million MWh.

The success of CHUEE in the energy efficiency space contrast with a recent loan of US\$ 400 million granted to Itau-Unibanco, a private Brazilian Bank interested in scaling up its portfolio in the green space. IFC was the lead arranger of a syndication, committing US\$ 100 million on its own account. The project was originally designed to support clean energy interventions in renewables, waste management, and energy efficiency. There were no individual targets for each segment. There was no significant technical assistance provided to help Itau originate, assess and monitor EE projects, undoubtedly the most difficult area to be scaled up. In the absence of a dedicated team with a clear target, and without the possibility of offering off-balance sheet financing (an important requirement expressed by industrial companies in Brazil), the syndicated loan invested most of its resources in large scale renewables projects under a project-financing scheme (the “easy path”), which have been sufficiently de-risked in the Brazilian market. Virtually no energy efficiency projects have been supported.

E.6.5. Key efficiency and effectiveness indicators

Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)

(a) Total project financing	US\$ 1,094M*
(b) Requested GCF amount	US\$ 186M*
(c) Expected lifetime emission reductions overtime	17.37 million tCO ₂ eq
(d) Estimated cost per tCO₂eq (d = a / c)	US\$ 63.0 / tCO₂eq
(e) Estimated GCF cost per tCO₂eq removed (e = b / c)	US\$ 10.7 / tCO₂eq

**note: this does not include GCF technical assistance (US\$4M) or grant funding (US\$5M), nor the World Bank's IPF contingent loan (\$200M), as the base-case scenario assumes it will not be utilized.*

Describe the detailed methodology used for calculating the indicators (d) and (e) above.

I. Public Street Lighting

Identification of the Baseline Scenario

The analysis assumes that, without this project, municipalities would slowly replace burned out lamps in the existing inventory with LEDs. However, given the municipal constraints on financing (under the Fiscal Responsibility Law, described in detail in the Lighting Brazilian Cities report), cities would be limited to using only surplus revenues available from their street lighting budget, after paying for all electricity and O&M costs. For many cities, the source of revenue to fund public street lighting is known as COSIP, a tax levied on end-users via their electricity bill.

Baseline Scenario

For the purposes of this analysis, electricity expenditures in the first year are estimated to be approximately R\$ 16.5 million and O&M costs for lamp replacement are R\$ 1.5 million, for a total expenditure of R\$ 18 million.

The Baseline Scenario assumes that the city has a COSIP surplus of approximately 10 percent of their total expenditures, after paying electricity and O&M costs for street lighting. In other words, COSIP collection will be R\$ 20 million, R\$ 18 million of which will be used to cover the current expenditures (including purchase of replacement lamps with conventional technologies such as HPS) with R\$ 2 million in surplus. With this surplus, the city will fund its annual system expansion investments, as well as modernization of the system through the purchase of more efficient technologies such as LEDs. The baseline assumes the city will use 25% of its total surplus to fund the modernization with LEDs, equal to R\$ 0.5 million in year 1.

In practice, this means that under the Baseline Scenario, the city would eventually fully convert its inventory to LEDs; however, due to its financing and budget constraints, the process will take place over a long period of time. By the end of 13 years, the Baseline Scenario assumes that 30 percent of the existing inventory has been replaced by LEDs.

Project Scenario

The Project Scenario, as described above, involves the establishment of a PPP that will finance the upfront cost of modernization of the entire inventory of public street lights with LEDs. By bringing in private sector financing through the FinBRAZEEC facilities, the project is able to overcome the one of the key barriers for implementing a LED modernization project: upfront financing costs under constrained budget and balance sheet conditions of municipalities. The project also overcomes other barriers, including improving municipal capacity and reducing the transaction costs involved in the preparation of projects through standardized approaches, and identifying investors willing to invest in these projects through a de-risked facility.

GCF
core
indicators

For the average city used in this analysis (42,500 points of light), this modernization is expected to take place over a two-year period.

GHG savings

GHG savings are calculated based on the comparison of electricity used for public street lighting in the baseline and project scenarios described above.

The estimated annual electricity consumption in the Baseline Scenario over 13 years is 415 GWh. In the Project Scenario, this number decreases to 222 GWh, a reduction of 46 percent. These savings are due to the efficiency differential between the LEDs and the inventory assumed in the baseline, as described in the assumptions section above.

The annual GWh per year are multiplied by the assumed grid emission factor (GEF) of Brazil using the IFI Harmonized Database. Based on a 50/50 combined margin, the GEF is 0.44 tCO₂e/MWh.

Since public street lighting is partially peak-coincident, and because demand in Brazil is increasing being met with fossil fuels, the actual GEF for public street lighting could be higher based on a time-of-use calculation¹⁹. The few studies available on the impact of climate change on hydropower due to changes in rainfall patterns demonstrate increased use of gas and fuel-oil-based thermal power plants to meet shortfalls. Thus, the annual GHG emissions of the sector in a dry year are four times higher than in years with favorable hydrology²⁰.

The electricity savings are multiplied by the GEF to determine the estimated GHG emission reductions, resulting in approximately 85,000 tCO₂e reduced over the life of the project.

Facility Level Impact

At the facility level, we assume approximately 36 “average” sized public street lighting projects will receive investment. Thus, the total GHG reductions by the EE Facility in the public street lighting sector is 3.1 million tCO₂e.

II. Industry Energy Efficiency

It was assumed that each Project-Portfolio has an abatement potential of 136.5 thousand tons of CO₂e per year and 14.9 million tons of CO₂e over the life of the sub-programme. (see Annex 2 for the definition of a Project Portfolio and Annexes 3.2 and 14 for detailed calculations) The following table presents a breakdown per technology/process intervention.

¹⁹ Public street lighting hours are assumed to be from 6pm to 5am. The first four hours are assumed to be peak coincident during weekday operation.

²⁰ In “dry year” such as 2014, GHG emissions increase up to 60MtCO₂, compared to only 15MtCO₂ on a wet year. (de Gouvello et al, 2017: “Low Hydrology scenario for the Brazilian Power Sector 2016-2030 – Impact of Climate on Greenhouse Gas Emissions”, the World Bank, 138p.)

Technologies within a "Project-Portfolio"	Expected Number of Occurrences	Combined emission factor (electricity) tons CO ₂ e/MWh	Emission Factor (natural gas) tons CO ₂ e/MMBTU
		0.440	0.054
	#	CO ₂ eq/yr.	CO ₂ eq/yr.
1. Installation of Variable Frequency Drivers (VFDs) for large scale pumping systems	5	10,413	0
2. Installation of regenerative burners in natural gas furnaces	3	0	27,037
3. Heat recovery in large high-pressure steam boilers	3	-7,630	21,020
4. Installation of state-of-the-art burners in large furnaces	1	0	30,754
5. Energy optimization - software for energy real-time optimization (ERTO)	1	13,200	0
6. Variable Speed Drivers for Furnace Fans*	0		
7. Optimization of compressed air systems	5	24,913	0
8. Conversion from electrical resistance to natural gas for ageing/homogenizing furnaces of aluminum	3	6,510	2,722
9. Replacement of HVAC and motors of low efficiency	5	6,336	0
10. Improvements in cooling and refrigeration systems	5	1,267	0
TOTAL	31	55,009	81,533
(*) Technology formerly referred as # 6 was eliminated from the Project-Portfolio as originally defined. It will include in the future investment and sanitation efficiency motor and pumping systems. Project identification and studies being carried out.			

The abatement potential for the industrial energy efficiency sub-programme is 14.3 million tons of CO₂. This is a conservative figure, which excludes savings (and GHG) benefits after year 8, in spite of a longer expected life for the IEE intervention. The cut-off point of 8 years was established taking into account a baseline scenario whereby the existing equipment would be replaced after this period, and therefore the benefits would not be additional.

III. Facility Level

Combining the investments of the Facility in the two sectors, we estimate 17.4 MtCO₂e over the life of the facility. The cost per tCO₂e of the total investment (US \$1.094 billion in debt + equity) is \$ 63.0.

The GCF share of the debt investment will be US\$186 million per project. Given that GCF funds are essential to make the project viable, the real impact of the GCF loan can consider the GCF investment per total tCO₂e reduced by the project, which is equal to US\$ 10.7/tCO₂e²¹.

²¹ GCF will also contribute US\$4 million in Technical Assistance and US\$5 million as a grant to support the liquidity of the EE Facility. In the base case scenario, the grant will not be used (it will only be used if the project default rates

	Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only)													
	For every dollar invested by the GCF, the project leverages 5.7 in private capital (55% in debt at the Facility level, and 45% in equity at the sub-project level). <table> <tr> <td>(a) Total co-financing mobilized (b+c+d+e)</td><td>US\$ 1,109 million ²²</td></tr> <tr> <td>(b) - From private sector (Concessionaires- equity)</td><td>US\$ 328 million</td></tr> <tr> <td>(c) - From private sector (Investors- debt)</td><td>US\$ 400 million</td></tr> <tr> <td>(d) - From public sector (WB)</td><td>US\$ 201 million</td></tr> <tr> <td>(e) - From public sector (CEF)</td><td>US\$ 180 million</td></tr> <tr> <td>(f) Requested GCF amount</td><td>US\$ 195 million</td></tr> <tr> <td>(g) Expected leverage factor (g = a / f)</td><td>5.7</td></tr> </table> All figures above are consistent with the financing table presented in Section B.1. Please refer to Section B.1 for more detailed breakdown of co-financing.	(a) Total co-financing mobilized (b+c+d+e)	US\$ 1,109 million ²²	(b) - From private sector (Concessionaires- equity)	US\$ 328 million	(c) - From private sector (Investors- debt)	US\$ 400 million	(d) - From public sector (WB)	US\$ 201 million	(e) - From public sector (CEF)	US\$ 180 million	(f) Requested GCF amount	US\$ 195 million	(g) Expected leverage factor (g = a / f)
(a) Total co-financing mobilized (b+c+d+e)	US\$ 1,109 million ²²													
(b) - From private sector (Concessionaires- equity)	US\$ 328 million													
(c) - From private sector (Investors- debt)	US\$ 400 million													
(d) - From public sector (WB)	US\$ 201 million													
(e) - From public sector (CEF)	US\$ 180 million													
(f) Requested GCF amount	US\$ 195 million													
(g) Expected leverage factor (g = a / f)	5.7													
	Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/programme)													

exceed expectations) and the TA funds will not go directly to project investments, these amounts are not included in this calculation. However, if included, the GCF cost per tCO₂e would be 11.2 \$/tCO₂e.

²² This figure includes the debt of the EE Facility (\$766), less the GCF loan amount (\$186), plus the sub-project equity (\$328) = \$1,094 million (cited above as the project financing. Since the table is the "total co-financing mobilized," it also includes the value of the WB IFP contingent loan (US\$200M) and WB Technical Assistance (US\$1M). The leverage ratio without these additional values would be \$909 / \$195 = 4.8.

* The information can be drawn from the project/programme appraisal document.

F.1. Economic and Financial Analysis

An economic and financial analysis of the project has been conducted at two levels: (1) at the Project level (i.e., benefits and costs arising from the implementation of the energy efficiency projects), and (2) at the facility level impact (i.e. benefits and costs associated with the establishment of the FinBRAZEEC financing facility). For each level, the analysis will be based on a comparison of two scenarios: one without the project and one with the project.

a) Project level analysis

At the project level, the primary economic benefits that will be realized include the energy savings (MWh multiplied by the cost of energy) and climate change mitigation benefits (estimated in terms of tCO₂e avoided, or MWh multiplied by the relevant emissions factors). Other social beneficial externalities are expected, but would be more difficult to quantify a priori. The analysis will be conducted for an average project or an average portfolio of projects for each of the two main categories considered at this stage: (i) LED-based street lighting PPPs and (ii) Energy Efficiency measures in urban industry.

(i) Street lighting

A summary of the major results from the financial and economic analysis can be found in Table 3 below.

Table 3: Summary of Financial and Economic Analysis

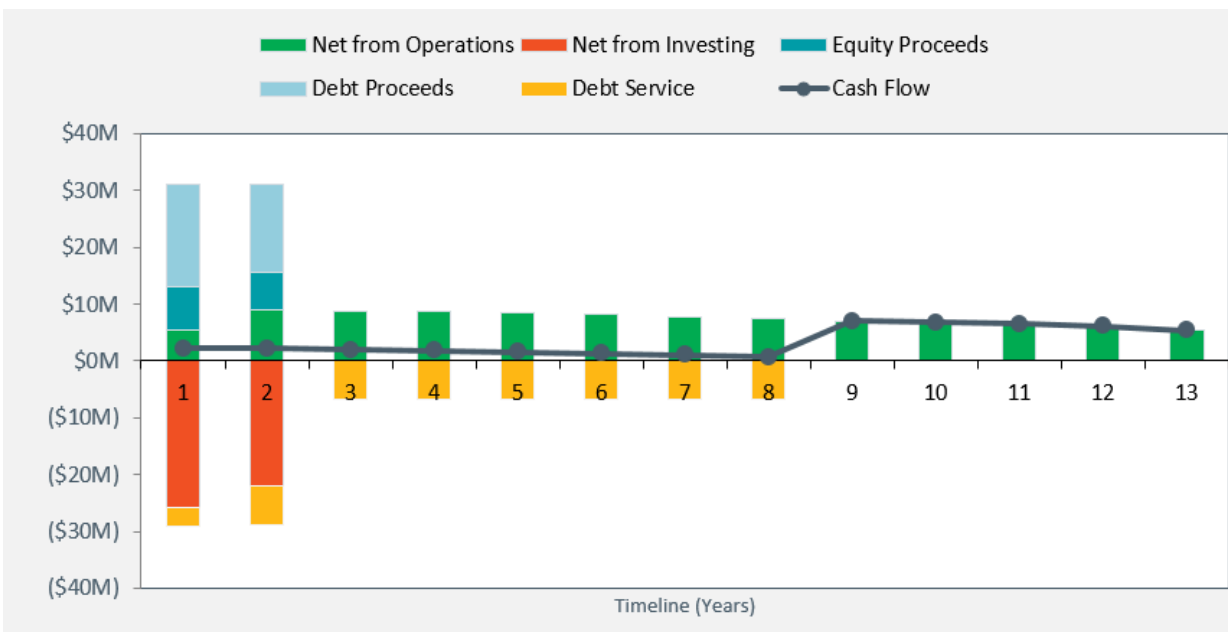
Metrics	Financial Analysis	Economic Analysis
Total investment	R\$ 48 million (US\$15.2 million)	R\$ 30 million (US\$9.5 million)
Total revenue	R\$ 112 million (US\$35.4 million)	R\$ 144 million (US\$45.6 million)
Total electricity savings	193 GWH	193 GWH
Annual GHG reductions	84,952 tons CO ₂ e	84,952 tons CO ₂ e
Total net present value (NPV)	R\$ 6.3 million (US\$2.0 million)	R\$ 59 million (US\$18.7 million)
Internal rate of return (IRR, net)	15%	45%
Payback period, non-discounted	5.5 years	2.7 years
Return on investment	1.3	3.8
WACC (net of inflation)	10.9%	3.3%

The total electricity savings from the sub-projects are estimated at 193 GWh over the 13-year life of the project. The GHG reductions associated with the energy saving are equal to approximately 85,000 tCO₂e over the 13-year period.

The financial returns of the project demonstrate that the sub-projects are financially viable under the assumptions used. The real expected internal rate of return (IRR) of 15 percent (compared to the real WACC of 10.9 percent) and a NPV of R\$6.3 million (US\$2.0 million). The expected payback period is 5.5 years. The total investment is R\$ 48 million (US\$15.2 million) and the expected non-discounted savings on electricity and O&M expenditures (limited to those associated with lamp replacement only) is R\$112 million (US\$ 35.4 million).

Figure 14 below shows the expected cash flows of the project.

Figure 5: Financial cash flows of the project (R\$)



However, it is possible that the market will require higher returns on equity than offered by this project (approximately 21 percent net) given that these are investments in a new asset class with little track record in Brazil. Moreover, debtors may require a higher Debt Service Coverage Ratio (DSCR) than is current forecast for these projects – in this case, the minimum is 1.10.

The results of the economic analysis show that the subproject has an expected economic internal rate of return (EIRR) of 45 percent with a payback period of 2.7 years. The EIRR is very attractive, primarily due to the assumptions for the unsubsidized cost of electricity and the inclusion of the social benefits of carbon emission reductions (see Carbon Accounting section below for details on GHG emissions calculations, and the Assumptions above for the value of carbon emission reduction benefits). The expected economic cost of the investment (excluding taxes) is R\$ 30 million (US\$34.5 million); expected non-discounted economic savings on energy and O&M expenditures are R\$ 144 million (US\$45.6 million).

The strong economic returns of the project demonstrate the value-for-money proposition that investments in public street lighting modernization with LEDs offer to society as a whole.

Financial risk assessment

The analysis identified two key financial risks to the project returns:

- Increase in CAPEX due to higher than expected price of LEDs.
- Decrease in the price of electricity.

The switching values assessment is shown in Table 4. The hurdle rate is assumed to be the WACC.

Table 4: Switching Value Assessment

	Unit	Baseline	Switching Value	
Cost of equipment (LEDs and smart controls; CAPEX)	R\$	1,067	1,286	A 21% increase in the cost of LEDs would lead to a NPV of zero for this project. The baseline price is deemed to be conservative, considering that larger scale projects (such as Belo Horizonte) are expected to achieve substantially lower prices (stakeholder consultations indicate approximately R\$ 870 per point). Continued increase in the scale of LED deployment in the country and competition through the PPP tender processes are expected to put downward pressure on LED prices. However, smaller scale projects may face higher prices due to lower economies of scale in the procurement of LEDs.
Electricity tariff	\$R/kWh	0.44	0.37	A 20% decrease in the cost of electricity for public street lighting (in real terms), would lead to an NPV of zero for this project. Given the recent crises in the Brazilian power sector and the resulting increase in dispatch of expensive thermal generation, a decrease in electricity prices of this magnitude is deemed to be unlikely.

Please see Annex 2 for more details on assumptions and market context.

Other benefits of these projects not quantified here include improved disposal of existing inventory of lamps will result in reduction in mercury pollution. Social benefits are expected to include improved quality of life for urban dwellers, increased perception of safety on streets due to better illumination (which could, in term, improve economic opportunities, particularly for women who otherwise would not feel safe leaving home at night). Finally, the projects will result in increased capacity of stakeholders to prepare high quality projects (resulting from the provision of a toolkit, example projects, etc.), which can benefits stakeholders not directly associated with the EE Facility.

(ii) **Industrial sector projects**

Financial analysis

The Concept of Project Portfolio

For simplification, it was assumed that the implementation a large IEE program will involve the scaling up of a series of notional "Project Portfolios". Each Project Portfolio encompasses 10 different technology/process interventions (e.g. HVAC, energy efficient motors, heat recovery, variable speed pumps, etc.), based on audited projects among industrial clients. Each the Project Portfolio represents a typical set of interventions in the industrial energy efficiency space which may be "scaled up" in modules depending on the availability of financial resources.

For this analysis, very high-capital intensive projects, such as co-gen facilities, were not included in the Project-Portfolio. The idea was to start with faster payback projects, able to generate sufficient returns to both the SPV and the host client, and still generate cash that can be used to reinvest in new projects.

Each Portfolio Project represents investments of about R\$ 123.7 million (approx. US\$ 39.1 million). The type of projects (referred to as 'sub-projects' of the Project Portfolio), expected number, and investment costs are described in the table below.

Technologies within the Project Portfolio (*)	Expected Number of Occurrences	Capital Investments
	#	(R\$ 000)
1. Installation of Variable Frequency Drivers (VFDs) for large scale pumping systems	5	21,578
2. Installation of regenerative burners in natural gas furnaces	3	62,413
3. Heat recovery in large high-pressure steam boilers	3	4,725
4. Installation of state-of-the-art burners in large furnaces	1	13,283
5. Energy optimization - software for energy real-time optimization (ERTO)	1	1,510
6. Variable Speed Drivers for Furnace Fans	0	
7. Optimization of compressed air systems	5	9,656
8. Conversion from electrical resistance to natural gas for ageing/homogenizing furnaces of aluminum profiles	3	2,370
9. Replacement of HVAC and motors of low efficiency	5	6,750
10. Improvements in cooling and refrigeration systems	5	1,400
TOTAL	31	123,685
(*) Technology formerly referred as # 6 was eliminated from the Project-Portfolio as originally defined. It will include in the future investments in water and sanitation efficiency motor and pumping systems. Project identification and studies being carried out.		

Financial and economic analyses were carried out for each Project-Portfolio and for the IEE facility as a whole. It was assumed that, based on the EE Facility's capital available and the expected allocation of capital to IEE²³, the EE Facility can finance approximately 14 Project Portfolios.

Each Project-Portfolio represents savings of 125 GWh/yr. in electricity and 1.5 million MMBTU in natural gas. At current prices (including taxes) of R\$ 312/MWh and R\$ 38/MMBTU (unless otherwise stated in the calculations), energy savings represent R\$ 24.2 Million/yr. in electricity and R\$ 58.6 Million/yr. in natural gas. It is assumed that the net-operating income of the SPV will be split equally between the SPV itself and the host customer – therefore providing the necessary incentives for the industry to breakeven in the initial years of the project (e.g. savings = fixed payments) and reap all the benefits after a certain period (assumed to be after 8 years – when project has already reached its payback and loans have been amortized).

The results of the financial analysis are shown in the following table.

Financial Analysis ('000s BRL)										
Year		0	1	2	3	4	5	6	7	8
Investments: 1 Project Portfolio		74,211	49,474							
Total Savings			41,382	82,765	82,765	82,765	82,765	82,765	82,765	82,765
Electric Energy = MWh @ R\$/MWh			12,085	24,169	24,169	24,169	24,169	24,169	24,169	24,169
Natural Gas MMBTU/yr @ R\$/MMBTU			29,298	58,595	58,595	58,595	58,595	58,595	58,595	58,595
Revenues for FZE (Assumed 50% of savings for FZE and 50% for host company; also assume that changes happen mid-way through the first year of implementation)			20,691	41,382	41,382	41,382	41,382	41,382	41,382	41,382
Gross Taxes (30%)			6,207	12,415	12,415	12,415	12,415	12,415	12,415	12,415
Tax Shield on Interest			-1,496	-2,493	-2,244	-1,829	-1,413	-997	-582	-166
Net Taxes			4,711	9,921	10,171	10,586	11,002	11,417	11,833	12,248
Net Cash Flow			15,980	31,461	31,212	30,796	30,381	29,965	29,549	29,134
Net Cash Flow After Investments		-74,211	-33,494	31,461	31,212	30,796	30,381	29,965	29,549	29,134
IRR (Real)	17%									
NPV (@10.9% real)	23,161									
Number of Projects Portfolio	14.5									
Total NPV FZE (R\$)	336,708									
IRR on Equity (Real)	19%									
Debt (70% of Investments)	86,580	51,948	34,632							
Interest		0	-4,987	-8,312	-7,480	-6,095	-4,710	-3,325	-1,939	-554
Amortization		0	0	-8,658	-14,430	-14,430	-14,430	-14,430	-14,430	-5,772
Total Debt Service		0	-4,987	-16,970	-21,910	-20,525	-19,140	-17,755	-16,369	-6,326
DSCR			3.20	1.85	1.42	1.50	1.59	1.69	1.81	4.63
Equity Contribution (30% of Investment)	37,106	22,263	14,842							
Free Cash Flow to Equity Holders		-22,263	-19,829	14,491	9,301	10,271	11,241	12,210	13,180	22,808

(*) All figures in constant BRL (R\$) unless otherwise stated

The Project-Portfolio is attractive, with an IRR (real) of 17 percent and a NPV of R\$ 23.16 Million (US\$7.3 million). The entire IEE sub-programme presents the same IRR and a NPV of R\$ 336 million (US\$106 million).

The Project-Portfolio is also bankable at the market rates charged by the facility. The lowest Debt Coverage Ratio, happening in year 3, is about 1.42. The real return on equity (IRR on equity) is 19 percent, which should make the project attractive to several classes of equity holders, such as pension funds, hedge funds and other investors, which are looking for attractive investments in the green space.

Economic analysis

²³ It is assumed that IEE will represent about 52% of expected investments of FinBRAZEEC facility.

The economic analysis builds upon the financial analysis developed in the previous section, with the typical adjustments, such as eliminating taxes from CAPEX and energy sources, excluding financing, depreciation and tax impacts, including a carbon price for CO₂e, and adjusting the discount rate.

The Project-Portfolio is attractive from an economic standpoint, with a real IRR of 37.4% and a NPV of R\$ 2,855 Million (US\$ 904 million). Details are provided in the following table and in Annex 3.2.

Economic Analysis ('000s BRL)											
Year			1	2	3	4	5	6	7	8	
Investments: 1 Project Portfolio (net of IPI = Imposto Sobre Produtos Industrializados)			66,790	44,527							
Total Savings (net of Value Added Taxes = ICMS = Imposto Sobre Circulação de Mercadorias e Serviços)				33,106	66,212	66,212	66,212	66,212	66,212	66,212	66,212
Electric Energy = MWh @ R\$/MWh			9,668	19,335	19,335	19,335	19,335	19,335	19,335	19,335	19,335
Natural Gas MMBTU/yr @ R\$/MMBTU			23,438	46,876	46,876	46,876	46,876	46,876	46,876	46,876	46,876
Revenues for F2E (Assumed 50% of savings for F2E and 50% for host company; also assume that changes happen mid-way through the first year of implementation)			16,553	33,106	33,106	33,106	33,106	33,106	33,106	33,106	33,106
GHG reduction (tons CO ₂ e/yr)			62,511	125,021	125,021	125,021	125,021	125,021	125,021	125,021	125,021
Social cost of carbon (R\$/ton) (Please refer to Annex 3.1)			106	109	112	115	119	122	125	129	
GHG reduction revenue (assuming 100% for F2E)			6,629	13,635	14,022	14,421	14,830	15,252	15,685	16,131	
Total Revenues (w/GHG reduction revenues)			23,182	46,740	47,128	47,526	47,936	48,358	48,791	49,237	
Net Cash Flow			23,182	46,740	47,128	47,526	47,936	48,358	48,791	49,237	
Net Cash Flow After Investments			-66,790	-21,345	46,740	47,128	47,526	47,936	48,358	48,791	49,237
IRR (Real)		37.4%									
NPV (@3% real)		196,397									
At Facility Level, for entire Program											
Number of Projects Portfolio		14.5									
Total NPV F2E (R\$)		2,855,215									
GHG CO ₂ e/eq reduction/year		1,817,553									
Total GHG reduction (cut off point at year 8)		13,631,648									
Cost (in US\$) per CO ₂ abated		41.7									

Other benefits not quantified here including an improvement industrial competitiveness and additional environmental benefits beyond climate change mitigation through the reduction of other local air pollutants or the substitution of refrigerant gases by ones with less GHG impact.

While rebound effect (increase in consumption due to suppressed demand) is typically an important consideration for EE projects, it is expected to be relatively minimal for the projects targeted by this project. In the case of street lighting, consumption is largely determined by lighting needs which are related to timing of sunrise and sunset and, therefore, it is highly inelastic to energy expenditures. In fact, if anything, the modernization with energy efficient LEDs will allow for the use of dimmer controls that can minimize the amount of lighting provided during sunrise and sunset hours, thereby reducing the total amount of lighting demanded by the city. In the case of Industrial EE, the cost of energy is only one of many costs that factor into the decision to increase production, thus mitigating the incentive to increase production solely based on a reduction in energy costs due to installation of EE equipment. Moreover, for a period of time, the industrial entities will use (most of) excess operating revenues to pay the investor for the cost of equipment, reducing the incentive to increase production in the short- and medium-term.

b) Facility level analysis

Please see Sections D.1 for the financial analysis of the facility.

Qualitative benefits of the creation of the FinBRAZEEC facility also include the increase in capacity of CEF and other investors to conduct due-diligence and assess risks of urban EE projects. This will be done through the deployment of tools, trainings and on-the-job learning through the implementation of the facility.

F.2. Technical Evaluation

Energy efficient street lighting

The World Bank has prepared and is publishing an in-depth report on the market for energy efficient investments in public street lighting in Brazil – “Lighting Brazilian Cities: Business Models for Energy Efficient Public Street Lighting” (Meyer et al, forthcoming, 2017). This report is included as Annex 12 to the Funding Proposal, and provides more detail on the technology, barriers to investment, market analysis, and investment needs.

In brief, Public lighting services represent an important energy consumption source for cities. In fact, in Brazil, public lighting represents more than 4% of the country’s total energy consumption, and the energy cost for public lighting already represents the second most expensive item of most municipalities’ budget, surpassed only by payroll expenditures. Therefore, energy efficiency projects in public lighting sector can play an important role for cities’ emission reductions, in addition to offer benefits for the municipality.

Within global scope, participant countries of the 21st Conference of Parties (COP-21) held in November 2015, submitted their contribution proposals to combat climate change. Among other objectives, Brazil committed itself to increase energy efficiency in the electric sector by 10 percent by 2030. Near one-fifth of this goal could be obtained by implementing energy efficient technologies in the Brazilian municipal street lighting sector.

Lamps using the new technology LEDs or Light-Emitting Diodes emerge as a feasible technical and commercial option for investments in energy efficiency in the public lighting sector. LEDs lamps are 40 to 60 percent more efficient in terms of energy consumption than the current technologies installed in the Brazil, in addition to offering important reductions in operational and maintenance costs and the capacity to integrate efficient lamps infrastructure with smart monitoring systems, creating the ‘backbone’ infrastructure for “smart cities”. In addition to this, the higher quality of LEDs has the positive effect of reducing criminality and increasing citizens’ perception of safety.

In 2012, an international non-governmental organization, The Climate Group, was one of the first organizations to conduct studies in cities with public lighting projects with LEDs technology to confirm the performance and acceptance of this new technology. Using the example of twelve cities the study showed that the technology based on LEDs reaches levels of energy savings ranging from 50 to 70 percent reaching up to 80 percent, when combined with smart systems of management and control. In addition to energy savings and reduced maintenance costs, LEDs lighting resulted in a wide range of socioeconomic benefits, such as improvement in lighting quality, improve security and increased local economy activity. Moreover, the adoption of smart control systems allowed greater flexibility in terms of lighting options with increased focus on people.

Urban EE industries

EE technologies vary by industrial subsectors but typical energy-saving measures include:

- **Energy systems.** Upgrading boilers and switching fuels, using electric-driven systems, including compressed air systems, electric chillers, machinery and lighting.
- **Process technology.** Upgrading and replacing equipment, machinery, and facilities.
- **Waste heat and waste use.** Use of waste heat (of hot/warm gases, liquids and solids) and burning combustible waste (gases, liquids, solids).

Energy intensive industrial sectors include cement, steel, textile, pulp and paper, food processing, bricks, and ceramics. Many of these industries are present in Brazilian urban industrial districts. Potential EE measures for cement industry include using roller mill for grinding raw material, improving furnace fans, installation of variable speed drivers, using waste fuel and waste heat recovery. The cement industry is one of the energy-intensive sectors, producing a huge volume of heat exhaust at a temperature up to 350°C. Therefore, the key measure to improve EE is the utilization of heat exhaust for power generation. This is a well-tested solution achieving multiple purposes, including utilizing heat

waste without consuming more fuel for power generation at a sizeable capacity scale, reducing emissions of CO₂, reducing heat and dust emissions into the environment, and increasing efficiency of equipment production line.

EE technologies for small urban industries would include waste heat recovery, construction of closed production lines, replacement of low-performance air compressor, and installation of inverter for motors to operate at low or fluctuating load. Waste heat recovery uses the heat from flue gas (exhaust heat from electric arc furnaces, furnaces, kilns, and so on) to preliminarily heat up the steel scraps before putting into the electric arc furnace for reducing the time in the furnace, saving energy, and improving capacity. It could help heat the steel bars before entering the kiln to save oil for furnaces. Furthermore, it also helps heat up the oil instead of the drying resistors to reduce electricity consumption.

A guidance note will detail eligible criteria and industry specialists and energy audit experts will be included in the subproject due diligence teams. They will ensure that the subprojects are in compliance with Brazilian industrial and technical policies and regulations, fully satisfy the technical eligibility criteria, and are technically feasible. Technical assistance to build capacity for project technical due diligence will also be provided to CEF during project implementation. This will ensure that the design of the sub-projects will be technically sound.

F.3. Environmental, Social Assessment, including Gender Considerations

Environmental and Social risk rating: The project aims to unlock private financing for urban EE projects in Brazil by reducing the credit risk of (i) urban LED street lighting service-providing companies; and (ii) industrial efficiency projects of private sector companies. The project will create an EE Facility managed by CEF, which will offer concessional loans and potentially equity to EE projects in the two priority sectors. The EE subprojects to be financed will be within the existing premises of privately owned industrial facilities and public spaces of middle-sized and large Brazilian cities. Due to this scope of project activities, the social risk for the project is rated as low. The activities are likely to generate environmental and social risks and impacts that are considered low to moderate and therefore the project is considered as having a medium-level intermediation (I2) consistent with the risk category definitions of the GCF.

Environmental Assessment: World Bank OP/BP 4.01 Environmental Assessment is triggered due to the potential adverse environmental and social impacts associated with the financed subprojects under Public Street Lighting and Industrial Energy Efficiency sectors. At this stage, additional sectors may be considered for future development, namely public buildings and water and sanitation services, although the sub-projects to be financed in these sectors are not yet clear. However, the adverse impacts are not expected to be significant, but site-specific, restricted and directly related to the technologies currently being considered as eligible for financing.

The Project (and the World Bank Loan) will support financing for subprojects under energy efficient street lighting, using LED technology and remote controls. Under industrial EE, the technologies envisioned are related, inter alia to:

- a. pumping systems (variable frequency drivers);
- b. heat recovery in furnaces;
- c. improvements in combustion systems;
- d. automation and energy integration;
- e. optimization and energy efficiency in water and pumping systems
- f. compressed air;
- g. energy matrix and fuel switching;
- h. heat, ventilation and air conditioning (HVAC); and
- i. advanced cooling systems.

Category A high-risk subprojects will not be financed by the EE facility and a positive list of projects that can be financed is being developed. It will be ensured that all subprojects/technologies are considered Category B or C. All subprojects will entail replacement or quality enhancement of existing infrastructure/manufacturing facilities.

On the positive side, the project will bring about important benefits to street lighting, urban industries and the environment by contributing to the reduction of GHGs and pollutants, increase energy savings, and encourage the promotion of environmentally good street lighting and industry practices. On the negative side, the project may cause potential environmental impacts during the construction and operation of sub-projects. These impacts include noise, dust, labor safety, and disposal of old parts and equipment which may contain hazardous waste. The possible impacts during the operation of new equipment and facilities will include worker and health safety issues, air pollution, generation of solid waste, wastewater, and disposal of hazardous substances from industries, even though in a less and lower toxicity than those generated from the old technologies and equipment.

It is anticipated that the subprojects will be category B and C, with typical impacts which are assessed as localized, varying from small to moderate scale, and which can be mitigated via good management practices and readily designed mitigation measures.

In addition, the TA component of this project is not expected to have significant environmental and social consequences. In the industrial energy efficiency sector, the project will essentially support technical studies and early operational costs, for which no bank or private investor is willing to invest during the initial labor and investment-intensive months of operation, before revenues are sufficient to support these functions. In the public street lighting sector, technical assistance activities supported by the project will work directly with municipalities and private sector companies helping them to prepare the studies and build the capacities needed to (i) launch successful tenders for PPPs to provide energy efficient street lighting services and (ii) create a new asset class for energy efficiency that can attract the interest of a wide range of investors and bondholders. Helping to draft a finance strategy for public-private-partnerships, these activities may be classified as Type 2 Technical Assistance activities.²⁴ Considering the sector of investment of these PPPs, TA activities may have an indicative EA category C. Therefore, they do not require the preparation of safeguards instruments prior Appraisal or during implementation.

The principal FI for this operation will be CEF, which adopted the Equator Principles in 2009 and is a current Equator Principles Financial Institution (EPFI) member. As such, CEF has in place an Environmental and Social (E&S) Management System that applies the World Bank Group/IFC Performance Standards and EHS Guidelines in its credit review procedures for lending activities. It has an established E&S sustainability unit - National Management of Sustainability and Social and Environmental Responsibility (GERSA) – and publishes annual reports on its E&S performance. General information regarding CEF's E&S Management System can be found at the following links:

- Equator Principles Reporting 2016: http://www.CEF.gov.br/Downloads/sustentabilidade/Equator_Principles_Reporting_2016.pdf
- CEF Sustainability Report 2016: http://www.CEF.gov.br/Downloads/CEF-relatorio-sustentabilidade/Relatorio_de_Sustentabilidade_CEF_2016_EN.pdf

An E&S Management Framework would be the appropriate safeguard instrument for this project, which will be developed by CEF to comply with the World Bank's Safeguard Policies, taking as reference its existing E&S Management System. The Bank will carry out an assessment of CEF's E&S Management System to verify its equivalence and acceptability in view of OP 4.01 and other relevant World Bank operational policies. The scope of such assessment will consider the capacity of the existing E&S Management System to deal with risks and impacts, the guidelines used to select and approve projects, the mechanisms and procedures for monitoring and evaluating risk and impact management processes in the projects it finances, as well as how it discloses information. The team responsible for managing the risks and social and environmental impacts of these projects needs to be evaluated as

²⁴ World Bank Interim Guidelines on the Application of Safeguard Policies to Technical Assistance (TA) Activities in Bank-Financed Projects and Trust Funds Administered by the Bank

well. An E&S Management Framework will be prepared to address any gaps found and ensure compliance with World Bank policies.

Until more information is available on scope of activities and location of project interventions and since additional sectors may be considered for future development, the triggering of Natural Habitats OP/BP 4.04, Pest Management OP 4.09, Physical Cultural Resources OP/BP 4.11, Forests OP/BP 4.36, Safety of Dams OP/BP 4.37 and Projects on International Waterways OP/BP 7.50 will be deferred.

Social Assessment: The project is expected to have overall positive social benefits because it promotes energy efficiency and reduces GHG emissions and other pollutants into the atmosphere. There is growing evidence that improved street lighting is positively associated with (i) increased urban pedestrian and motorized mobility as well as (ii) crime and violence prevention and reduction. The latter effect results from the facts that improved lighting:

- (a) leads to increased surveillance of potential offenders;
- (b) signals community investment in the area and that the area is improving, leading to increased community pride, community cohesiveness, and informal social control; and, then,
- (c) generates the perception that public places are safe.²⁵ Crime and violence are major challenges faced by Brazilian cities.

In short, improving street lighting tends to reduce crime and perception of insecurity. In Brazil, unsafe urban neighborhoods and victimization hamper the most the life and job opportunities open to poor people and, particularly, poor women. The project has also a huge potential to generate jobs in the lighting industry, which currently comprises 604 factories which generate more than 37,000 jobs. These factories are mostly located in the Southeast and South regions and 65 percent of them are small manufactures.²⁶

Social safeguard related impacts: The EE sub-projects to be financed will be within the existing premises of privately owned industrial facilities and public spaces of middle size and large Brazilian cities.

OP/BP 4.12 Involuntary Resettlement. This operational policy is not triggered because these activities will neither require land acquisition, nor cause restriction of access to natural resources in protected areas. Unlikely disruptions of business activities during replacement of lamp bulbs in public areas will be treated under OP/BP 4.01 and addressed in the project's E&S Management Framework (ESMF).

OP/BP 4.10 Indigenous Peoples. Among the Brazilian population, 38 percent of the people who have identified themselves as Indigenous Peoples live in towns and cities.²⁷ They have migrated to these places mostly by reasons related to access to education, health services, and jobs. In some cities (such as Manaus, Campo Grande and Porto

²⁵ B.C. Welsh, B. D.P. Farrington (2008): **Effects of improved street lighting on crime.** Campbell Systematic Review, v. 13, n. 1, 2008, available at: <http://www.campbellcollaboration.org/>. Soper, M. (2015): **Evidence regarding the impact of the street lighting on crime and antisocial behavior.** Castel Hill: The Cambridgeshire Research Group; K. Painter, D.P. Farrington (1999): **Street Lighting and Crime: Diffusion of Benefits in the Stoke-on-Trent Project.** *Crime Prevention Studies*, volume 10: *Surveillance of Public Space: CCTV, Street Lighting and Crime Prevention*, pp. 77-122; K. Painter, D.P. Farrington (1997): **The Crime Reducing Effect of Improved Street Lighting: The Dudley Project.** R.V. Clarke (ed.) *Situational Crime Prevention: Successful Case Studies.* Criminal Justice Press, pp. 209-226.

²⁶ Data from the Brazilian Association of Lighting Industry (ABILUX), available at <http://www.abilux.com.br/portal/conjuntura/1/diagnosticotico>.

²⁷ In Brazil, there are 817,963 persons who are self-identified as Indigenous Peoples. They count for 0.4 percent of the country's population. Most of them (61.5 percent) live in rural areas and demarcated indigenous lands (57.5 percent). They are present in 80.5 percent of the 5,565 Brazilian municipalities. In 64 percent of the municipalities the Indigenous population living in urban areas is smaller than 100 people. Only 0.9 percent of the municipalities have more than 1,000 persons who have identified themselves as Indigenous Peoples.

Alegre), this group of the Brazilian population people occupy a few neighborhoods; however, in most cities and towns, they are scattered across many neighborhoods, which host just a few number of them. Contingent upon the findings of the Social Impact Assessment that will be carried out as part of the project's ESMF, OP/BP 4.10 Indigenous Peoples may be or may be not triggered by the project. This Operational Policy applies to Indigenous Peoples that have "collective attachment to geographically distinct habitats or ancestral territories in the project area and to the natural resources in these habitats and territories" or have experienced forced severance" during their life time.²⁸ Most of the Indigenous Peoples living in Brazilian cities have moved there voluntarily looking for access to education and health services as well as job opportunities. Therefore, the majority of Indigenous Peoples living in Brazilian towns and cities do not fit the criteria of collective attachment to the territory of these cities or "forced severance". There is only a small number of towns in which they have faced "forced severance". Upon the selection of cities and neighborhoods to benefit from improvements in street lighting, CEF will communicate to the World Bank if activities are envisaged in those very few and exceptional cities, towns, and/or neighborhoods into which Indigenous Peoples have been driven in "forced severance". In this unlikely situation, OP/BP 4.10 will be triggered and measures will be devised to ensure proper consultation and participation of Indigenous Peoples.

Non-social safeguard related impacts: The project ESMF developed under the framework of OP 4.01 will address the project's social impacts. The sub-loan agreements will specify that participating industrial enterprises and street lighting concessionaires must fully comply with the existing national labor laws, including those related to children, women and forced labor. It will also include appropriate mitigation measures for the unlikely cases in which the works of replacement of street lighting lamp bulbs may provoke disruption of formal and informal business.

Grievance Redress Mechanism: The ESMF will also set principles and procedures for operationalization of a grievance redressing mechanism, developed from the channels, processes and procedures that CEF has in place. In addition, communities and individuals who believe that they are adversely affected by a World Bank-supported project may submit complaints to existing project-level grievance redress mechanisms or the World Bank's Grievance Redress Service (GRS). The GRS ensures that complaints received are promptly reviewed in order to address project-related concerns. Project affected communities and individuals may submit their complaint to the World Bank's independent Inspection Panel which determines whether harm occurred, or could occur, as a result of World Bank noncompliance with its policies and procedures. Complaints may be submitted at any time after concerns have been brought directly to the World Bank's attention, and World Bank Management has been given an opportunity to respond. For information on how to submit complaints to the World Bank's corporate Grievance Redress Service (GRS), please visit www.worldbank.org/grs. For information on how to submit complaints to the World Bank Inspection Panel, please visit www.inspectionpanel.org.

Gender. As part of the Social Impact Assessment, the World Bank is conducting a gender sensitive analysis. The analysis aims to understand key gender issues in project areas and potential entry points and opportunities to address them through the interventions financed by the project. The gender analysis will assess potential impacts at community and individual levels once energy efficiency investments – particularly, street lighting – under the proposed project are introduced. The specific objectives of the assessment are to identify whether the implementation of street lighting interventions in areas with high levels of victimization and perceptions of insecurity will contribute to a higher perception of safety, especially among women, decreased crime rates for selected types of crimes, and an overall perceived improvement in quality of life. The assessment will also look at potential benefits of the financing of EE in urban industry and its accompanying TA component on the selection and treatment of women employees in the industry. There is growing evidence that (i) women victimization, fear of victimization and perception of insecurity may hinder their overall mobility and influence their work choices and options, their ability to take on extra work activities and (ii) street lighting

²⁸ "Forced severance" refers to the loss of collective attachment to geographically distinct habitats or ancestral territories occurring during the concerned group members' lifetime because of conflict, government resettlement programs, dispossession from their lands, natural calamities or incorporation of such territories into an urban area.

can have a specific impact on women's perception of safety leading to common recommendations of improvements in lighting near bus stops and near public toilets are some of the most common recommendations.²⁹ Since 50 percent of the investments in the CEF EE facility component will be targeted at street lighting projects, it is expected that the project will have a significant positive impact on women's perception of safety, mobility and work choices.

F.4. Financial Management and Procurement

Financial Management: A preliminary financial management (FM) assessment was undertaken and concluded that the project, upon observation of the action plan to be agreed, has adequate FM arrangements acceptable to the World Bank. The overall arrangements in place for implementing the project provide reasonable assurance that the proceeds will be used for the purposes intended. CEF has sufficient capacity to fulfill its FM responsibilities., and the project will be executed by a dedicated team, with adequate experience in World Bank funded operations and respective financial management arrangements, with moderate satisfactory performance. CEF is required to have annual financial statements prepared in accordance with relevant acceptable standard and audited by independent external audit firm, and Brazil oversight auditing entities, including the General Controller of the Union (CGU), that will be responsible to audit the program. Given the innovative project design involving multiple agencies, funding sources and the criticality of maintenance of appropriate accounting records with appropriate details, including the loans to the sub-projects and other financing instruments, the FM risk rating is appraised as 'Substantial' after mitigation.

Financial management assessment

A preliminary Financial Management Assessment of CEF was performed in accordance with World Bank OP/BP 10.00 and the Financial Management Manual for World Bank-Financed Investment Operations (effective March 1, 2010 and Revised February 10, 2017). The scope of the assessments included:

- (i) an evaluation of the existing financial management systems in place to be used for project monitoring, accounting and reporting;
- (ii) a review of staffing requirements;
- (iii) a review of the flow of funds arrangements and disbursement methodology;
- (iv) a review of the internal control mechanisms in place;
- (v) format and content of Interim Financial Reports (IFRs); and,
- (vi) a review of internal and external audit arrangements.

The conclusion is that CEF has sufficient capacity to fulfill its FM responsibilities; is adequately exposed to the Bank's financial management procedures having adequately implemented Bank funded operations (such as Integrated Solid Waste Management and Carbon Finance Project, P106702). At this stage, the FM risk is assessed as Substantial, mainly due to:

- project design involving multiple financing agencies;
- need appropriate adjusting its accounting and MIS (management information system) to provide details for the respective operations and timely information to monitor the program and run agreed reports; and,
- possible delays to implement the TA component.

Based on previous project lessons learned, to mitigate implementation delays, at least two dedicated qualified FM staff, identified by appraisal, but no later than the project signing, and responsible to support the multiple agencies and entities involved throughout project implementation. CEF's corporate system (SIAPF) should be adjusted to properly reflect, account project transactions (loans and guarantee operations) and generate reports.

Implementation arrangement

²⁹ "Better lighting, wider pavements: steps towards preventing sexual violence in New Delhi". May 6, 2013. Available at <http://www.unwomen.org/en/news/stories/2013/5/better-lighting-wider-pavements-steps-towards-preventing-sexual-violence-in-new-delhi>.

In order to implement the program, an EE Facility will be created to offer financing (concessional loans and potentially equity) to EE projects. The EE facility will be managed by CEF. The project will be composed by a variety of financing instruments to capitalize the EE facility and reduce its credit risk.

CEF is a financial institution constituted by Decree-Law 759/1969, in the form of a public company and linked to the Ministry of Finance. As part of the national financial system, it assists in the execution of the federal government's credit policy, and is subject to rules and decisions of the federal government competent bodies. CEF implements its main banking activities through the collection and application of resources in commercial, foreign exchange, real estate and rural operations, in the provision of banking services, including the administration of investment funds and portfolios, as well as to implement social public programs. In addition to the intermediation of securities, debit and credit cards, insurance, private pension, capitalization and administration of consortia through equity interests in CEF Seguridade and CEF Participações S/A (CEFPAR).

CEF will implement the program with two main functions:

- (a) to support and supervise EE facility investment lending; and,
- (b) to manage all TA activities being financed under the project.

A specific unit, to be identified under current CEF structure, will host dedicated staff (to work exclusively for the project) who will be responsible for the project financial management including preparation and submission of quarterly IFRs for program expenditures within 45 days of the end of the period covered, and also responsible for the whole program consolidated annual financial statements and audit. The operations manual, which will be developed after project approval, will set out project-specific principles and procedures that are additional to CEF's own current procedures.

External audit

CEF is subject to the Federal Government internal and external control oversight bodies, CGU and Supreme Audit Institution (TCU) policies and procedures. Its internal audit department follows international standard practices issued by The Institute of Internal Auditors (IIA). CEF adopts the Brazilian Corporation Law no. 6.404 / 76, modified by Law no. 11.638/07 and the guidelines of the Brazilian Securities Commission (CVM/SNC/SEP), the Central Bank of Brazil and specific regulations applicable to the finance institutions, under the National Monetary Council. The consolidated financial statements are under the responsibility of its Board of Directors and have been prepared in accordance with International Financial Reporting Standards (IFRS), issued by the International Accounting Standards Board (IASB). Ernst Young Auditores Independentes issued an unmodified opinion over the latest available (FY2015) consolidated Financial Statements; however, due to ongoing investigation by the federal public authorities of possible non-compliance with laws and regulations by CEF employees and administrators, it is not possible to predict their effects on the consolidated financial statements.

For the purpose of the project, the external audit will be conducted by CGU due to its mandate to audit externally financed projects at the Federal level. The CGU will follow agreed Terms of Reference (TORs) acceptable to the World Bank, and will conduct the audit in accordance with International Standards on Auditing (ISAs), issued by The International Auditing and Assurance Standards Board (IAASB) of the International Federation of Accountants (IFAC) or national auditing standards if, as determined by the World Bank, these do not significantly depart from international standards.

The auditors will be required to issue an opinion on the project annual financial statements, and produce a management letter, where any internal control weaknesses will be identified, contributing to the strengthening of the control environment. The auditor's report will be submitted to the World Bank no later than six months after the end of the fiscal year. The World Bank will review the audit report and will periodically determine whether the audit recommendations are satisfactorily implemented. The World Bank also requires that the borrower disclose the audited project financial statements in a manner acceptable to the World Bank and following the World Bank's formal receipt of these statements

from the borrower, the World Bank will also make them available to the public in accordance with The World Bank Policy on Access to Information.

Disbursement arrangements

The following disbursement methods will be available: Advance, Reimbursement and Direct Payment. Disbursements will be primary based on the use of Advances. The Bank will advance funds into a segregated Designated Account, maintained exclusively for management of loan proceeds, opened in US dollars or Brazilian Real at CEF, in the name of the EE facility. The ceiling of the Designated Account will be fixed as US\$196 million equivalent. The CEF will report on the use of Advances and Reimbursement requests through IFRs, and Direct Payments will be documented by records – copy of the invoices.

Other investors funds will be managed separately from the Designated Account and will be properly accounted, monitored and reported by CEF.

Retroactive financing for an aggregate amount up to be defined at negotiations will be available for eligible expenditures incurred between January 1, 2018 and signing date of the Loan Agreement and Financing Agreement, provided that relevant World Bank procurement guidelines are followed. The disbursement deadline date will be four months after the closing date of the project. Supporting documentation will be required for documenting eligible expenditures paid from the designated accounts and for reimbursements will be IFRs, and if applicable, a list of payments against the contracts that are subject to the World Bank's prior review, together with records.

The GCF and World Bank IPF contingent Loan project will finance 100 percent (inclusive of taxes) of respective eligible expenditures. Eligible expenditures of EE investment lending will be defined in the Loan Agreement as eligible sub-loans to the industries and to the SPVs, based on approved sub-projects. Eligible expenditures for the technical assistance component are in accordance with the World Bank standard condition.

An additional possible grant of US\$25 million (US\$5 million from GCF and US\$20 million from the Clean Technology Fund (CTF), is being discussed and aim to act as a first loss/liquidity facility for the EE facility in order to reduce CEF and the private sector investors' risk. Details on respective disbursement arrangements and funds flow will be assessed during appraisal.

IPF contingent loan drawdown triggers. The drawdown triggers for this IPF contingent loan will have to be formulated carefully, to strike the right balance between addressing the expected needs and providing enough flexibility to address unanticipated needs without allowing excessive latitude for drawdown. There are no precedents for drawdown triggers for the IPF contingent loan for this operation. The DPF DDO has no drawdown triggers, while the CAT DDO may be drawn down provided a pre-specified trigger (typically the Member Country's declaration of a state of emergency) has been met. The only previous IPF DDO had parametric triggers linked to weather conditions. As this IPF contingent loan would provide similar credit enhancement as a guarantee, the drawdown triggers would be similar to the conditions that would trigger a call on the guarantee.

Funds Flow

The total program amount of FinBRAZEEC EE Facility is \$906M (\$649.7M for the Street Lighting PPP and \$251.3 million for the industrial sub-projects, plus \$5M of technical assistance).

The GCF Loan will flow through the World Bank to the CEF's EE Facility. CEF will leverage the concessional funds and the IPF contingent loan instrument to raise additional debt as well as equity from the public and private sectors. These funds will be managed by the same facility, and then be used to finance EE projects in the industrial and street lighting sectors. The GCF non-reimbursable grants funds of US\$5 million are expected to be used along with the World Bank IPF contingent loan to support CEF's credit enhancement product(s) offered to commercial lenders.

The EE Facility will charge risk-based fees for each product that it offers to potential PPPs or aggregators. Project receiving loans from the facility will need to comply with eligibility criteria to benefit from the IBRD and GCF lending support.

Disbursement to Industrial EE sub-projects: CEF's EE facility will manage US\$251.3 million, funded from different investors, including GCF loan of US\$54.4 million for sub-projects, backstopped by a World Bank IPF contingent loan of US\$54.4 million. Agreed amounts will be disbursed up-front to CEF as seed capital/corpus to finance or provide eligible sub-loans to the EE industries, and will need to comply with eligibility criteria. These disbursements will be treated as eligible expenditure. The World Bank IPF contingent loan of US\$54.4 million (contingent finance) will essentially cover any eventual debtors' shortfall. Disbursement will be contingent upon CEF filing a claim the IPF contingent loan guarantee as specified in the OM.

Disbursement to municipal street lighting PPP sub-projects: CEF's EE facility will manage US\$649.7 million, funded from different investors, including GCF loan of US\$141.6 million, backstopped by a World Bank IPF contingent loan of US\$141.6 million. Agreed amounts will be disbursed up-front to CEF as seed capital/corpus to finance or provide eligible sub-loans to the SPVs that will need to comply with eligibility criteria. These disbursements will be treated as eligible expenditure. The World Bank IPF contingent loan of US\$141.6 million (contingent finance) will essentially cover any eventual default of the SPVs and/or EE facility to its debtors. Disbursement will be contingent upon CEF filing a claim the IPF contingent loan guarantee as specified in the operations manual.

Disbursement for the technical assistance grant: CEF's EE facility will manage US\$4 million, funded from GCF loan of US\$4 million. The amount will flow to CEF as advances to a specific Designated Account, on the basis of quarterly Interim Unaudited Financial Report, submitted to the Bank within 45 days from the end of each quarter in the prescribed agreed format.

Control over the CEF EE facility funds and its utilization will be exercised through specific dedicated ledger accounts in the existing accounting system of CEF, which will be subject to both internal and external audit.

Procurement

The procurement capacity review and risk assessment. During project preparation, the Bank will carry out a detailed assessment of CEF to implement the procurement activities under the US\$4M technical assistance component of the project.

Mitigation measures. It is anticipated that CEF will not be sufficiently familiar with the Bank's procedures for the selection of consultants, so the following key actions will be proposed to the borrower to be implemented throughout project preparation and implementation to mitigate the major procurement-related risks and to strengthen CEF's procurement implementation capacity.

Mitigation Measures

No.	Actions	Responsibility	Date of Completion
1	Appoint procurement officer or hire a procurement specialist with adequate qualifications and experience.	CEF	To be completed before effectiveness
2	Prepare and adopt a project OM, including a chapter on procurement.	CEF	Completed before effectiveness
3	Procurement training for project management unit staff, including initial procurement training at project launch and in-depth procurement trainings during project implementation	CEF/WB	Continued during implementation

Applicable procurement procedures. For contracts financed in whole or in part by the IBRD Loan or IDA Credit, procurement would be carried out in accordance with the Bank's Procurement Regulations for IPF Borrowers dated July 2016. However, the Procurement Regulations do not apply to the procurement of goods, works, non-consulting services, and consulting services financed by the Bank through loans made by eligible financial intermediaries to private borrowers.

Project Procurement Strategy for Development. CEF will prepare a detailed Project Procurement Strategy for Development (PPSD) for the procurement activities under the TA. The PPSP shall address how procurement activities will support the development objectives of the project and deliver the best value for money under a risk-based approach. It shall also provide adequate justification for the selection methods in the procurement plan and for the prior review thresholds.

Retroactive financing: Procurement activities that require retroactive financing will be identified in the procurement plan and will be carried out in accordance with the Procurement Regulations.

Procurement Plan (to be agreed latest by negotiation). A draft Procurement Plan for the first 18 months of implementation, acceptable to the Bank (latest by project negotiation), will be prepared by CEF through the Bank's STEP (Systematic Tracking of Exchanges in Procurement) system. The Procurement Plans will be updated at least annually (or as needed) by the CEF to (a) reflect project implementation; (b) accommodate changes that should be made; and (c) add new packages necessary for the project. Each update will be subject to Bank prior review. Procurement Plans will be published on the Bank's website.

Procurement supervision and post review by the Bank. Contracts not subject to prior review will be subject to post review. The Bank will carry out procurement post reviews on an annual basis with an initial sampling rate commensurate with the risk rating of the project. This rate will be adjusted periodically during project implementation based on CEF's performance. The Bank will also carry out regular procurement supervision missions on a semiannual basis. CEF shall upload all procurement and contract information IN STEP, which will be used to provide the Bank with a consolidated list of all contracts for goods, works, and consultants' services awarded under the whole project. The post review contract sample will be selected from STEP.

G.1. Risk Assessment Summary

The overall project risk is considered as moderate. The main financial, technical, operation, social and environmental risks are described in Section G2. Lack of commitment from the financial intermediary or lack of interest in the proposed financial facility are the most important risks to be addresses. Environmental and social risks are considered very limited given the nature of the proposed project, but will be evaluated prior to negotiations.

G.2. Risk Factors and Mitigation Measures

Please describe financial, technical and operational, social and environmental and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures.

Selected Risk Factor 1

Description	Risk category	Level of impact	Probability of risk occurring
Lack of commitment from the financial intermediary to implement the proposed mechanism: The financial intermediary could eventually resist implementation of an innovative financing mechanism that departs from its usual lines of business; this might derive from the change of commitment and support from the top management and intermediary level and/or from the lack of capacity of staff.	Technical and operational	High (>20% of project value)	Low

Mitigation Measure(s)

CEF, the financial intermediary, has been selected jointly with the PPI secretariat because of the commitment and previous experience with innovative projects involving multilaterals. CEF has demonstrated strong interest in the proposed project concept; technical assistance activities will ensure continued engagement with CEF, including participation in workshops and training sessions with the participation of IFC.

Selected Risk Factor 2

Description	Risk category	Level of impact	Probability of risk occurring
Lack of EE projects willing to benefit from the proposed financing mechanism: For more than a decade, municipalities and industries were granted access to subsidized loans from public banks (BNDES, CEF, BdB). However, with the macro-fiscal crisis, this former financing model has supposedly come to an end. However, in case the pipeline of bankable projects is limited and public banks prefer to attract these few projects for direct financing rather than leverage private resources, stakeholders may revert to the status quo of direct financing from public banks and the proposed mechanism might not receive as many project proposals as expected.	Financial	Medium (5.1-20% of project value)	Medium

Mitigation Measure(s)

In the context of the high-level dialogue organized by the newly created PPI secretariat of the Presidency, which champions public-private partnerships, the main public banks have confirmed their interest in leveraging private resources in PPP solutions rather than only investing their own resources in direct loans; they also expressed interest for the instrument proposed, which has been extensively discussed with them.

Selected Risk Factor 3

Description	Risk category	Level of impact	Probability of risk occurring
Insufficient quality of the projects applying to the facility: There is a general consensus among financing institutions that the quality of projects proposed by project sponsors is insufficient, making them unable to sustain standard due diligences. There is therefore a risk to generate an influx of sub-projects but of insufficient quality.	Technical and operational	Medium (5.1-20% of project value)	Medium

Mitigation Measure(s)

Supported by the GIF, ESMAP and GCF technical assistance resources, the project preparation and the project itself include grant based-assistance activities which will develop a toolkit to improve the preparation and quality of sub-projects. As part of this toolkit, the World Bank conducted and disseminated several pre-feasibility studies and developed user-friendly tool (CityLED) under the Bank executed BRAZEEC projects. Further technical assistance activities will complete the toolkit (i.e. standardized contract templates and guidance for designing PPPs for public street lighting in Brazil) and support the preparation of projects for investment.

Selected Risk Factor 4

Description	Risk category	Level of impact	Probability of risk occurring
Lack of private investors willing to invest at facility or project level: Commercial banks are not used to project finance approach in Brazil and institutional investors (pension funds, insurance companies, etc.) typically invest in government bonds. Thus, there could be a lack of buy-in from these financiers, in particular in case they consider that the residual risk, even with the credit enhancement mechanism implemented by CEF) is still too high.	Financial	Medium (5.1-20% of project value)	Low

Mitigation Measure(s)

The team has carried out extensive consultations and will continue to engage with potential private investors to validate the concept. The level of coverage proposed (as a percentage of the total investment) also offers a margin of maneuver to adjust to market perception.

Selected Risk Factor 5

Description	Risk category	Level of impact	Probability of risk occurring
Insufficient energy conservation and GHG mitigation performance: Sub-projects might deliver less energy savings and emissions reduction than expected. Sub-projects efficiency might be overestimated as a result of uncertainties that are inherent to such ex-ante assessment of performance of simulated technology basket to be implemented by	Technical and operational	Low (<5% of project value)	Low

sub-projects. Energy and GHG savings could underperform due to failure of the installed technologies to perform as expected.			
Mitigation Measure(s)			
Technology risk will be mitigated by establishing eligibility criteria that includes only commercially proven technologies. LEDs as well as the positive list of 10 technologies for IEE are all highly proven technologies. Eligibility criteria will be established to ensure that only certified equipment from reputable manufacturers are procured. For this reason, the probability of the risk occurring is considered low. CEF will also perform in-depth due-diligence on the projects prior to investment. This due-diligence will be performed to assess technology risk and other risks (e.g., credit risk). The syndicated commercial lenders will also conduct due diligence themselves, in parallel to CEF; thereby helping to ensure the due diligence process is very robust. CEF will also be responsible for periodic monitoring of the projects to ensure they are performing as expected. The team will mitigate this by supporting CEF with technical assistance, and providing the necessary resource to evaluate and monitor the sub-projects to reduce the likelihood of technology risks materializing that could reduce energy savings and GHG emissions reductions.			
Other Potential Risks in the Horizon			
<i>Please describe other potential issues which will be monitored as “emerging risks” during the life of the projects (i.e., issues that have not yet raised to the level of “risk factor” but which will need monitoring). This could include issues related to external stakeholders such as project beneficiaries or the pool of potential contractors.</i>			

* Please expand this sub-section when needed to address all potential material and relevant risks.

H.1. Logic Framework.

Please specify the logic framework in accordance with the GCF's [Performance Measurement Framework](#) under the [Results Management Framework](#).

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level³⁰

Paradigm shift objectives

<i>Shift to low-emission sustainable development pathways</i>	The proposed Project is expected to finance about 36 public street lighting and around 14 industrial EE Project-Portfolios, with an estimated lifetime emissions reduction of 17.4 million tons of CO ₂ e. The project will help create new asset classes for street lighting and industrial energy efficiency, putting the right mechanisms in place and building confidence among private sector to invest in EE. It has the potential to be scaled up involving a larger number of municipalities and small and medium-sized industrial companies, helping shift Brazil to low-emission sustainable development pathways.
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Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	

Fund-level impacts

<i>M3.0 Reduced emissions from buildings, cities, industries and appliances</i>	Tonnes of carbon dioxide equivalent (tCO ₂ eq) reduced over lifetime	CEF M&E Reports ³¹	0	5 million	17.4 million	Assumes 36 street lighting projects with 85 thousand tCO ₂ e per project (3.1 MtCO ₂ e total), plus 14.3 MtCO ₂ e for the entire portfolio of IEE projects in the off-balance sheet facility. See Annex 2 (Feasibility Study) , Annex 3.1 and Annex 3.2 for emission reductions calculations of each sector. Baseline figure of 0 is relative to the project scenario
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³⁰ Information on the Fund's expected results and indicators can be found in its Performance Measurement Frameworks available at the following link (Please note that some indicators are under refinement):

[http://www.greenclimate.fund/documents/20182/239759/5.3 -
Performance Measurement Frameworks PMF .pdf/60941cef-7c87-475f-809e-4ebf1acbb3f4](http://www.greenclimate.fund/documents/20182/239759/5.3_-_Performance_Measurement_Frameworks_PMF_.pdf/60941cef-7c87-475f-809e-4ebf1acbb3f4)

³¹ Regular Monitoring & Evaluation reporting from Caixa Economica Federal (CEF) that will be gathered as part of the standard reporting required for an IBRD loan. Assumed frequency every 6 months.

						only – project scenario already takes into account gradual update of EE technologies without the project (only differential impact included in final target)
<i>M3.0 Reduced emissions from buildings, cities, industries and appliances</i>	Volume of finance leveraged (US\$ million)	CEF M&E Reports	0	581+ 350 = 931	575 + 730 = 1306	Public sector: \$195M GCF, \$201M IBRD, \$180 CEF Private sector: US\$ 400 M in private sector debt at EE Facility level and US\$ 330 M in private sector equity at sub-project level

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level

Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Project/programme Outcomes	Outcomes that contribute to Fund-level impacts					
M5.0 Strengthened institutional and regulatory systems	M5.1 institutional and regulatory systems that improve incentives for low-emission planning and development and their effective implementation [number of financial intermediaries strengthened]	CEF Reports	0	0	1	See Procurement Plan for the proposed list of training and capacity building activities
M7.0 Lower energy intensity of buildings, cities, industries and appliances	7.1 Improved energy efficiency of cities, industries and appliances – PSL (GWh/yr)	CEF M&E Reports	0	175 GWh/yr	534 GWh/yr	Annual energy savings from electricity directly attributable to the project, converted to GWh. The baseline value is 0. Projected savings calculated against baseline. Please see assumptions for savings for street lighting in the Feasibility Study (Annex 2).
M7.0 Lower energy intensity of buildings, cities, industries and appliances	7.1 Improved energy efficiency of cities, industries and appliances – IEE (million MMBTU/yr)	CEF M&E Reports	0	0.5 Million MMBTU/yr.	21.15 Million MMBTU/yr.	Annual energy savings from fuel directly attributable to the project, converted to million MMBTU. The baseline value is 0. Projected savings calculated against baseline.

						Please see assumptions for savings for the industrial sector in the Feasibility Study (Annex 2).
M7.0 Lower energy intensity of buildings, cities, industries and appliances	7.1 Improved energy efficiency of cities, industries and appliances - IEE (TWh/yr)	CEF M&E Reports	0	900 GWh/yr	1,747 GWh/yr	Annual energy savings from electricity directly attributable to the project, converted to GWh. The baseline value is 0. Projected savings calculated against baseline. Please see assumptions for savings for the industrial sector in the Feasibility Study (Annex 2).
Project/programme outputs	Outputs that contribute to outcomes					
1a. EE Street-Lighting sub-projects financed	Number of projects	CEF M&E Reports	0	15	36	Sub-projects supported by the FinBRAZEEC Facility
1b. Industry EE sub-projects financed	Number of Project Portfolios	CEF M&E Reports	0	5	14	Sub-project Project Portfolios supported by the FinBRAZEEC Facility
2a. CEF capacity strengthened	Number of trainings conducted or consultants hired	CEF M&E Reports	0	5	6	Estimate 2 consultants hired and 4 trainings conducted
2b. EE street lighting projects prepared	Number of projects	CEF reports, IBRD reports (using GIF funds)	0		10	Estimated number of projects reaching PPP structuring completion using GIF funds and GCF TA funds

2c. Number of IEE projects identified and audited	Number of projects	CEF reports	0		20	Estimated number of projects undergoing investment grade EE audits using GCF TA funds
2.d. Establishment of EE Facility	Number of off-balance sheet funds for IEE established	CEF reports	0		1	At least one F2E-like fund established for IEE off-balance sheet investments
2.e. Establishment of off-balance Sheet Fund for Energy Efficiency projects in urban industries (F2E) – or alternatively lending to existing ones if set up on time by the private sector	Number of off-balance sheet funds for IEE established	CEF reports	0	1	4	At least one F2E-like fund established/funded for IEE off-balance sheet investments
Activities	Description		Inputs		Description	
1.i. Private sector capital mobilized for the FinBRAZEEC Facility	The FinBRAZEEC Facility capitalized by IBRD/GCF loan, WB IPF contingent loan, CEF loan and private sector capital		WB US\$200m, GCF US\$191m, and CEF US\$180m under Component 1			
1.ii. Lending to sub-projects	Debt financing provided by the FinBRAZEEC Facility to eligible sub-projects		Resources mobilized under Component 1			
2.a.i. Technical Assistance to the Financial Intermediary (CEF Economica Federal) for due diligence and facility management	Build capacity within Financial Intermediary (CEF); technical and Legal Support to CEF to develop A/B loans (risk allocation, cost sharing, responsibilities of lenders, legal documents, role of CEF, etc; capacity building to CEF staff on EE technologies, EE project structuring and evaluation; Investment grade audits		TA Budget under Component 2		Consultancy and; legal services; trainings, see procurement plan	
2.b.i Develop the pipeline of bankable EE Street-Lighting sub-projects	Feasibility studies for street-lighting sub-projects		TA Budget under Component 2		Consultancy services, see procurement plan	
2.b.ii.. Increase market capacity to prepare EE Street- lighting through PPP	Standardization of documents and processes for PPPs, capacity building for municipalities		TA Budget under Component 2		Consultancy and; legal services; trainings, see procurement plan	

2.c.i. TA Activities to develop the pipeline of bankable Industry EE sub-projects	Complementary Industry Sub-sector market studies, ESCO market analysis, market sounding, fiscal studies	TA Budget under Component 2	Consultancy and; legal/fiscal services; see procurement plan
2.d.i. . TA support to establish the EE FinBRAZEEC Facility	Structuring and operationalizing the EE FinBRAZEEC Facility within the Financial Intermediary (CEF Economica Federal))	TA Budget under Component 2	Financial and Legal Studies and Services, see procurement plan
2.e.i. Establishment the Off-balance Sheet Fund for Energy Efficiency projects in urban industries (F2E)	Structuring and operationalizing the F2E; contracting of a Fund Manager	TA Budget under Component 2	Legal, administrative, staff and logistic costs, see procurement plan

H.2. Arrangements for Monitoring, Reporting and Evaluation

Besides the arrangements laid out in the AMA, the project implementation involves (a) monitoring of performance indicators in Section H; periodic progress reports; and (b) midterm review of implementation progress. CEF will be responsible for overall monitoring and evaluation (M&E), including collection of project performance information and reporting on projects impacts and results. For the sub-project level activities, each borrower should enter in an acceptance of conditions agreement with CEF, so that each borrower will be responsible for collecting the required information to report to CEF and the World Bank. The World Bank will follow its standard monitoring and reporting guidelines, which will include semi-annual reports from the beneficiaries to the PIU established within CEF. CEF will aggregate the information and report to the World Bank, who will then share the information with the GCF as per the agreed reporting requirements.

It is noted that the end targets of the project are best estimates as it is not known upfront what subprojects will be financed and the targets may be revised at mid-term review as necessary. By the end of the project, the Bank will conduct jointly with CEF implementation completion assessment and prepare and implementation completion report (ICR) which will review the performance of the project, assess effectiveness and efficiency of project implementation, the achievement of the development objective, and provide relevant lessons learned.

I. SUPPORTING DOCUMENTS FOR FUNDING PROPOSAL

- ☒ Annex 1. NDA No-objection Letter
- ☒ Annex 2. Feasibility Study
- ☒ Annex 3.1. Integrated Financial Model that provides sensitivity analysis of critical elements (xls format)
- ☒ Annex 3.2. CITYLED model
- ☒ Annex 3.3. IEE model
- ☒ Annex 5. Project/Programme Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.) – *see the Accreditation Master Agreement, Annex 1*
- ☒ Annex 6. Initial Environmental and Social Impact Assessment (ESIA)
- ☒ Annex 7. Appraisal Report
- ☒ Annex 8. Map indicating the location of longlisted cities
- ☒ Annex 9. Timetable of project implementation
- ☒ Annex 10. Gender Action Plan
- ☒ Annex 11. TA Procurement Plan
- ☒ Annex 12. ESMAP-supported Study - Lighting Brazilian Cities: Business Models for Energy Efficient Public Street Lighting
- ☒ Annex 13. Project Budget
- ☒ Annex 14. GHG Emissions calculations

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.

No-objection letter issued by the national designated authority

18/01/2018

SEI/MF - 0291557 - Ofício



MINISTRY OF FINANCE
Secretariat for International Affairs
Esplanada dos Ministérios, Bloco P, 2º andar, sala 225
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Ofício SEI nº 1/2018/CODIN/SUEFI/SAIN-MF

Brasília, January 18th 2018.

Mr. Howard Bamsey
Executive Director
Green Climate Fund
175 Art center-daero
Yeonsu-gu, Incheon 22004, Republic of Korea

Re: Funding proposal for the GCF by the World Bank regarding Financial Instruments for Brazil Energy Efficient Cities - FinBRAZEEC.

Reference: Processo nº 12120.100029/2018-41.

Dear Mr. Bamsey,

1. We refer to the project FinBRAZEEC in Brazil as included in the funding proposal submitted by the World Bank to us on October 19th 2017.
2. The undersigned is the duly authorized representative of the Secretariat for International Affairs, Ministry of Finance, the National Designated Authority/focal point of Brazil.
3. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project FinBRAZEEC as included in the funding proposal.
4. By communicating our no-objection, it is implied that:
 - a. The government of Brazil has no-objection to the project FinBRAZEEC as included in the funding proposal;
 - b. The project FinBRAZEEC as included in the funding proposal is in conformity with Brazil's national priorities, strategies and plans;



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c. In accordance with the GCF's environmental and social safeguards, the project FinBRAZEEC as included in the funding proposal is in conformity with relevant national laws and regulations.

5. We also confirm that our national process for ascertaining no-objection to the project FinBRAZEEC as included in the funding proposal has been duly followed.

6. We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Documento assinado eletronicamente

MARCELLO ESTEVÃO

Secretary for International Affairs



Documento assinado eletronicamente por **Marcello de Moura Estevão Filho, Secretário(a) de Assuntos Internacionais**, em 18/01/2018, às 18:10, conforme horário oficial de Brasília, com fundamento no art. 6º, § 1º, do [Decreto nº 8.539, de 8 de outubro de 2015](#).



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Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	FinBRAZEEC: Financial Instruments for Energy Efficiency in Brazil
Accredited entity	World Bank
Environmental and social safeguards (ESS) category	Intermediation 2 (I2)

Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Not applicable
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	2018-01-26
Language(s) of disclosure	Portuguese and English
Link to disclosure	English: http://documents.worldbank.org/curated/en/902691516976014147 Portuguese: http://documents.worldbank.org/curated/en/692321516975881251 An ESMS consistent with the requirements for a category I-2 project is contained in the Environmental and Social Management Framework (ESMF)
Other link(s)	
Resettlement Action Plan (RAP) (if applicable)	
Date of disclosure on accredited entity's website	Not applicable
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Not applicable